Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
And
KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Direct Testimony

of

Climate Leadership and Community Protection Act Panel

Dated: April 28, 2023

Testimony of Climate Leadership and Community Protection Act Panel

Table of Contents

I. Introduction and Qualifications 1

II. Purpose of Testimony 4

III. National Grid's Commitment to New York's Net Zero Future 8

IV. Context of These Proceedings 18

V. Advancing the Goals of the CLCPA 22

1. GHG Emissions Impacts 24
2. Consistency with the Companies' CLCPA Study 36
3. Billed Gas Usage Reductions 45
4. Disadvantaged Communities 47

VI. Earnings Adjustment Mechanisms 53

1. Energy Efficiency EAM: Gas Share the Savings ("STS") 55
2. Energy Efficiency EAM: Gas LMI 57
3. System Efficiency EAM: Gas Demand Response ("Gas DR") 59
4. LMI Engagement EAM: EAP Enrollment 64

VII. Conclusion 70

I. Introduction and Qualifications

Q. Please introduce the members of the Climate Leadership and Community Protection Act ("CLCPA") Panel ("Panel").

A. The Panel consists of Karsten A. Barde, Patricia J. Dorsch, and Meghan McGuinness.

Q. Mr. Barde, please state your name and business address.

A. My name is Karsten A. Barde. My business address is 170 Data Drive, Waltham, Massachusetts 02451.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid USA Service Company, Inc. ("National Grid Service Company" or "Service Company"), a subsidiary of National Grid USA ("National Grid"), as a Director on the National Grid US Policy and Regulatory Strategy team. In that role, I provide services to both The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, the "Companies"). My primary responsibilities include analyzing public policy and advising National Grid's operating companies on the immediate and potential future impacts of public policy, with a particular focus on clean energy transition issues. I also support the

businesses in determining their positions on public policy topics and engaging with policy stakeholders in emerging areas of interest.

Q. Please describe your educational background and business experience.

A. I have a Bachelor of Arts from Dartmouth College in Government and Geography and a Master of Business Administration with a focus on Strategy and Sustainability from the Tuck School at Dartmouth. I have worked at National Grid for nine years in a variety of roles, including new product development, customer strategy, and regulatory strategy. I previously worked at Pacific Gas & Electric in San Francisco, and at a social venture fund in Boston.

Q. Ms. Dorsch, please state your name and business address.

A. My name is Patricia J. Dorsch. My business address is 1650 Islip Ave, Brentwood, New York 11717.

Q. By whom are you employed and in what capacity?

A. I am employed by the Service Company as Director of New York Environmental Sustainability. My primary responsibilities include working with National Grid's New York operating companies, including KEDNY

and KEDLI, to facilitate delivery of National Grid's net zero commitments and oversight of emissions calculations and external disclosures.

Q. Please describe your educational background and business experience.

A. I have a Bachelor of Engineering in Mechanical Engineering from Stony Brook University and a Master of Science in Management Engineering from Long Island University. I have worked for National Grid for 36 years with experience in Load Research, Gas Sales, Technical Support for energy efficiency, customer renewables including solar and small wind, and sustainability.

Q. Ms. McGuinness, please state your name and business address.

A. My name is Meghan McGuinness. My business address is 170 Data Drive, Waltham, Massachusetts 02451.

Q. By whom are you employed and in what capacity?

A. I am employed by the Service Company as Director of US Regulatory Strategy. My primary responsibilities include development of regulatory and policy strategy related to clean energy, climate policy, and innovative regulatory frameworks.

Q. Please describe your educational background and business experience.

A. I received a Bachelor of Arts in Economics and Environmental Studies from Middlebury College in 2000, and a Master of Science in Technology and Policy from Massachusetts Institute of Technology ("MIT") in 2008. Prior to joining National Grid in 2016, I worked on energy and environmental policy and regulatory issues affecting utilities for a number of organizations, including the Bipartisan Policy Center, NERA Economic Consulting, MIT's Center for Energy and Environmental Policy Research, the United States Environmental Protection Agency, and Resources for the Future. I was previously a Principal Analyst in my current department and was promoted to my current role in January 2022.

II. Purpose of Testimony

Q. What is the purpose of the Panel's testimony?

A. The purpose of the Panel's testimony is to demonstrate how the Companies' proposals in these proceedings comport with the requirements of the CLCPA. In the 2019 KEDNY and KEDLI Rate Cases, the Public Service Commission ("Commission") found that Sections 7(2) and 7(3) of the CLCPA were applicable to rate cases. Under Section 7(2), the Commission is to determine whether any administrative approvals issued in these cases would be "inconsistent with or will interfere with the

attainment of the statewide greenhouse gas emissions limits established" in the CLCPA. Under Section 7(3), the Commission is to determine whether administrative approvals issued in these cases would "disproportionately burden disadvantaged communities" and shall also "prioritize reductions of greenhouse gas emissions and co-pollutants in disadvantaged communities . . . ."

As we summarize below, the proposals presented by the Companies are projected to achieve cumulative greenhouse gas ("GHG") emissions reductions for KEDNY of at least 886,000 metric tons of carbon dioxide equivalent (MT CO_2e) and for KEDLI at least 793,000 MT CO_2e over the proposed four-year rate plan presented in the filings. By the end of the four-year period, and not accounting for growth from customer additions or increased demand, KEDNY will have reduced GHG emissions from pre-rate case levels by 374,000 MT CO_2e and KEDLI by 324,000 MT CO_2e.

The Companies' filings also include new and continuing initiatives to benefit customers in Disadvantaged Communities. These include programs to enhance access to the Companies' energy efficiency programs, thereby saving customers money and reducing emissions. They also include initiatives to increase participation in the Companies' Energy

Affordability Program ("EAP"), making customers eligible for other benefits and savings. The Companies' infrastructure programs also include consideration of effects of projects on Disadvantaged Communities.

In addition to describing how the Companies' filings are consistent with and advance the goals of the CLCPA, the Panel also describes the Companies' four proposed earnings adjustment mechanisms ("EAMs").

Q. Is the Panel sponsoring any exhibits as part of the testimony?

A. Yes. The following exhibits were prepared or compiled under the Panel's direction and supervision:

(i) Exhibit___(CLCPA-1): National Grid's *Responsible Business Charter*.

(ii) Exhibit___(CLCPA-2): National Grid's *Clean Energy Vision*.

(iii) Exhibit___(CLCPA-3): Assessment of KEDNY Estimated Greenhouse Gas Emissions Impacts; Rate Year – Data Year 3.

(iv) Exhibit___(CLCPA-4): Assessment of KEDLI Estimated Greenhouse Gas Emissions Impacts; Rate Year – Data Year 3.

(v) Exhibit___(CLCPA-5): Proposals supporting Disadvantaged Communities.

(vi) Exhibit___(CLCPA-6): Gas Share the Savings Metrics and Targets Calculations.

(vii) Exhibit___(CLCPA-7): Earnings Adjustment Mechanism ("EAM") Targets.

(viii) Exhibit___(CLCPA-8): Gas Demand Response EAM Methodology.

(ix) Exhibit___(CLCPA-9): EAM Basis Points and Incentives.

(x) Exhibit___(CLCPA-10): EAM Net Benefits Calculations.

Q. How is the remainder of your testimony structure?

A. The remainder of the testimony:

(1) Describes National Grid's support for New York's clean energy transition.

(2) Explains how these proceedings fit in the context of the various ongoing CLCPA-related proceedings and activities.

(3) Summarizes how the proposals in these proceedings advance the State's energy transition, achieve results that are consistent with the goals of the CLCPA, and satisfy the requirements of the Commission's orders on CLCPA implementation. This includes summarizing the Companies assessments of GHG impacts from the proposals in the cases, as well as how the rate filings may affect Disadvantaged Communities.

(4) Finally, the Panel describes four earnings adjustment mechanisms ("EAMs") the Companies propose for the period covered by these proceedings.

III. National Grid's Commitment to New York's Net Zero Future

Q. Please describe National Grid's commitment to a net zero GHG future.

A. National Grid's vision is to be at the heart of a clean, fair, and affordable energy future. The importance of moving to a low-carbon future cannot be overstated. Economies around the world must reduce their carbon emissions and implement sustainable solutions if we are to avoid the worst impacts of climate change. At the same time, the transition must be fair and equitable, ensuring all customers, particularly the most vulnerable, have the opportunity to benefit from the transition; it should avoid harm to Disadvantaged Communities; and it must be affordable so that no customers are left behind.

The clean energy transition also must reliably meet customers' energy needs. If the process of transitioning to a decarbonized energy system erodes reliability, it will adversely impact the health and economic wellbeing of the State and its citizens and reduce the public's trust in the

benefits of the transition, which will slow the ability to reach our collective decarbonization goals.

Q. What is National Grid's Responsible Business Charter?

A. The *Responsible Business Charter* (Exhibit __ (CLCPA-1)) describes National Grid's approach and commitment to business operations. It recognizes that businesses in society—and especially businesses such as ours that provide essential services to customers and communities—must work towards better ways of doing things for the wellbeing of people and the planet. The Charter describes National Grid's groupwide commitments in the areas we believe we can have the most impact on society: the environment, the communities we serve, our people, the economy, and our governance.

Q. How does the National Grid *Responsible Business Charter* inform the Companies' filings in these proceedings?

A. The *Responsible Business Charter* describes National Grid's commitment to reduce its own GHG emissions (typically referred to as Scope 1 and 2 emissions) to net zero by 2050. The Companies propose a number of initiatives in these proceedings to advance this commitment, including reducing fugitive methane emissions from its gas distribution system, converting fossil-fuel powered vehicles to an electric fleet, and improving

the energy efficiency of its offices and operations centers. The *Charter* also recognizes that most of the markets in which National Grid operates, including New York, have adopted net zero carbon emissions reduction targets. Significant reductions in emissions from our customers' use of natural gas must be achieved if we are to meet these net zero targets. That is why the Companies' proposals in these proceedings include enhancements to our energy efficiency programs, projects to decarbonize the gas supply through low-carbon and renewable fuels, and targeted electrification.

The *Responsible Business Charter* also emphasizes that the clean energy transition should ensure no one gets left behind. Protecting vulnerable customers is central to that commitment, and the *Charter* emphasizes the importance of helping customers understand and manage their energy use, and recognizes the connection between the Companies' infrastructure and other investments and the economic health of the communities we serve.

Q. What is National Grid's *Clean Energy Vision*?

A. The decarbonization of heat is uniquely challenging. Solutions are needed that are cost-effective, minimize customer disruptions, and advance the goal of GHG emissions reductions. In April 2022, National Grid issued its *Clean Energy Vision* (*see* Exhibit __ (CLCPA-2)), which provides a

plan for achieving fossil-free heat by 2050. The *Clean Energy Vision* rests on four pillars: First, accelerating insulation and energy efficiency improvements to buildings; Second, supporting cost-effective, targeted electrification on our gas network, including piloting networked geothermal solutions, to electrify as much as 50 percent of the heating load by 2050; Third, in areas where full electrification may not be practical or cost-effective, providing customers with the tools to pair electric heat pumps with their gas appliances; Fourth, and finally, eliminating fossil fuels from our existing gas network no later than 2050 by delivering renewable natural gas ("RNG") and green hydrogen to customers. This portfolio approach is intended to support affordability of achieving net zero by moderating the amount of required new electric system infrastructure and magnitude of up-front costs to customers, support customer choice, and support energy system resilience through the complementary use of decarbonized electric and gas systems.

Q. How does the National Grid's *Clean Energy Vision* compare to the Scoping Plan issued by the New York State Climate Action Council in December 2022?

A. The Climate Action Council's Scoping Plan recommends sector-specific and economy-wide actions to achieve the goals of the CLCPA and examines different potential pathways to achieving those goals. As the Council notes

in the Scoping Plan, the pathway analyses were "governed by foundational principles of ensuring reliability of the energy system as fundamental to New Yorkers' welfare, safety, and prosperity and the cost-effectiveness of the approaches to achieving the required emissions limits." (Scoping Plan, p. 3). Among the key findings from the Scoping Plan's integration analyses for achieving the CLCPA's emissions reduction goals are that energy efficiency and greater end-use electrification are essential parts of all pathways. National Grid's *Clean Energy Vision* likewise recognizes the need to substantially accelerate and expand energy efficiency in buildings, and to promote greater electrification of heating to substantially reduce delivered gas volumes by 2050. The Scoping Plan calls for "a substantial reduction in fossil natural gas use and strategic downsizing and decarbonization of the gas system." (Scoping Plan, p. 350). It also recognizes the role that renewable fuels may play "to meet customer needs for space heating or process use where electrification is not yet feasible or to decarbonize the gas system as it transitions." (Scoping Plan, p. 361). These findings are also consistent with the *Clean Energy Vision*, which emphasizes decarbonization of the gas network through renewable fuels, as well as targeted electrification to avoid gas network investment where feasible and cost-effective.

Importantly the Scoping Plan also recognizes the obligation of gas utilities to provide ongoing safe and reliable service to customers, and the need for ongoing system investment to meet this obligation. The Commission similarly has stressed that failure to maintain safe and adequate electric and gas systems throughout the State during the energy transition would undermine the intent of the CLCPA.

Q. What are the Companies doing to encourage electrification of heat?

A. As gas utilities, the Companies have limited ability to directly incentivize electrification, other than in certain situations such as implementation of non-pipes alternatives ("NPAs"). As part of the Joint Proposal in the 2019 KEDNY and KEDLI Rate Cases, the Companies committed to a process by which they would refer prospective gas customers who were potential candidates for electrification to Consolidated Edison Company of New York, Inc. ("Con Edison") and PSE&G Long Island and LIPA ("PSEG-LI/LIPA") as part of those companies' heat pump programs. The process has resulted with the Companies referring nearly 800 potential customers to their peer utilities in 2022 – well above the targets established in the Joint Proposal. The Companies propose to continue the referral process in this rate plan, and to work with Con Edison and PSEG-LI/LIPA to better understand the efficacy of these efforts.

The Companies are also working to implement NPAs, which have the potential to enable targeted electrification as an alternative to leak-prone pipe ("LPP") replacement. The Companies will continue to look for opportunities to advance NPAs in the Rate Year and Data Years and propose to continue current NPA commitments, such as annually submitting a request for proposal for NPA solutions and for each company to identify five LPP projects that could be replaced with NPAs.

The Companies also are proposing utility thermal energy network pilots (one for KEDNY and one for KEDLI) in Case 22-M-0429. The pilots, which are planned for Disadvantaged Communities, would evaluate the potential for using geothermal networks to meet customer needs. Among other things, the pilots will evaluate customer adoption, rate designs, and cost recovery.

National Grid also has been monitoring legislative activities aimed at advancing the State's climate goals and implementing recommendations from the Scoping Plan. There currently are legislative proposals that would require most new residential construction to be electric-only. If enacted during the period covered by these proceedings, such legislation would further accelerate end-use electrification beyond what is reflected in the Companies' proposals.

Q. What other steps has National Grid undertaken to date to support the clean energy transition and achievement of CLCPA targets?

A. Specific to reducing overall GHG emissions from their natural gas systems, the Companies have been actively implementing and promoting several initiatives and programs.

- Since 2013, the Companies have replaced more than 1,500 miles of LPP (514 miles at KEDNY and 1,060 miles at KEDLI), with corresponding system methane emissions reductions of 28 percent at KEDNY and 32 percent at KEDLI.

- National Grid's energy efficiency programs reduced over 1,800,000 MMBtu of gas that would otherwise have been consumed by New York customers in 2021 and in 2022; the American Council for an Energy Efficient Economy ("ACEEE") ranked New York's energy efficiency program 3rd in the U.S.[1]

- To address growing customer heating demand, KEDNY and KEDLI have developed alternatives to new infrastructure such as building weatherization programs, NPAs, and industry-leading gas demand response programs to reduce customers' peak period gas use, among other potential solutions.

[1] ACEEE, 2022 State Energy Efficiency Scorecard New York. https://www.aceee.org/sites/default/files/pdfs/State_Scorecard/2022/one-pagers/New_York.pdf

- The Companies actively evaluate opportunities to advance NPAs as part of their capital planning process, including an obligation to submit a request for proposals for NPA solutions and evaluate opportunities to replace LPP projects with NPAs. In August 2022, the Companies submitted filings, individually and with the State's other gas utilities, in the Commission's Gas Planning Proceeding (Case 20-G-0131) regarding NPA evaluation criteria, incentive mechanisms, and cost recovery. The Companies will incorporate any requirements and guidance regarding NPAs from that proceeding into their capital planning processes.

- National Grid is working with the New York City Department of Environmental Protection and Waste Management on the Newtown Creek Biogas Facility. The project is in service, and capable of injecting up to 760 dth/day of biogas into the gas distribution system when operating at full capacity.

- As discussed in more detail in the Gas Supply Panel testimony, National Grid issued a Request for Information in April 2022 that identified potential renewable natural gas ("RNG") and hydrogen projects throughout North America.[2]

[2] National Grid, 2022. https://www.nationalgridus.com/News/2022/08/National-Grid-8217-s-first-of-its-kind-renewable-energy-RFI-validates-a-fossil-free-energy-strategy-for-the-Northeast/

- National Grid is working with the Town of Hempstead, New York, to expand an existing clean hydrogen facility, which will blend hydrogen that is created from renewable energy into an isolated portion of KEDLI's gas distribution system.[3]

- National Grid is a partner with the New York State Energy Research and Development Authority ("NYSERDA") in the development of the Northeast Regional Clean Hydrogen Hub proposal.[4]

- National Grid received a $1 million cost-share award from the U.S. Department of Energy to partially fund a pilot by KEDNY to test gas demand response using hybrid gas-electric heating systems (*see* Matter No. 23-00775).

In addition to efforts to reduce gas system emissions, the Companies' upstate New York affiliate, Niagara Mohawk Power Corporation ("Niagara Mohawk") is making substantial investments in initiatives to enable the State to reach its clean energy goals. These include programs and projects to enable the interconnection and delivery of increasing levels of renewable electric generation, expand the network of electric vehicle

[3] National Grid, 2021. https://www.nationalgridus.com/News/2021/12/National-Grid-and-Town-of-Hempstead-to-Develop-One-of-the-First-Green-Hydrogen-Blending-Projects-in-the-Country/

[4] NYSERDA, 2022. https://www.nyserda.ny.gov/About/Newsroom/2022-Announcements/2022-03-24-Governor-Hochul-Announces-Multi-State-Agreement-on-Hydrogen; and https://www.governor.ny.gov/news/seven-states-northeast-regional-clean-hydrogen-hub-announce-submission-362-billion-proposal-us.

charging infrastructure across the state, and effectively implement energy storage.

As the foregoing shows, National Grid is working hard to reduce GHG emissions and advance the transition to a clean energy future.

IV. Context of These Proceedings

Q. What CLCPA-related proceedings is the Panel aware of that relate to the Companies' rate filings?

A. Since the CLCPA was enacted and became effective, the Commission, the New York State Department of Environmental Conservation ("DEC"), and other State entities have taken steps to fulfill the various mandates and advance the law's objectives. For example, the DEC adopted regulations effective December 30, 2020, establishing statewide GHG emissions limits: a 2030 limit equal to 60 percent of 1990 GHG emissions levels, and a 2050 level equal to 15 percent of 1990 GHG emissions levels. The DEC is required to issue additional regulations by January 2024 to ensure compliance with the statewide GHG emissions limits.

The CLCPA also created the 22-member Climate Action Council charged with developing a scoping plan to "identify and make recommendations on regulatory measures and other State actions that will ensure the attainment of the [2030 and 2050] statewide greenhouse gas emissions

limits established" in the CLCPA. The Council approved the Scoping Plan on December 19, 2022, which outlines strategies to achieve CLCPA goals while prioritizing climate justice and Disadvantaged Communities. National Grid continues to actively participate in upcoming processes and rulemakings that were recommended by the Scoping Plan, such as the statewide cap-and-invest program. The DEC and NYSERDA are working to create the program framework for the cap-and-invest program and to implement corresponding regulations by the beginning of 2024.

The Commission has also proactively engaged in advancing CLCPA initiatives affecting electric and gas utilities in the State. On May 12, 2022, the Commission initiated the CLCPA Implementation Proceeding (Case 22-M-0149) to assess implementation of and compliance with the CLCPA by utilities. The Commission's May 12, 2022, Order in the CLCPA Implementation Proceeding requires gas utilities to:

- Develop a proposal for a GHG emissions inventory report that includes an inventory of total gas system-wide emissions, which the utilities filed December 1, 2022.
- Develop a proposal for a GHG emissions reduction pathway study, which the utilities filed March 31, 2023.

- Include in future rate case filings, descriptions of the investments, capital programs, and initiatives necessary to achieve the objectives in the GHG emissions reduction pathway study proposal.
- Include in future rate case filings an assessment of the GHG emissions impacts of each specific investment, capital expenditure, program, and initiative included in the rate filings.

In addition to the CLCPA Implementation Proceeding, the Commission issued an order on May 12, 2022, in the Gas Planning Proceeding (Case 20-G-0131). Under the Gas Planning Proceeding Order, KEDNY and KEDLI are required to:

- File initial long-term gas plans no later than May 31, 2024.
- File depreciation studies that reflect the effects of accelerating depreciation of new and existing gas assets, as well as different assumptions of customer counts and gas throughput. KEDNY and KEDLI filed such studies November 8, 2022, and supplemented those filings January 31, 2023, and March 17, 2023.
- File an NPA screening and suitability criteria, as well as an NPA incentive mechanism and cost recovery proposal. KEDNY,

KEDLI, and the other utilities filed the NPA proposals on August 10, 2022.

Q. How do the Companies' rate case filings relate to the ongoing CLCPA proceedings?

A. It is the Panel's understanding that the Commission intends to consider and address broad policy issues that may affect all of the State's gas utilities in the context of the ongoing statewide proceedings. For example, the issue of whether gas utilities should be adopting significantly accelerated depreciation methodologies for their gas system assets is being addressed in the context of the Gas Planning Proceeding. Although the proposals presented in this case relate to many of the same issues being considered in the generic proceedings, the Companies have not attempted to presuppose the outcome of those proceedings. Instead, as discussed throughout the Companies' pre-filed testimony, the Companies are proposing to advance targeted, "no-regrets" solutions for addressing the State's carbon reduction and other energy policy goals as part of these proceedings; deferring the larger policy decisions to the generic proceedings.

V. Advancing the Goals of the CLCPA

Q. Do the Companies' rate proposals provide information that would permit the Commission to make the findings required by the CLCPA?

A. Yes, they do. The CLCPA is landmark legislation enacted to lead New York State to net zero GHG emissions by 2050. The law provides for 70 percent renewable electricity by 2030, 100 percent zero-emission electricity by 2040, and reductions in statewide GHG emissions from 1990 levels of 40 percent by 2030 and 85 percent reduction by 2050, with net zero emissions statewide by 2050. The CLCPA also emphasizes that the energy transition must be just and equitable, including a requirement that Disadvantaged Communities receive at least 35 percent of the overall benefits of spending on clean energy and energy efficiency programs, projects, or investments. As the Panel explained previously, the Commission has determined that CLCPA Sections 7(2) and 7(3) apply to rate cases.

In the CLCPA Implementation Order (Case 22-M-0149), the Commission directed utilities in future rate case filings to include an assessment of the GHG emissions impacts of each specific investment, capital expenditure, program, and initiative included in such rate filings, and to describe the investments, capital programs, and initiatives necessary to achieve the objectives in the GHG emissions reduction pathways study.

The projects, programs, and initiatives proposed in these proceedings would promote substantial GHG emissions reductions compared to emissions that would occur absent such measures and provide for continued safe and reliable service to customers. Measures proposed in these proceedings also would provide substantial benefits to Disadvantaged Communities as provided for in the CLCPA and would not disproportionately burden such communities.

Q. What information are the Companies presenting to address issues raised by the CLCPA and the CLCPA Implementation Order?

A. Regarding GHG emissions impacts, the Companies have identified projects, programs, and initiatives included in the cases that are anticipated to have a material impact on emissions, and the Companies have estimated the associated emissions impacts. The Panel also briefly describes the pathways study the Companies performed pursuant to the requirements approved in the 2019 KEDNY and KEDLI Rate Cases and how the projects and programs proposed in this case are consistent with achieving the objectives of that study. Regarding effects on Disadvantaged Communities, the Companies identify projects or programs that are located in or otherwise would affect customers in those communities and provide an overview of the impacts and benefits.

1. GHG Emissions Impacts

Q. How does the Panel present the GHG emissions impacts associated with proposals in the cases?

A. To the extent a project or initiative proposed in these proceedings is expected to have a material impact on GHG emissions, the respective workstream sponsoring that measure (*e.g.*, Gas Infrastructure and Operations Panels, Gas Safety Panel, or the Customer Panel) describes the GHG impact associated with the project or initiative. Our testimony compiles and summarizes the collective GHG emissions impacts identified by each of the separate workstreams.

Q. What process or methodology did the Companies use to estimate the GHG emissions impacts?

A. Working with subject matter experts from each of the involved workstreams, the Panel determined the emissions change expected to result from the implementation of a particular measure. The details of how those amounts were estimated are dependent on the nature of the proposed program or initiative. For example, emissions impacts from LPP retirements are based on reduced fugitive methane emissions and are calculated using accepted emissions factors and the length of pipe replaced; whereas emissions impacts from energy efficiency programs are based on projected reductions in gas consumed by customers. To provide a common

basis for assessment, the GHG emissions impacts are presented in carbon dioxide equivalent ("CO_2e") and a 20-year global warming potential ("GWP") using the DEC's accounting protocols for GHG.

The DEC emission factors were sourced from the November 2022 New York State Oil and Gas Emissions Inventory. In the cases where a DEC emission factor was not available, accepted emission factors from the United States Environmental Protection Agency ("EPA") were utilized. For example, the EPA Emissions & Generation Resource Integrated Database ("eGRID") emission factors were applied to calculate emissions reductions associated with electricity savings.

Q. What are the results of the Companies' assessment of the estimated GHG emissions impacts of the rate cases?

A. In the Rate Year, the proposed projects and programs are estimated to result in a reduction of at least 84,000 MT CO_2e for KEDNY, and at least 77,000 MT CO_2e for KEDLI as compared to emissions immediately preceding the Rate Year, not accounting for growth from customer additions or increased demand. By the end of Data Year 3, KEDNY is projected to have reduced emissions compared to pre-rate period levels by at least 374,000 MT CO_2e and KEDLI by 324,000 MT CO_2e, again not accounting for growth from customer additions or increased demand. Over the proposed four-year rate

period, the Companies project to achieved estimated, cumulative emissions reductions of 1.69 million MT CO_2e (886,000 MT CO_2e at KEDNY and 793,000 MT CO_2e at KEDLI).

Table 1, below, provides a summary of the estimated GHG emissions reductions by major program area by year for KEDNY.

Table 1 -- KEDNY GHG Emissions Reductions (MT CO_2e)

Program Area	Rate Year	Data Year 1	Data Year 2	Data Year 3	Total
Gas Infrastructure & Operations	15,573	32,071	49,184	67,400	164,228
EE/Demand Response	67,857	135,714	203,571	271,428	678,571
RNG/hydrogen	-	-	-	28,593	28,593
Facilities/Fleet	899	2,404	4,416	7,078	14,797
Total	84,330	170,189	257,171	374,499	886,190

Table 2, below, provides a summary of the estimated GHG emissions reductions by major program area by year for KEDLI.

Table 2 -- KEDLI GHG Emissions Reductions (MT CO_2e)

Program Area	Rate Year	Data Year 1	Data Year 2	Data Year 3	Total
Gas Infrastructure & Operations	20,038	40,768	61,670	80,680	203,156
EE/Demand Response	57,128	114,256	171,384	228,512	571,281
RNG/hydrogen	-	-	-	11,288	11,288
Facilities/Fleet	444	1,320	2,311	3,734	7,809
Total	77,611	156,343	235,364	324,214	793,533

Information on the specific programs, initiatives and emissions reductions reflected in the estimates is provided in Exhibits __ and __ (CLCPA-3 and CLCPA-4).

Q. Explain what the Panel means that the Companies' estimated GHG emissions reductions do not account for growth from customer additions or increased demand.

A. The projects and programs the Companies are proposing to reduce GHG emissions are independent of the amount of overall gas demand the Companies serve. In other words, if the Companies did not implement these initiatives, total emissions would be higher by the estimated amounts shown. While the Companies' efforts to reduce energy use and gas emissions will reduce overall gas consumption, the underlying customer demand for gas is the primary driver of overall gas use and any related emissions.

As the Gas Load Forecasting Panel describes in its testimony, each of the Companies is projecting moderate increases in overall gas usage over the four years presented in the filings. The forecasts include estimated impacts of approved energy efficiency programs, reflect anticipated impacts of recent New York City legislation (*i.e.*, Local Laws 97 and 154), include projections of customer counts and use per customer. How actual gas

deliveries compare to the forecasts will depend on how actual conditions compare to the Companies' projections. Actual gas demand will also depend highly on weather, and can be influenced by emergent factors, such as new laws or policies, and the economy. The Companies' forecasts reflect the anticipated impact of known policies and programs to reduce demand, promote efficiency, and support electrification. Although the Companies are actively pursuing additional policies and initiatives to reduce GHG emissions, the Companies must plan based on currently known policies and activities to ensure safe and reliable service.

Nevertheless, the emissions reductions initiatives included in these cases and that are summarized in Exhibits __ and __ (CLCPA-3 and CLCPA-4) are generally independent of, and not expected to be impacted materially by, the overall level of gas usage within the typical margins of forecast error.

Q. Do the estimates above reflect all of the GHG emissions reductions the Companies anticipate that the projects, programs, and initiatives in these proceedings will generate?

A. No. The emissions reflected in Tables 1 and 2, above, are associated with measures whose GHG emissions impacts can be quantified using relatively straightforward and accepted methodologies. An example of this is the proactive LPP program, where established emissions factors are based on

the types of pipe material involved, and changes in emissions are dependent on the length of pipe being replaced. Energy savings from efficiency measures also can be readily expressed in avoided GHG emissions using emissions factors associated with gas and electric energy.

Other programs and investments the Companies are proposing also will directly reduce GHG emissions; however, they are less amenable to quantification with any reasonable degree of confidence. Therefore, although such investments and initiatives would deliver direct GHG emissions reductions, quantifying those reductions at this time would be uncertain. An example of this type of initiative is the Advanced Leak Detection program presented in the Gas Safety Panel's testimony.

Still other investments and initiatives will have more indirect impacts on GHG reductions. That is, they may provide a foundation to enable the Companies to deliver GHG emissions reductions in the future. For example, the Advanced Communication Infrastructure spending is expected to enable the Companies to operate more efficiently and improve the ability to detect leaks in the future; however, this investment will not directly reduce GHG emissions in the near term. Similarly, as discussed in the Customer Panel testimony, the Companies are proposing new customer initiatives such as an enhanced language access program and a weatherization health and

safety program. Although spending for these programs will not directly reduce GHG emissions right away, they will provide necessary foundations for enrolling greater numbers of non-English speakers in the Companies' energy efficiency programs and enable more income-eligible customers to participate in home weatherization, which will drive future GHG emissions reductions.

Still other investments may be needed to ensure the Companies can continue to provide safe and reliable service and are expected to have little or no impact on GHG emissions (*e.g.*, Operator Qualification training, as described in the testimonies of the Gas Infrastructure and Operations Panels and the Gas Safety Panel).

Finally, some of the projects or programs supported by the Panel's testimony may actually increase emissions slightly on a temporary or sustained basis, but are necessary to enable the Companies to continue to provide safe and reliable gas service. Such investments and their associated emissions impacts are reflected in Exhibits __ and __ (CLCPA-3 and CLCPA-4).

Q. Please describe further the programs and investments the Companies are proposing in these proceedings to reduce GHG emissions.

A. The Companies propose several programs and investments that the Panel summarizes below. Additional details on the programs and investments can be found in the respective sponsoring testimonies.

Energy Efficiency; Customer Programs to Enhance Participation

The Companies' Customer Panel describes energy efficiency programs based on savings targets established in the New Efficiency: New York ("NE:NY") Order (Case 18-M-0084), that are projected to deliver significant GHG emissions reductions. The Companies anticipated these programs may directly reduce CO_2e emissions by approximately 67,857 metric tons annually for KEDNY and 57,128 metric tons annually for KEDLI, such that by the end of Data Year 3, GHG emissions are projected to be more that 271,000 MT CO_2e less for KEDNY and over 228,000 MT CO_2e less for KEDLI than if the programs had not been implemented. To the extent the Companies' energy efficiency programs are revised in connection with the NE:NY Interim Review currently underway in Case 18-M-0084, the projected emissions reductions would be affected accordingly.

The Customer Panel also describes the Companies' proposal for funding for several incremental full-time equivalent ("FTE") positions to increase and

accelerate the reach and adoption of the Companies' energy efficiency programs.

Other program proposals, including expanded outreach to Disadvantaged Communities, an energy efficiency language access initiative, and a digital building replica pilot, will also tend to reduce GHG emissions—albeit less directly—and will pave the way for more widespread adoption of building efficiency across communities and customer segments necessary to achieve CLCPA goals.

Infrastructure Investments and Operations

The KEDNY Gas Infrastructure and Operations Panel presents a number of investments that will directly reduce emissions by nearly 96,000 MT CO_2e annually by the end of Data Year 3, while the KEDLI Gas Infrastructure and Operations Panel presents projects that will directly reduce annual emissions by nearly 92,000 MT CO_2e over that same period. A substantial portion of these reductions are from the retirement of LPP. Since 2013, these programs have enabled KEDNY and KEDLI to reduce system methane emissions by 28 percent and 32 percent, respectively. Other projects that contribute to these emissions reductions include innovative projects to facilitate the interconnection of RNG into the network to reduce upstream emissions and displace fossil natural gas. As described in the

testimonies of the Companies' Gas Infrastructure and Operations Panels, the RNG projects will reduce local emissions to the atmosphere at the locations where the biogas is being captured, and displace fossil gas consumption. However, for purposes of the analyses in these cases, the Companies are only reflecting the avoided upstream emissions associated with the locally produced RNG and are not claiming the additional emissions reductions benefits associated with the typical life-cycle analysis of RNG.

Gas Safety

The Gas Safety Panel describes several projects and programs that will directly reduce emissions. Such programs include Advanced Leak Detection, Inside Service Line Inspection, Advanced Communications Infrastructure program, which will potentially minimize the impact and duration of emerging leaks, and enhancements to the Companies' Damage Prevention programs. These programs are specifically designed to reduce or avoid methane emissions, but the amounts of such reductions are not amenable to direct quantification. Therefore, although it is without question that such initiatives are aligned with achieving CLCPA goals, the Companies' do not provide specific estimates of the amount of GHG emissions reductions that the programs would deliver.

Facilities and Fleet

The Companies' Gas Infrastructure and Operations Panels described several facilities and fleet projects that will directly reduce GHG emissions. The projects include replacing existing lighting with highly efficient LEDs, replacing decades-old roofs, and replacing old heating, ventilation, and air conditioning ("HVAC") systems with new, high efficiency systems. The Companies also are proposing to fully electrify their light duty fleet by 2030 and install associated charging infrastructure at their facilities. By the end of Data Year 3, the Companies' facilities and fleet investments together are expected to deliver emissions reductions of approximately 7,100 MT CO_2e for KEDNY and 3,700 MT CO_2e for KEDLI.

Economic Development

As described in the Customer Panel testimony, the Companies are proposing to discontinue two existing economic development programs: the Capital Investment Incentive Program and the Industrial Building Redevelopment Program. Both programs included considerable focus on helping customers develop or upgrade end-use gas infrastructure; closing the programs to new customers will reduce incentives to expand gas usage. The Companies' remaining proposed economic development initiatives include three programs intended to reduce emissions: Economic Development and the Future of Heat, Sustainable Gas and Economic

Development, and Cleantech Incubation. Each of these programs enables the development and deployment of new technologies as alternatives to traditional natural gas upgrades and support clean tech entrepreneurship. Finally, the Companies are redesigning their economic development programs to prioritize benefits to Disadvantaged Communities. The Companies' proposed program modifications provide the coordinated approach to prioritizing Disadvantaged Communities and stimulating growth in the clean energy economy that is required under the CLCPA.

Information Technology

The Information Technology and Digital ("IT&D") Panel supports a number of projects that provide the necessary foundation and technology infrastructure needed to realize many of the Companies' clean energy initiatives. As the Customer Panel describes, IT&D investment is needed for adoption of the Clean Energy 2.0 system to enable the Companies to implement their energy efficiency programs more effectively and efficiently, which is particularly important given the critical role increased energy efficiency will play in achieving the State's CLCPA goals. Incremental IT&D investment is also needed for the Green Button Connect information-sharing framework. The Green Button Connect program will enable customers to easily share energy usage information with authorized third parties, increasing access to products and services to help customers

manage their energy use and reduce overall emissions. Although the effect on GHG emissions from such investments is generally indirect, such modernizing investments are nevertheless foundational and essential to the Companies' abilities to deliver on emissions reductions targets.

More detail on the projects listed above can be found in the respective testimonies of the Gas Infrastructure and Operations Panels, Gas Safety Panel, and Customer Panel and corresponding exhibits.

2. Consistency with the Companies' CLCPA Study

Q. Please describe the GHG emissions reduction pathways study the Companies performed to analyze the impacts of the CLCPA on their gas business.

A. The Joint Proposal adopted (with modifications) in the 2019 KEDNY and KEDLI Rate Cases required the Companies to complete a CLCPA Study to "analyze[] the scale, timing, costs, and customer bill impacts of achieving significant, quantifiable reductions in carbon emissions from the use of gas delivered by the Companies in their service territories" and "provide further analysis on the projects and programs needed to achieve specific decarbonization goals." (p. 77). The Joint Proposal also provided that the Study would "identify potential barriers to achieving the targeted carbon emissions reductions and recommended solutions" and

"incorporate and respond to any findings or guidance of the New York State Climate Action Council." (p. 77-78). The Study analyzed impacts and barriers based on three decarbonization pathways: two electrification-focused scenarios identified in the Climate Action Council's Integration Analysis and a third pathway consistent with National Grid's *Clean Energy Vision*. The Companies submitted the study in the dockets for the 2019 KEDNY and KEDLI Rate Cases on March 17, 2023.

Q. How do the scenarios considered in the CLCPA Study compare to the Companies' gas demand forecasts?

A. The scenarios considered in the CLCPA Study are not forecasts. Instead, they reflect different combinations of decarbonization measures that achieve CLCPA targets by 2050, and project their implications for gas demand. These include varying degrees of expanded and accelerated energy efficiency measures, rapid acceleration of electrification, and integration of renewable fuels, and the CLCPA Study evaluates the implications of different technology mixes, without consideration as to whether the necessary economic, market, behavioral, or policy conditions exist to support the outcomes. All scenarios considered in the CLCPA Study project declines in delivered gas volumes beginning between 2025 and 2030. The Companies' forecasts, on the other hand, are intended to inform the Companies' supply and infrastructure requirements to safely

and reliably serve the demand that can reasonably be expected over the period covered by these filings. Forecasts consider the impacts of economic and weather trends on gas demand, while also reflecting the impacts of enacted policies and established programs – including ongoing efficiency and electrification efforts on gas demand. The forecasts project modest, albeit relatively flat, demand growth for the proposed rate period over which the Companies must plan to serve customers safely and reliably.

Q. Are the Companies participating in another GHG emissions reduction pathways analysis process?

A. Yes. In the CLCPA Implementation Order (Case 22-M-0149), the Commission directed the State's gas utilities to develop a proposal for a GHG emissions reduction pathways study (Study Proposal). Like the Companies' CCLPA Study, the pathways Study Proposal directed by the CLCPA Implementation Order is to include an analysis of the "scale, timing, costs, risks, uncertainties (translated into sensitivity analyses around key cost and availability assumptions), and customer bill impacts of achieving significant and quantifiable reductions in GHG emissions from the use of gas delivered by the Utilities." (CLCPA Implementation Order, p. 26). The Study Proposal also is to include three specific components:

- A coordinated long-term gas sector decarbonization pathway analysis through 2050.
- A coordinated near-term plan to address actions needed to achieve statewide decarbonization targets through 2030.
- Individual, long-term utility decarbonization plans to achieve each utility's share of statewide decarbonization targets through 2050.

The Companies, together with the State's other gas utilities, engaged in the stakeholder process set forth in the CLCPA Implementation Order and filed their Study Proposal on March 31, 2023, in Case 22-M-0149.

Because the Study Proposal does not include specific, actionable initiatives at this point, and has not yet been considered by the Commission, the Companies' proposals in these proceedings are informed by the CLCPA Study filed on March 17, 2023.

Q. What are the main findings of the Companies' CLCPA Study?

A. Key findings from the CLCPA Study include:

- Many essential next steps to enable decarbonization are common across all decarbonization pathways. These include increasing funding to support energy efficiency and customer-side technologies, policies to support development of supply and integration of renewable fuels, and

ensuring improved coordination and planning across gas and electric networks.

- The extent of customer-side and energy system investments needed to achieve building decarbonization means affordability and equity considerations must be prioritized and addressed. Policy and regulatory decisions should provide particular support for low- and moderate-income ("LMI") customers and Disadvantaged Communities, and minimize adverse cost impacts.
- Achieving net zero will be costly under any scenario, but the strategies identified in National Grid's *Clean Energy Vision* support a more affordable pathway to net zero through lower energy system and total customer costs compared to alternatives that electrify the vast majority of heating demand. Mitigation of electric peak demand growth through the utilization of hybrid heating systems is a major source of energy system cost savings.
- In all scenarios, gas customer counts and delivery volumes are projected to decline, though to a lesser degree under the *Clean Energy Vision*. This decline raises challenges to long-term gas network affordability that must be addressed through regulatory and policy reforms. Reforms to depreciation will be necessary, potentially in combination with additional approaches to support recovery of gas network costs, while protecting Disadvantaged Communities and low-income customers.

Q. Did the CLCPA Study recommend specific actions?

A. The CLCPA Study identifies a number of next steps for the Companies based on the decarbonization pathways analysis. Given the commonalities across decarbonization pathways, these next steps are not dependent on selection of a specific pathway, and they address no-regrets focus areas: (1) affordability and equity, (2) infrastructure, (3) technology and workforce, and (4) demand reduction.

With respect to affordability and equity, recommended next steps include programs and initiatives to target incentives to LMI customers and Disadvantaged Communities, consideration of opportunities to prioritize delivery of infrastructure benefits to Disadvantaged Communities, and consideration of modified depreciation approaches to balance near and long-term gas network affordability.

With respect to infrastructure, recommended next steps include LPP reduction strategies to prioritize near-term safety and emissions benefits, while assessing potential options to avoid infrastructure investment (*e.g.*, NPAs); support for development of in-state RNG production; planning for development of hydrogen blending infrastructure; and development of community-scale network geothermal systems to retire LPP or reduce gas demand.

With respect to technology and workforce, recommended next steps include advancing network geothermal and hydrogen blending pilots; and initiatives to support clean energy workforce development across decarbonization technologies.

With respect to demand reduction, recommended next steps include expanded energy efficiency and gas demand response programs and incentives; supporting customer education pertaining to program offerings and technology options; supporting efforts to expand the energy efficiency workforce; exploring innovative customer financing options; and continued evaluation of NPAs to avoid infrastructure investment and address capacity constraints.

Q. Are the Companies' proposals in these proceedings consistent with these recommended next steps?

A. Yes. As described in this testimony, and as discussed in more detail in the testimonies of the Customer Panel and the Gas Infrastructure and Operations Panels, the Companies are proposing several initiatives designed to support affordability and benefit low-income customers and Disadvantaged Communities. These include additional personnel focused on serving the needs of the diverse groups of customers that comprise these customer segments, increasing the number of the Consumer Advocates,

and creating the new Clean Energy Advisor position, which would be responsible for helping low-income customers navigate the various clean energy programs available to them, including coordination with NYSERDA's clean energy hubs. The Companies also are proposing new access and outreach initiatives, such as a language access program to reach non-English speaking customers—many of whom are located in Disadvantaged Communities—to help customers participate in energy efficiency offerings that can save them money and reduce GHG emissions.

The Companies' LPP programs described in the Gas Infrastructure and Operations Panels also would retire aged facilities in Disadvantaged Communities, resulting not only in reduced GHG emissions, but also improved local air quality and enhanced safety. That panel also describes proposals by the Companies to facilitate the interconnection of RNG from food waste and wastewater treatment sources, reducing local emissions from those wastes, reducing upstream emissions, and helping decarbonize the gas system by capturing RNG and injecting it into the gas delivery system.

In addition, KEDLI is proposing to expand an existing green hydrogen project in the Town of Hempstead, New York, to assess the potential for hydrogen to be produced effectively using renewable energy and blended

into the natural gas system to reduce its carbon intensity. The project will also further advance the development of the clean energy workforce, and the Companies will continue to work with other skills and workforce development initiatives, such as an ongoing contractor training program that targets employees of minority and/or women-owned business enterprise, or employees of veteran-owned businesses, in National Grid's service territory, as well as individuals living in Disadvantaged Communities who are employed in or wish to enter the energy services field.

As described by the Customer Panel describes, the Companies also are proposing substantial energy efficiency initiatives consistent with the NE:NY programs, as well as several new initiatives designed to reduce gas demand and promote long-term energy efficiency. These include, among other things, a digital building replica proposal and a weatherization health and safety program.

In summary, the Companies' proposals in these proceedings are consistent with and would substantially advance the recommendations in the CLCPA Study.

3. Billed Gas Usage Reductions

Q. Please describe the Companies' current commitment to reducing billed gas usage.

A. In the 2019 KEDNY and KEDLI Rate Cases, the Companies agreed to operate their gas networks with the objective of reducing billed gas usage, normalized for temperature, in their service territories over the term of the rate plans. For purposes of measuring the goal, the Companies endeavored to achieve weather-adjusted reductions of one half of one percent in billed gas usage as compared to the forecast levels of non-generator gas use.

Q. Are the Companies proposing to continue the billed gas usage reduction goal?

A. Yes. The Companies believe such a goal is laudable and propose to continue it during the proposed four-year rate plan. Specifically, the Companies will seek to achieve a weather-adjusted, one-half of one percent reduction in forecast levels of gas usage (excluding non-core and electric generation) as included in the direct testimony of the Gas Load Forecasting Panel Exhibit ___ (GLF-4A) for KEDNY and Exhibit ___ (GLF-4B) for KEDLI. The Companies propose to report on progress relative to these goals on an annual basis.

Q. How will the Companies endeavor to meet these goals?

A. There are a few levers within the Companies' control that can be used to achieve temperature-normalized reductions in non-generator billed gas usage, namely energy efficiency, NPAs, and geothermal/district heating projects. More detailed descriptions of those efforts can be found in the testimony of the Customer Panel and the Gas Infrastructure and Operations Panels, as well as the Companies' filings in the Energy Efficiency Proceeding (Case 18-M-0084), the Gas Planning Proceeding (20-G-0131), and the Utility Thermal Energy Networks Proceeding (22-M-0429). Electrification of heat is another method to reduce billed gas usage. Although the Companies will continue to refer customers requesting new and upgraded gas connections to Con Edison's and PSEG-LI's heat pump programs, participation in those programs is ultimately up to the customer and their electric utility provider. Importantly, other factors outside the Companies' control also can influence billed gas usage, including economic growth and development; state and local policy and regulation regarding decarbonization, gas connections, building energy codes, and non-normalized weather factors. Lastly, the forecast levels in the Gas Load Forecasting Panel's testimony are not precise predictors, but rather the Companies' best estimates of future usage.

4. Disadvantaged Communities

Q. Does the Panel address how the Companies' proposed projects, programs and initiatives affect Disadvantaged Communities?

A. Yes, many of the projects and programs proposed in the case are located in Disadvantaged Communities. The Companies' approach to considering Disadvantaged Communities impacts and benefits in these proceedings reflects the Companies' commitment to enabling the clean energy transition for all customers, and is informed by the specific targets and requirements articulated in the CLCPA. As the Companies work to enable the clean energy transition we are working to ensure that customers in Disadvantaged Communities benefit from improved infrastructure, expanded outreach to support participation in energy efficiency and affordability programs that can help customers manage their bills, and specific community economic benefits through programs such as workforce development grants as well as our shareholder-funded community initiatives. The Companies' proposed investments and programs are intended to ensure the continued safety and reliability of service for all customers, as well as support GHG reductions and improved environmental outcomes and will directly benefit customers in Disadvantaged Communities in doing so.

In addition, the Companies have sought to prioritize Disadvantaged Communities in the development of customer programs included in this

case, which seek to both expand resources for delivering programs to these communities, while also working to address certain foundational issues (*e.g.*, language barriers, health and safety prerequisites needed for weatherization). The Companies are also proposing to modify their existing economic development programs to focus more on how they can best deliver benefits to Disadvantaged Communities. These initiatives reflect the Companies' commitment to delivering the environmental, health, and energy savings benefits of its programs in support of the targets established under the CLCPA.

Q. What gas programs and investments are the Companies proposing in these proceedings to benefit Disadvantaged Communities?

A. The Companies present several programs and investments that will provide direct benefits to Disadvantaged Communities. Additional details on each proposal can be found in the respective testimony where the proposal is presented.

Gas Infrastructure and Operations and Gas Safety Panels

Large portions of the Companies' service territories consist of areas designated as Disadvantaged Communities. The gas network infrastructure investments proposed in this case, including LPP retirement, will lead to improved infrastructure in these communities, and provide multiple

benefits, including reduction of leaks and local GHG emissions, improved local air quality, and improved safety and reliability to customers in these communities who depend on gas service to meet their everyday energy needs.

The Companies are also proposing the electrification of their fleets, with the largest number of electric vehicles ("EVs") planned to be deployed at facilities in two communities within the boundaries of Disadvantaged Communities, as well as facilities located in two other communities that serve Disadvantaged Communities in their immediate vicinities. Because of the Companies' EV program rollout, customers will directly benefit from reduced ambient noise, improved air quality, and a reduction in GHG emissions.

In addition, KEDNY's proposed Greenpoint Master Plan seeks to incorporate an overall design that will help mitigate negative public health effects, poor air quality, and other adverse environmental factors that affect Disadvantaged Communities in the area. To the extent possible, the Company incorporate sustainable construction materials into the design and will seek community stakeholder input to aid in the overall planning effort.

Customer

The Companies view energy efficiency as a key driver of an equitable energy transition for our customers. Toward that end, they are actively working to support achievement of the CLCPA's targets for benefitting Disadvantaged Communities and embed consideration of Disadvantaged Community impacts across energy efficiency program development, including through dedicated resources to support access and delivering benefits. Among the specific incremental energy efficiency proposals relevant to Disadvantaged Communities included in this filing are the weatherization health and safety pilot program, Disadvantaged Community outreach program, the energy efficiency language access program, and the digital building replica pilot, which will model school buildings within Disadvantaged Communities to facilitate adoption of coordinated energy efficiency enhancements.

Also as discussed above and in the testimony of the Customer Panel, the Companies are proposing modifications to their Economic Development programs to prioritize delivery of benefits in Disadvantaged Communities. Relevant modifications include adjusting eligibility requirements for the Brownfield Redevelopment Assistance and Cinderella programs to prioritize and provide additional funding for eligible projects located in Disadvantaged Communities. In addition, for the Cinderella program, the

Companies propose to provide matching grants for commercial corridor improvements in Disadvantaged Communities, and to support the provision of economic opportunities to business districts within those areas.

Earnings Adjustment Mechanisms

As discussed in Section VI, below, the Companies also are proposing two EAMs in this proceeding intended to encourage further investment and innovation as they work to prioritize beneficial outcomes in Disadvantaged Communities: (1) a continuation of the current LMI Customer Energy Efficiency Savings EAM, and (2) a new Energy Affordability Program ("EAP") Enrollment EAM. The LMI EAM aims to encourage the Companies to exceed the LMI energy efficiency targets set by NE:NY and to direct energy efficiency savings towards Disadvantaged Communities in support of the benefits targets established in the CLCPA.

The EAP EAM is intended to accelerate increased participation in the existing EAP program to provide a bill discount to more income eligible customers. Enrollment in the EAP serves as a gateway for identifying eligible customers for LMI energy efficiency and other programs. This EAM would encourage the Companies to increase investment and advance innovative approaches to identify, reach, and enroll customers in

EAP. These EAMs are described in more detail in Section VI of the Panel's testimony.

A summary of projects and initiatives located in or that support Disadvantaged Communities is provided in Exhibit __ (CLCPA-5). Additional details on each project and initiative can be found in the respective sponsoring testimony.

Q. Are the Companies proposing to report on their efforts to deliver benefits to Disadvantaged Communities?

A. Yes. The Companies are participating in ongoing efforts being coordinated by Department of Public Service Staff pursuant to the Commission's CLCPA Implementation Order (Case 22-M-0149) regarding tracking and reporting of compliance with the CLCPA's Disadvantaged Communities mandates. The CLCPA Implementation Order provides for annual reporting in that proceeding, and is expected to include information on the State's investment in Disadvantaged Communities, which would include customer funds that the Companies spend for the benefit of customers in Disadvantaged Communities. The Companies will continue to participate in the existing process under Case 22-M-0149 and adopt the tracking and reporting protocols that result from that effort.

VI. Earnings Adjustment Mechanisms

Q. Please describe the Companies' approach to developing EAMs for this proceeding.

A. Fundamentally, EAMs are intended to drive performance for the benefit of customers and to support achievement of policy objectives. Well-designed incentives can encourage utilities to innovate to provide value to customers in many areas, including through the delivery of services and programs at costs below historic or projected baselines, the identification of opportunities to avoid incremental capital investment, and in driving customer participation in programs to levels that exceed expectations or historical performance. As the Companies work to advance CLPCA targets, EAMs can provide an opportunity to deliver outcomes at levels that may not be sufficiently incented by the policies themselves. In addition, encouraging utilities to identify and drive new efficiencies beyond what is mandated will be particularly important given anticipated costs of achieving net zero.

In developing the proposed EAMs that follow, the Companies looked to build upon existing energy efficiency EAMs and consider additional EAMs that provide an opportunity to create system or customer benefits beyond what would otherwise be expected. The Companies' proposed incentives are intended to maximize the customer value of current energy efficiency

budgets, reduce risks to reliability and incremental system costs via demand response, and expand the reach and impact of the Companies' EAP.

Q. Please summarize the EAMs the Companies are proposing in this proceeding and the basis for this proposal.

A. The Companies are proposing to continue two of the current EAMs established in the 2019 KEDNY and KEDLI Rate Cases, and to add two new EAMs, for a total of four EAMs. The EAMs proposed are:

(1) Energy Efficiency EAM: Gas Share the Savings (existing)

(2) Energy Efficiency EAM: Gas LMI (existing)

(3) System Efficiency EAM: Gas Demand Response (new)

(4) LMI Engagement EAM: EAP Enrollment (new)

The Companies recognize that EAMs related to energy efficiency performance of all New York electric and gas utilities may be addressed in the NE:NY interim review. If that happens, the Companies will incorporate any changes made in the Interim Review as part of the Companies' Corrections and Updates filing.

Q. Please describe the Companies' current EAMs.

A. The Companies current rate plan consists of three EAMs for each Company: Gas Peak Reduction, Energy Efficiency Share the Savings ("STS"), and LMI Customer Energy Efficiency Savings.

1. Energy Efficiency EAM: Gas Share the Savings ("STS")

Q. Please summarize the Share the Savings EAM metric.

A. The Gas STS metric encourages the Companies to exceed non-LMI Gas energy efficiency targets set in NE:NY while driving unit cost efficiencies. The Companies can achieve incentives under this EAM by reducing total gas portfolio expenditures and/or by increasing the overall achievement above the approved baseline level of lifetime energy savings. The realized difference in the expected and actual unit cost per dekatherm is shared with customers by being reallocated within the portfolios, allowing the delivery of further savings within the original budgets. The savings are split 70 percent and 30 percent between customers and the Companies, respectively.

Q. Please describe how the Companies will measure this EAM.

A. Gas STS will continue with the same methodology from the 2019 KEDNY and KEDLI Rate Cases. The EAM is designed to simultaneously incent both higher levels of energy savings and greater efficiency in the unit cost to deliver those savings for non-LMI energy efficiency programs as per the incentive formula illustrated in Exhibit__(CLCPA-6).

The Companies propose a sharing percentage of 30 percent based on the acquired energy efficiency savings, excluding LMI savings, once the

Companies have met the energy efficiency threshold savings targets as shown in Exhibit ___ (CLCPA-7). This metric will be based on Verified Gross Savings.

Q. How will the Companies calculate a baseline for this EAM?

A. The Gas STS EAM is formulaic, as shown in Exhibit___(CLCPA-6). The Companies must meet their respective annual NE:NY energy efficiency savings targets, excluding LMI savings, set forth in the NE:NY Order. Additionally, targets will include cumulative over or under achievement of all NE:NY prior years savings. and will be distributed across the remaining years of the NE:NY period through 2025. If the Commission authorizes additional budgets and targets, the EAM calculation, including baseline targets shall be updated accordingly.

Q. How will the Companies calculate minimum, midpoint and maximum targets for this EAM?

A. Minimum, midpoint, and maximum targets are not applicable to this EAM. Instead, earnings are determined formulaically once the threshold obligation is met, as shown in Exhibit___(CLCPA-6).

2. Energy Efficiency EAM: Gas LMI

Q. Please summarize the Gas LMI Energy Efficiency EAM metric.

A. The LMI energy efficiency metric aims to encourage the Companies to exceed the LMI energy efficiency targets set in the NE:NY order by finding innovative ways to achieve more energy savings with the allocated funding. This would also direct energy efficiency savings towards Disadvantaged Communities further incentivizing the Companies to identify ways to exceed the 35 percent of investment benefits minimum mandated in the CLCPA. This EAM is also aligned with CLCPA's goals of achieving an 85 percent reduction in GHG emissions by 2050 and reducing 22 million tons of carbon through energy efficiency and electrification.

Q. Please describe how the Companies will measure this EAM.

A. Consistent with the methodology from the 2019 KEDNY and KEDLI Rate Cases, the Companies will measure the LMI energy efficiency EAM achievement by calculating the annual LMI energy efficiency savings that are acquired through the LMI programs. This metric aims to reward meeting or exceeding targets set out by the NE:NY Order. If the Commission authorizes additional budgets and targets, the EAM calculation, including baseline targets shall be updated accordingly.

Q. How will the Companies calculate a baseline for this EAM?

A. The LMI EAM metric is based on the sum of annual MMBtu savings targets for the Companies' LMI energy efficiency portfolios over a four-year term. Annual targets for the LMI portfolios have been established in the 2020 NE:NY Order.

Where:

LMI Baseline = the sum of the Company's annual LMI savings targets for calendar years 2024-2027 as outlined in the NE:NY Order. As with other targets beyond 2025, these are subject to adjustment after the NE:NY Interim Review; and

Acquired LMI MMBtu Savings = the sum of acquired annual verified gross energy savings (measured in MMBtu) for calendar years 2024-2027 of the Company's LMI portfolio.

Q. How will the Companies calculate a minimum, midpoint and maximum for this EAM?

A. Minimum, midpoint, and maximum targets will be set using the following methodology, which is consistent with the 2019 KEDNY and KEDLI Rate Cases:

LMI Min = LMI Baseline * 105%

LMI Mid = LMI Baseline * 115%

LMI Max = LMI Baseline * 145%

3. System Efficiency EAM: Gas Demand Response ("Gas DR")

Q. Please summarize the new Gas DR EAM metric

A. The Gas DR EAM would measure performance across the Companies' portfolio of gas demand response programs. Because the programs' primary goal is to provide region-wide gas system relief during peak hours, the EAM would measure performance across both Companies (*i.e.*, achieving above the minimum target would lead to earnings for both KEDNY and KEDLI, but missing it would lead to no incentive for either Company).

Q. Please describe how the Companies will measure this EAM.

A. The EAM will measure the performance of the Companies' firm gas demand response portfolio, which is currently composed of the Load Shedding, Load Shifting, and Bring Your Own Thermostat ("BYOT") programs. That performance will be assessed in terms of units of dekatherm ("dth")/hour. The EAM baseline will be determined using a logarithmic regression derived from actual performance during the preceding three winter seasons. Achievement will be measured using actual performance during events.

Q. Why do the Companies propose to measure performance in units of dth/hour?

A. The gas demand response programs are designed to reduce peak hour demand but measure enrollment in different units (dth/day, dth/hour, and number of devices enrolled, for Load Shedding, Load Shifting, and BYOT, respectively). Therefore, achievement in the programs needs to be converted into one common metric for purposes of measuring the EAM. Utilizing dth/hour will (i) enable like-for-like comparisons of events that differ in length and (ii) encourage the Companies to focus on providing the maximum event reductions needed to maintain reliable service and avoid the unintended incentive to call longer-duration events.

Q. Why do the Companies propose using a logarithmic regression to set baseline values?

A. The growth rate of customer volume enrolled in the Load Shedding program has diminished and will continue to diminish over time, because (i) most of the customers who enroll in the Load Shedding program participate in load shedding events by switching to backup sources of fuel, and the pool of customers who have the ability to switch to those backup sources or have discretionary non-heating loads is limited, and (ii) due to prior overachievement in enrollments the remaining customers in that pool are, on average, smaller in size than customers who have enrolled to date. By

contrast, the Companies believe that the potential pool of customers for the Load Shifting and BYOT programs remains larger. However, because the per-customer peak hour reductions in those two programs are much smaller than in the Load Shedding program, customers enrolled in the Load Shifting and BYOT programs tend to be much smaller in size. By program design, customers in the Load Shifting and BYOT programs also curtail only a portion of their load rather than their full load, resulting in the growth rate of the overall portfolio diminishing over time. Therefore, it is appropriate to use a regression model such as a logarithmic regression that accounts for that diminishing potential, rather than a linear regression, which would result in an unrealistic and unachievable representation of the growth potential of the firm gas DR portfolio.

Q. How many years of historical data do the Companies propose to use to calculate the baseline, and why?

A. Because the Companies' gas DR programs were still in a nascent stage in winters prior to 2021-22, utilizing those winters would lead to non-meaningful targets in future years. Therefore, the Companies propose to use the previous three winters' performance to set the baseline for the Rate Year, and to carry that methodology forward into the Data Years. Thus, the baseline for the Rate Year would be set using performance during winters 2021-22, 2022-23, and 2023-24; the baseline for Data Year 1 would be set

using performance during winters 2022-23, 2023-24, and 2024-25; and so forth.

Q. What methodology do the Companies propose to use to compute achievement for the historical values used to compute the baseline and the performance year achievement values?

A. First, the Companies propose to utilize the total aggregate performance across the entire gas DR program portfolio for each gas day on which an event occurs, regardless of event type. Events can be one of three types: (1) test events, generally called either at the start of the winter season to ensure customers are prepared to perform during events later in the season, or mid-winter if no actual events have occurred; (2) actual events, called on days where the forecast average temperature is below the program temperature event thresholds; or (3) emergency events, called in response to emergency conditions on the system such as constraints on upstream transmission pipelines. Second, the gas day on which the portfolio achieves its highest peak hour reductions during a given winter will be used to derive the baseline values and actual achievement for each winter. It may be worth noting that, due to program design, events with the highest peak hour reductions will likely correlate to the coldest days of the year. The highest peak hour reductions are calculated as: the total dth reductions achieved by each program during the respective gas DR event ÷ the total # of event hours

for each program during that gas day. See Exhibit __ (CLCPA-8) for an illustrative numeric and graphical representation of this methodology.

Q. Why are the Companies proposing to utilize actual performance rather than enrollment values?

A. Utilizing actual performance rather than enrollment to determine both the historical values used to calculate the EAM baseline and actual achievement will encourage the Companies to make efforts to improve customer performance during events and align with the goal of the gas demand response program portfolio.

Q. How is "performance" determined during a gas demand response event?

A. The methodologies by which performance is determined for the Load Shedding and Load Shifting programs during events are detailed in the Companies' tariffs. Performance for the BYOT program is determined using a randomized control trial methodology.

Q. If the Companies were to develop new gas demand response pilots or programs, would those be included in achievement on the Gas DR EAM?

A. Yes, they would, since doing so will incentivize the Companies to devise new and innovative programs and methods to reduce peak demand via gas demand response beyond that which they currently achieve or are otherwise required to achieve. As noted in the testimony of the Customer Panel, aside from the hybrid heating pilot being proposed outside of this proceeding, the Companies have no plans for such pilots or programs at this time. However, if any new programs are proposed that will contribute to achievement on this EAM, the Companies will consult with DPS Staff regarding the calculation methodology, and, if necessary, submit a filing in this docket that explains the proposed methodology.

Q. How do the Companies propose to set the EAM targets?

A. The minimum, midpoint, and maximum targets will be set at 1 percent, 2 percent, and 5 percent above the baseline, respectively.

4. LMI Engagement EAM: EAP Enrollment

Q. Please summarize the new EAP EAM metric.

A. The EAP EAM aims to encourage increasing participation in the existing EAP to provide a bill discount to more eligible low-income customers.

Enrollment in the EAP serves as a gateway for identifying eligible customers for LMI energy efficiency and other programs. This EAM would encourage further investment and innovation in identifying, reaching, and enrolling customers in EAP.

Q. Please describe how the Companies will measure this EAM.

A. The EAP EAM metric would measure the average number of customers enrolled in the EAP for the 12 months of the program year, which runs from December 1 to November 30. The metric for the Rate Year would measure the enrollment for the 2023-2024 EAP program year. The baseline for the Rate Year would be set as the average enrollment of the 12 months in program year 2022-2023. The metric for Data Year 1 would measure the enrollment for the 2024-2025 EAP program year. The baseline for Data Year 1 would be set as the average enrollment of the 12 months in program year 2023-2024. The metric for Data Year 2 would measure the enrollment for the 2025-2026 EAP program year. The baseline for Data Year 2 would be set as the average enrollment of the 12 months in program year 2024-2025. The metric for Data Year 3 would measure the enrollment for the 2026-2027 EAP program year. The baseline for Data Year 3 would be set as the average enrollment of the 12 months in program year 2025-2026. The average enrollment for the EAP program year will be based on the average

of the 12 monthly EAP reports in the designated program year filed as part of Case 14-M-0565.

The minimum, midpoint, and maximum will be set at 4 percent, 6 percent and 8 percent over the baseline. The natural variation in program enrollment over the period CY 2018 – CY 2022 is 1 to 2 percent in KEDNY and an average of 2 percent in KEDLI as shown in the table below. This was used to determine the percentage targets for increasing enrollment

Table 3: Demand Response program Enrollment CY 2018 to CY 2022

KEDLI	**Annual Enrollment**				
	2022	2021	2020	2019	2018
Q1	11,785	12,661	10,463	12,779	12,760
Q2	12,311	13,372	10,363	12,452	12,176
Q3	13,695	12,537	11,023	11,485	12,450
Q4	14,546	13,195	12,322	11,812	11,733
Annual Average	13,084	12,941	11,043	12,132	12,280
Enrollment increase over previous year	143	1,899	-1,089	-148	
% increase over previous year	1%	17%	-9%	-1%	
KEDNY	**Annual Enrollment**				
	2022	2021	2020	2019	2018
Q1	151,507	155,884	154,183	158,889	138,534
Q2	152,011	159,488	150,445	142,845	138,244
Q3	144,492	140,274	156,063	160,273	166,352
Q4	148,122	152,309	153,713	157,978	163,643
Annual Average	149,033	151,989	153,601	154,996	151,693
Enrollment increase over previous year	-2,956	-1,612	-1,395	3,303	
% increase over previous year	-2%	-1%	-1%	2%	

Q. Please describe how customer needs and benefits were involved in the development of the Companies' EAM proposal.

A. The Companies have built their EAM proposal with careful consideration of lessons learned from the existing EAMs, customer needs, and the State's energy and environmental goals. The Companies' proposed EAMs are designed to encourage the Companies to deliver benefits to customers such as dekatherm savings and GHG emissions reductions through the implementation of energy solutions that benefit customers above and beyond that which the Companies would otherwise be required to achieve. Along with the continuation of the STS EAM, the Companies have identified new initiatives and EAM metrics intended to deliver customer benefits and support the State's objectives. The Companies then set targets for those metrics at levels that will drive behavior to obtain positive and measurable outcomes. The Companies then assigned incentive levels calculated using basis points to each EAM based on the net societal benefits that each metric could deliver.

Q. Please describe the targets the Companies are proposing for each EAM?

A. The targets for each metric are shown in Exhibit___ (CLCPA-7).

Q. How do the Companies determine these targets?

A. For STS and LMI energy efficiency EAMs, where the programs are funded by customers through base rates, the Companies have set minimum targets such that each Company must first exceed the targets mandated by the Commission through NE:NY to be eligible for any incentives. For the enrollment-based EAP EAM, the Companies determined base figures would be established using the previous EAP program year's enrollment and targets would be calculated based on a percentage increase over the baseline program year. For the Gas DR EAM, the Companies derived baseline levels from performance during demand response events in the 3 winter seasons preceding each performance season, and targets are based on a percentage increase over baseline.

Q. How do the Companies propose to allocate basis points across EAMs?

A. The Company proposes allocating the basis points for EAMs as shown in Exhibit___ (CLCPA-9).

Q. When do the Companies propose the EAMs become effective?

A. EAM performance is typically measured and reported on a calendar year basis to align with annual energy efficiency program plans and implementation. The Companies propose that all EAMs become effective

January 1, 2024 and continue through the end of last calendar year of the rate plans.

Q. How do the Companies intend to report EAM results and recover incentives?

A. On or before April 15th of each year from 2025 through 2027, the Companies will make an annual compliance filing with the Commission. The annual filing will include a report on the Companies' prior calendar year performance relative to each EAM target, showing the savings, enrollment, and benefits achieved, as well as the calculations for the incentives earned. The Companies will also provide an explanation of targets not achieved. The yearly reporting approach is consistent with current filing timelines.

Q. When will the Companies begin recovering EAM incentives it earns?

A. Any incentive that is achieved would be recovered through a surcharge, as explained in the testimony of the Rate Design Panel.

Q. Did the Companies analyze the portfolio of benefits and costs associated with its proposed EAMs?

A. Yes. The Companies evaluated the portfolio of benefits and costs associated with the proposed EAMs. The Companies estimated the total value of

societal benefits that would be delivered at minimum, midpoint, and maximum target levels. Gas benefit accounting includes dth and GHG emissions savings values. From there, the Companies subtracted total estimated costs to deliver these results.

Q. What did the Companies analysis of the benefits and costs conclude?

A. The Companies' analysis concluded that the programs and products supporting the EAMs provide significant qualitative and quantitative benefits including increased customer choice and customer savings. For the Net Benefits Calculations for each EAM see Exhibit ___ (CLCPA-10).

VII. Conclusion

Q. Does this conclude the Panel's testimony?

A. Yes.

INDEX OF EXHIBITS

Exhibit __ (CLCPA-1): National Grid's *Responsible Business Charter*

Exhibit __ (CLCPA-2): National Grid's *Clean Energy Vision*

Exhibit __ (CLCPA-3): Assessment of KEDNY Estimated Greenhouse Gas Emissions Impacts; Rate Year – Data Year 3

Exhibit __ (CLCPA-4): Assessment of KEDLI Estimated Greenhouse Gas Emissions Impacts; Rate Year – Data Year 3

Exhibit __ (CLCPA-5): Proposals Supporting Disadvantaged Communities

Exhibit __ (CLCPA-6): Gas Share the Savings Metrics and Targets Calculations

Exhibit __ (CLCPA-7): Earnings Adjustment Mechanism ("EAM") Targets

Exhibit __ (CLCPA-8): Gas Demand Response EAM Methodology

Exhibit __ (CLCPA-9): EAM Basis Points and Incentives

Exhibit __ (CLCPA-10): EAM Net Benefits Calculations

Exhibit __ (CLCPA-1)

National Grid's *Responsible Business Charter*

Responsible Business Charter

nationalgrid



Responsibility at National Grid



John Pettigrew
Chief Executive,
National Grid PLC

At National Grid we're committed to being a responsible business in everything we do. It's enshrined in our purpose – to Bring Energy to Life.

We do this by delivering the electricity and gas that our customers and communities rely on. We will always do this reliably, safely, and in a fair and affordable way for all. At National Grid, we apply responsibility as a principle every day, by doing the right thing, finding a better way and making it happen. Society expects it of us, and we know we can make a positive impact in doing so.

We are committed to shaping a sustainable economy for all, in partnership with others in our industry and beyond.

Society's expectations

Society's expectations of business and the role business plays in the world are changing. We share the belief that business needs to stand for something more than profitability. Now, more than ever, we have a responsibility to demonstrate our contribution to society more broadly – whether that's by making sure young people today become the problem solvers of tomorrow, or in tackling climate change by reducing our own emissions and helping our customers use energy more efficiently.

Our response

Responsibility and transparency have always been important to us at National Grid. In 2019, we looked at where we can create the most positive impact on society. We adapted the way we manage our business responsibly, looking at everything from our strategic investment process to our role in the community and our business processes and policies. We applied this framework to shape our COVID-19 response and showed that it is fit for purpose.

To further our ambition, we are joining the B Team, a movement of global business leaders driving better ways of doing business for the wellbeing of people and the planet. The B Team principles of sustainability, equality and accountability will inform our actions and we will share our approach with others.

We support the United Nations Sustainable Development Goals (SDGs), focusing on the areas where we can make a material difference.

Our charter

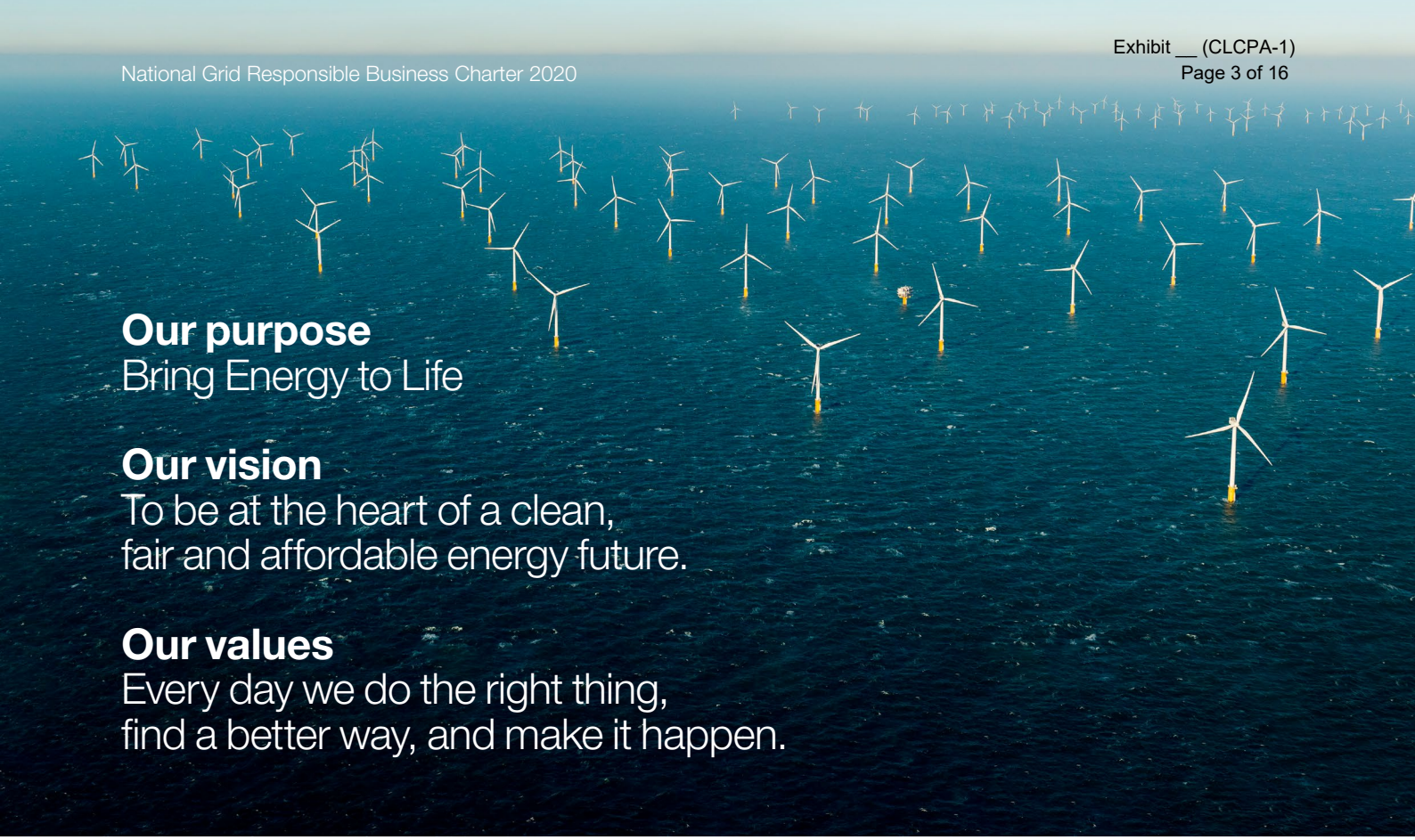
This charter is our articulation of what 'responsibility' means for us at National Grid.

We have identified where we can have the most impact on society: the environment, the communities we serve, our people, the economy, and our governance. For each, we have set out commitments and how we will achieve them.

We will report on our progress in our new annual Responsible Business Report from 2021.*



* All commitment dates refer to the end of the calendar year. We will report on progress on a financial year basis.



Our purpose
Bring Energy to Life

Our vision
To be at the heart of a clean, fair and affordable energy future.

Our values
Every day we do the right thing, find a better way, and make it happen.

Our responsible business approach

01 The environment
We will enable a fair and affordable transition to a clean energy economy, and reduce our own emissions.
Read more on page 4

02 Our communities
We will deliver sustainable energy safely, reliably and affordably, ensuring no one gets left behind.
Read more on page 8

03 Our people
We will develop the right skills to enable and accelerate the energy transition, and strive to build a diverse workforce and inclusive culture.
Read more on page 10

04 The economy
We will power and heat society, and partner with regulators, our business partners, suppliers and other key stakeholders.
Read more on page 12

05 Our governance
We will make sure our governance mechanisms reflect our commitments, and that the principles of responsibility guide us in everything we do.
Read more on page 14

01 Responsible business in the environment

Our journey so far

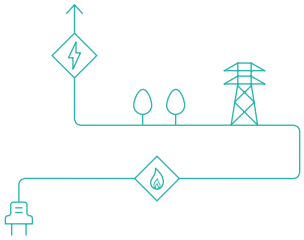
Climate change is the defining challenge of this generation. The decisions we take now will influence the future of our planet and life on earth. We must make significant changes to curb harmful emissions.

Since 1990, we've reduced our direct (Scope 1 and 2) emissions by 70% (as at March 2020). That's well in excess of our interim target, which was 45% by 2020. But there is much more we can do.

Most of the markets in which we operate have announced net zero carbon reduction targets, and we anticipate further legislation. We embrace the opportunity to play our role in helping achieve these challenging goals.

Understanding our greenhouse gas emissions

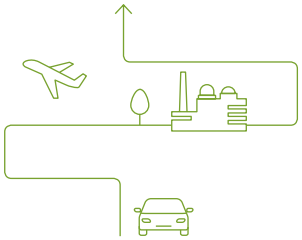
Scope 2: Indirect[1]



Upstream

- Line losses from our electricity transmission and distribution lines
- Energy purchased for use at our facilities
- Use of electric drive compressors in our gas business

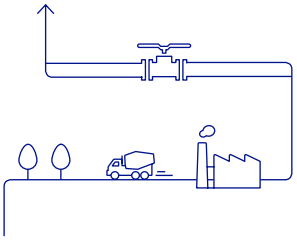
Scope 3: Indirect



Upstream

- Purchased goods and services
- Business travel
- Employee commuting

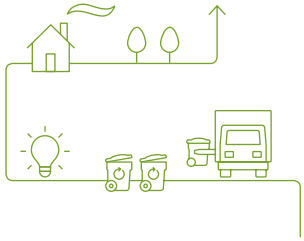
Scope 1: Direct[1]



Our operations

- Long Island Power Authority electricity generation
- Leaks and venting from our gas transmission and distribution systems
- SF6 leaks from our electric equipment
- Fleet vehicle use
- Gas-fired compressor use

Scope 3: Indirect



Downstream

- Use of 'sold product' or emissions from our customers' use of the gas and electricity we purchase on their behalf
- Waste management

[1] Included in our net zero target.

What we will do

Our commitments

This charter represents the next stage of our environmental commitments, setting out a significant change in what we will achieve.

- **Achieve net zero by 2050.** We will reduce Scope 1 and 2 greenhouse gas (GHG) emissions 80% by 2030, 90% by 2040, and to net zero by 2050 from a 1990 baseline.[2]

  Our Scope 1 and 2 target aligns to a well-below two degrees pathway consistent with the ambition requirements of the Paris Agreement and Science Based Targets initiative (SBTi).

- **Reduce Scope 3 GHG emissions for the electricity and gas we sell to our customers (making up 80% of our Scope 3 emissions) by 20% by 2030 from a 2016 baseline.**

  Our Scope 3 target aligns to SBTi's two degrees pathway. With SBTi moving to only recognise targets aligned to well-below two degrees and 1.5 degrees, we will continue to work closely with the team on this higher level of ambition, and work to have the Scope 1, 2 and 3 targets officially validated by the SBTi.

- **Reduce SF6 emissions from our operations 50% by 2030,** from a 2019 baseline.
- **Move to a 100% electric fleet by 2030 for our light-duty vehicles,** and pursue the replacement of our medium- and heavy-duty vehicles with zero carbon alternatives.
- **Reduce energy consumption in our offices 20% by 2030,** from a 2019 baseline.
- **Improve the natural environment by 10% on the land we own by 2030.**
- **Achieve zero carbon emissions from business air travel.** From this year (2020) onwards, we will reduce our annual air miles travelled by at least 50% from a 2019 baseline on an enduring basis, and we will offset any remaining emissions responsibly.

Our ambitions

- **Accelerate our net zero target wherever possible.** We will work to achieve net zero in each part of our business as fast as we can. Where we need to pursue carbon offsetting we will seek offsetting options that deliver multiple benefits.
- **Eliminate all SF6 gas from our assets by 2050.** Technology and solutions are not yet available to achieve this. Therefore, we will work with partners from across the sector to identify, develop and implement SF6-free solutions at the earliest opportunity.
- **Further reduce our Scope 3 emissions** from selling gas to our customers beyond 2030 to be consistent with the targets set by the markets in which we operate (e.g. net zero by 2050).
- **In the UK, the National Grid ESO (Electricity System Operator) will be able to operate a zero-carbon system by 2025.**



[2] Our interim 2030 target is science-based (80% by 2030 is the equivalent to 50% from a 2016 baseline).

We will enable a fair and affordable transition to a clean energy economy, and reduce our own emissions.

We will play a leading role in enabling and accelerating the transition to a clean energy system. The energy system will look very different in a net zero world, and we will work to accelerate the transition, while balancing decarbonisation, affordability and reliability.

Our aim is to be a leading utility globally in demonstrating the technical and commercial solutions that will help achieve net zero for the energy sector. Over 2020-21 we will work closely with the UK Government, Scotland and Wales, US partners and partners around the world to demonstrate these solutions as part of work for COP26[3].

Electricity

A fully decarbonised electricity grid. We expect demand for electricity to rise as transport and some heat becomes electrified. To meet this demand, we will connect renewables as quickly and efficiently as possible, and invest in grid modernisation. We will also make good use of demand-side management through our energy efficiency and demand-side response programmes in the US and UK.

We will continue to build interconnectors to bring low carbon energy to the UK, and we will grow our renewables business in the US. Our acquisition of Geronimo Energy, a leading developer of wind and solar generation assets, creates a solid foundation on which to develop and grow a large-scale renewable business.

Transport

The decarbonisation of transport. In a net zero world, almost all road transport will be decarbonised. We have a role to play here, making sure the right charging infrastructure is in place in the UK and US to enable an increase in electric vehicles.

We will also build electricity network flexibility and support low carbon alternatives, such as hydrogen, for heavy transport.

Heat

The decarbonisation of heat is uniquely challenging; we'll need to deploy cost-effective solutions for consumers and businesses, while minimising disruption to people's daily lives. While we don't yet have the answers, we are sure there is no one-size-fits-all solution. We will work with our customers and regulators to enable a range of potential solutions, including heat pumps, and renewable natural gas and hydrogen in our networks. In the US, we will continue leading on energy efficiency and helping residential, commercial and industrial customers switch away from the most polluting heating fuels, such as oil.

Natural gas

Natural gas makes up a large proportion of our own emissions. We'll continue to reduce fugitive emissions from our gas networks. In addition, our Long Island generation fleet, which has already cut emissions substantially[4], will transition to cleaner sources of generation and storage, in line with New York's climate goals[5]. We'll also continue our role in helping develop and deploy new technologies that have the potential to help us decarbonise at scale in the years ahead – such as hydrogen networks and carbon sequestration.

[3] The UK is hosting the 26th UN Climate Change Conference in November 2021 in Glasgow.
[4] Our generation business has reduced emissions by over 80% since 1990.
[5] New York has a goal of 100% clean generation by 2040.

Reducing our own emissions

We'll continue reducing our own direct emissions, replacing leak-prone gas pipelines with plastic ones, and by using robots traveling through our pipelines to detect and reduce leakage. We are committed to replacing SF6 with cleaner alternatives, reducing SF6 emissions by 50% by 2030, and have set ourselves an ambition to eliminate its use in our operations completely by 2050.

We will also reduce our business travel emissions by changing to alternative fuel vehicles and reducing business flights. We will encourage more flexible ways of working and use technology to reduce business travel, and incentivise our people to buy electric vehicles. We will reduce the energy consumed in our buildings and procure green energy where possible.

We will achieve net zero emissions from our Scope 3 construction activities in the UK by 2026. In the US, we need to do more work to understand our construction carbon intensity, but we will aim to develop a target for reduction by 2022.

To make sure we are making the right decisions, we're implementing carbon pricing on all major investment decisions by the end of 2020.

Waste

Adopting the principles of the circular economy, we'll design assets that can be recycled, refurbished and reused. We are committed to reducing the use of single use plastics and sending zero waste to landfill, where possible, in our main offices.

Natural environment

We must also address the challenge of restoring the natural environment. Using our own land and working with partners, we have an opportunity to cut carbon and restore nature at the same time. We will improve the natural environment on our own land; for example, by protecting habitats and increasing biodiversity. We'll do this by using best practice methods, such as natural capital evaluation, so we can make sure we create the most benefit. We'll also build on the work we've already done to increase the environmental value of more than 50 of our sites in the US and UK.

Improved air quality

We will reduce NOx and SOx emissions by improving the efficiency of our plant and equipment. Our initiatives to facilitate the electrification of transport and reduce our own travel emissions will also improve air quality in the areas we work.

Our contribution to the UN SDGs

We support the United Nations Sustainable Development Goals (SDGs), which are a universal call to action to end poverty, protect the planet and ensure all people enjoy peace and prosperity.

Relevant UN SDGs for our environments focused programmes and actions:



02 Responsible business and our communities

Our journey so far

As National Grid has evolved and grown in scale in the UK and US, we have become part of the fabric of the communities we serve. We keep the lights on, we keep them warm, we help economies thrive. We are an operator, an employer and a supplier. We support our communities with the time and expertise of our people, and through corporate giving programmes.

Our approach to responsibility in our communities has been to go beyond safely maintaining the resilient energy systems society expects. It's been about making sure our economic and social role in the community has the greatest possible positive impact.

That involves developing infrastructure and helping customers use energy more efficiently. It also includes partnering with charity organisations, encouraging our people to support social, economic and environmental development.

Helping our communities recover from the COVID-19 crisis is a key part of this. We will contribute to a clean and just recovery that will build a sustainable future for our communities.



What we will do

Our commitments

We will strive for affordability and fairness, and we will develop skills for the future, with a focus on lower income communities.

- **Deliver energy in a fair and affordable way to the communities we serve.**
- **Play our role in ensuring no-one is left behind in the transition to clean energy.** The associated benefits should be enjoyed by all.
- **Continue to reinvest in energy infrastructure at approximately £5 billion each year.**
- **Report transparently on energy costs throughout the energy transition** - on average costs per household for our UK transmission network and for our US electric and gas business.
- **Develop skills for the future, with a focus on lower income communities, providing access to skills development for 45,000 people by 2030.** Through upskilling young people with STEM[6] skills, we will positively impact lives and help to grow the future energy workforce needed to enable the energy transition.
- **Achieve 500,000 employee volunteering hours by 2030.** Through volunteering, our people will help equip the next and future generations to participate in the clean energy transition.

Our ambitions

- **Play a meaningful role in the COVID-19 recovery and enable people in our communities to gain employment in the clean and just recovery.**
- **Help customers reduce their energy usage through energy efficiency programmes and education.**

[6] Science, technology, engineering and mathematics.

We will deliver sustainable energy safely, reliably and affordably, ensuring no one gets left behind.

We believe a strong community is good for the people who live there, good for our business and good for the wider economy.

As we look ahead, we know our communities need and expect more from us. That's why we are committed to upgrading our networks to provide cleaner, greener and more affordable energy over the long term.

We will serve society fairly and affordably

We want to make sure that the technological and environmental benefits the clean energy transition will bring are felt by all. Protecting vulnerable customers will remain a priority for us as we aim to make sure that no community is left behind – the energy transition represents a shared challenge and opportunity for us all.

Throughout the transition, we have a role to play in helping customers better understand and manage their energy use – whether through energy efficiency, flexible load management or adopting clean solutions like electric vehicles. Our aim is to help them reduce their carbon emissions and total energy costs.

National Grid has been repeatedly acknowledged as the number one utility for energy efficiency in the US, and we will make sure lower income customers and communities get the benefit from more efficient energy consumption.

We will maintain reliability

Reliability and resilience of our networks are part of our regulatory duty, but also our social contract. As we transition to an increasingly complex electricity system, managing intermittent renewables and two-way power flows, we will maintain customers' expectations for reliable energy.

And, because of the effects of climate change, our networks will need to be more prepared to recover from extreme weather events. We're committed to making sure our systems are resilient and can play a leading role in disaster recovery.

Education, employment and infrastructure

We will continue to apply a responsible business framework to our investments in the communities we serve. This links our energy projects to local infrastructure development, including affordable housing. It will make sure the associated benefit of more accessible and affordable energy is enjoyed by those who need it most. We also place great importance on supporting communities through education and employment.

We will focus on the lower income communities we serve. We will aim to increase workforce diversity, and make sure the people most in need benefit from our investment.

We will listen and respond

We will continue to engage widely with the communities and customers we serve, responding to their needs, minimising disruption from our operations, and providing a service that satisfies them. Our overarching goal is to enhance the communities we serve.

Our contribution to the UN SDGs

Relevant UN SDGs for our communities focused programmes and actions:



03 Responsible business and our people

Our journey so far

The 23,000 people we employ in the UK and US are the lifeblood of our company. Their safety and wellbeing is a top priority for every one of us at National Grid – it underpins everything we do.

Our vision is to build and develop an inclusive culture and a diverse workforce that is fully representative of the communities we serve. We treat everyone fairly and equally, without discrimination on the grounds of race, age, role, gender, gender identity, colour, religion, country of origin, sexual orientation, marital status, dependents, disability, social class or political views.

This includes consideration for recruitment, redundancy, promotion, reward and benefits, training or retirement, all of which must be based on merit.



What we will do

Our commitments

- **Be as transparent as possible internally and externally on gender and ethnicity/ race.** This will include reporting on recruitment, promotion, progression and leaver rates by diverse groups. We will use this to spot trends and react accordingly.

  In order to increase the overall proportion of our workforce from diverse backgrounds, we will set annual targets to increase new hires and promotions from diverse backgrounds.
- **Maintain fairness across the organisation for pay and make sure our pay practices do not show bias. We will work until pay equity is achieved for our people.**
- **Achieve 50% diversity in our Senior Leadership group by 2025.**
- **Achieve 50% diversity in all our new talent programmes by 2025.** We will work with charities, diverse colleges and universities, and organisations in the community to attract those from under-represented backgrounds.
- **Provide unconscious bias training to all our people over the next year.** We will continue our programme of training to build awareness of cultural differences and understanding of the importance of diversity, and necessity of achieving equity and inclusion.

Our ambitions

- **Eradicate discrimination of any kind, and create a culture where our people feel confident to speak out safely.**
- **Among our people, reflect the diversity of the communities we serve.**
- **Develop the skills needed for the net zero workforce of the future by investing in developing our people,** so they can enable the clean energy transition.

We will develop the skills to enable and accelerate the energy transition, and strive to build a diverse workforce and inclusive culture.

An inclusive and diverse workforce for the energy industry

We believe that National Grid, and the wider energy industry, should be more representative of, and reflect, all aspects of diversity in the communities we serve.

Through industry-wide initiatives such as POWERful Women, we are committed to the professional development of women and gender diversity action plans to help us make possible the energy systems of tomorrow.

A diverse workforce and inclusive culture brings huge value to our business. We have many initiatives to encourage and promote diversity and inclusion, as well as dedicated employee resource groups. We are committed to building an inclusive company where everybody can achieve their potential.

We will build the net zero workforce

We are continually working within National Grid, as well as with our partners, suppliers and other stakeholders, to identify the jobs needed to help society achieve a clean energy future. We need forward-thinking, creative minds to help meet the challenges ahead. We anticipate a greater need for people skilled in data analytics and artificial intelligence to manage more complex grid flows, manage customer interactions needed to leverage demand-side management, and support piloting and scaling of new technologies such as transporting and using hydrogen for heat.

We'll develop the skills necessary to enable the clean energy future by investing in and developing our people and people across our communities. STEM education plays an important role in our skills development plan. It will help make sure that future employees of National Grid – and those joining our partners, suppliers and other organisations participating in the cleaner energy future – have the skills needed to make a real and positive difference.

Our contribution to the UN SDGs

Relevant UN SDGs for our people focused programmes and actions:



04 Responsible business and the economy

Our journey so far

Our economic contribution to society comes primarily through delivering safe and reliable energy. We make sure energy reaches homes and businesses safely, reliably and efficiently. But our contribution as a responsible, purpose-led business goes further – as an employer, a tax contributor, a business partner, a community partner and a B Team member.

We help national and regional governments formulate and deliver their energy policies and commitments. The taxes we pay help fund essential services.

Our geographic footprint means that our economic contribution is felt in lower-income communities. They can truly benefit from the ripple effect of our local presence, from rural communities in New England and New York to the UK, where most of our economic contributions are made outside London.



What we will do

Our commitments

- **Maintain reinvestment in our infrastructure and demonstrate the social benefits of our capital delivery programmes.**
- **Continue to invest in developing technologies and innovations that benefit our customers and wider society.**

  As part of our research and development activities, we established National Grid Partners (NGP), the utility industry's first Silicon Valley-based venture capital innovation firm. Through NGP, we have already committed to $250 million of investment, which is being allocated towards defining a smarter, more renewable energy future.
- **Work across our supply chains to ensure that, together, we reflect the diversity of the communities we serve and respond to the economic needs of those communities.**
- **Continue to influence our supply chain to operate as responsible businesses.**
  - We are fair to our suppliers and committed to paying them promptly.
  - We require our UK suppliers to pay the real living wage to their employees (as defined by the Living Wage Foundation).
  - We are engaging with our supply chain to set carbon reduction targets through the Carbon Disclosure Project supply chain programme.
  - We promote equal opportunities among our suppliers. Our expectations of them are in our Supplier Code of Conduct and Global Supplier Diversity Policy.
  - We encourage suppliers to sign up to industry standards, such as the FIR (fairness, inclusion and respect) commitment.
- **Meet or exceed the B Team Responsible Tax Principles[7].**

Our ambitions

- **Build partnerships with small and local businesses, and all suppliers who set clear ambitions related to environment, diversity, economic wellbeing and recovery, and governance.**
- **Help the economy and supply chain recover from the COVID-19 crisis,** looking for opportunities for investment together with suppliers in the lower income communities most in need.

[7] https://bteam.org/assets/reports/A-New-Bar-for-Responsible-Tax.pdf

We will power and heat society, and partner with regulators, our business partners, suppliers and other key stakeholders.

We'll make sure our working practices continue to support our role as a responsible employer, tax contributor, business partner and community partner.

We'll continue providing energy systems that help economies grow in a sustainable, reliable and affordable way. And we'll continue contributing significantly to the economies of the countries in which we operate through our people, our supply chains and our operations.

Working with key stakeholders

We work with our regulators and other key stakeholders to make sure the risks associated with the distribution and transmission of gas and electricity are matched by fair returns.

Our approach to regulatory consultation is to seek a framework that puts consumers at the centre of price controls and rate plans, while enabling the clean energy transition. Evolving that partnership to enable the clean energy transition and slow the pace of climate change before it can be reversed will be key in protecting future economic growth, and safety and wellbeing in society.

Working with our suppliers

We've set out in our commitments how we'll continue to work with our suppliers. We'll be fair to them and pay them promptly. We also recognise that the relationship we have with our suppliers can influence how they improve their own societal impact.

We require all businesses in our supply chain to share our commitment to respecting, protecting and promoting human rights. This includes alignment to the principles of the United Nations Global Compact, the International Labour Organization (ILO) minimum standards, the Ethical Trading Initiative (ETI) Base Code, the UK Modern Slavery Act 2015, the US Trafficking and Violence Protection Act 2000 and, for our UK suppliers, the requirements of the Living Wage Foundation.

We also engage with relevant industry initiatives to promote inclusion and diversity across our sector.

Tax

Our approach to tax is part of our commitment to being a responsible business, and is guided by our values.

We are committed to a coherent and transparent tax policy - meeting or going beyond the B Team Responsible Tax Principles, as part of our commitment to the B Team movement.

When making decisions on tax matters, we follow the CBI's Statement of Tax Principles. We do not enter into artificial arrangements or structures, or conduct operations in tax havens or low tax jurisdictions, where the sole purpose is to achieve tax savings. We'll also continue taking legislative and regulatory requirements into account, as well as their tax consequences, when we're considering business developments.

Sustainable financing

We published our green financing framework in 2019, and since then have issued bonds out of our UK and US electricity businesses, funding projects to enable the transition to clean energy. We've also issued the first green export credit agreement loans to fund the construction of our Viking interconnector.

We will continue to use the framework and are considering other forms of sustainable financing to support our strategy going forward.

Our contribution to the UN SDGs

Relevant UN SDGs for our economy focused programmes and actions:



05 Responsible business and our governance

Our journey so far

From transparency and culture, to making sure we have a diverse leadership team, we are committed to the highest standards of corporate governance and making sure we continually apply best practices.

The National Grid Board is collectively responsible for the effective oversight of the Company and its businesses. It determines the Company's strategic direction and objectives, business plan, viability and governance structure to help achieve long-term success and deliver sustainable shareholder value.



What we will do

Our commitments

- **Continue to review and adapt our governance policies to reflect and support our responsible business commitments and ambitions**. We will apply a corporate governance model that supports all our responsible business commitments and ambitions and applies our responsible business framework to everything we do.
- **Continually review the company culture to ensure it is inclusive.** The Board will continue to monitor progress throughout the Group by using a culture scorecard standard, including the FIR (fairness, inclusion and respect) commitment and via direct employee engagement.
- **Listen to our people and ensure their voices are heard at Group Executive and Board level.**
- **Ensure the voices of all our stakeholders are heard and understood by the National Grid Board.**
- **Achieve 50% diversity in our Group Executive Committee.**

Our ambitions

- **Aim for our responsible business actions and impact to differentiate us from our peer group, and to make us a clear choice for ESG[8]-minded investors.**
- **Meet and ultimately exceed the Hampton-Alexander and Parker diversity review standards and achieve 50% diversity in our Board.**
- **Build a company where everybody can achieve their potential.**

[8] Environment, social and corporate governance

We will make sure our governance mechanisms reflect our commitments, and that the principles of responsibility guide us in everything we do.

Our approach to corporate governance plays an important role in helping us develop our culture – one that embraces diversity and inclusion and creates an environment where everybody can fulfil their potential. Our Board will continue to play a vital role in setting the tone from the top.

Company culture

The Board has been on a journey to create the right culture throughout our company. This means making sure our people are embracing positive and inclusive behaviours and values in everything we do. A culture scorecard helps the Board monitor progress.

Getting input directly from our people is an important part of tracking progress. The Board hears their views through the employee engagement survey, employee participation in Board events and through internal communication channels. It engages with employees on a broad array of issues such as gender pay, culture and putting responsibility into action.

Inclusion and diversity

Board diversity is important in tackling today's challenges and setting the right direction for the company. Leadership diversity, in the Board, the Group Executive Committee and our Senior Leadership group, is a priority for us.

The Board and its committees will continue to evolve to ensure the right balance of skills, experience, independence and incentives to reach our responsible business goals.

Stakeholder engagement

We prioritise our responsibilities to our different but interrelated stakeholder groups and wider society. We endeavour to understand the interests of our stakeholders and reflect them in the decisions we make.

We recognise that in balancing those different perspectives, it isn't always possible to achieve each stakeholder's preferred outcome. Stakeholder engagement will continue to play an important role in how our Board ensures responsibility in governance. This includes listening to our stakeholders' views, inviting external guests to meetings, and using independent research to bring the voice of the customer and other stakeholders into the boardroom.

Managing risk

The Board oversees our risk management and internal control systems. As part of this, the Board conducts an annual strategic review of corporate social responsibility to map relevant issues to the company's risk register.

We have rigorous policies in place that support our approach to corporate responsibility, and we report on a number of non-financial performance measures.

We have committed to implementing the recommendations of the Financial Stability Board's Task Force on Climate-related Financial Disclosures in full, following on from our initial disclosure in 2017/18. We were awarded the Carbon Disclosure Project A list for the fourth year running in 2019 for our performance in reporting and mitigating the impact of climate change..

Our contribution to the UN SDGs

Relevant UN SDGs for our governance focused programmes and actions:



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October 2020. NG10594/WS

Exhibit __ (CLCPA-2)

National Grid's *Clean Energy Vision*



Our clean energy vision

A fossil-free future for cleanly heating homes and businesses



nationalgrid

We are committed to playing a leading role enabling and accelerating the transition to a clean energy future, while ensuring all customers and communities continue to have affordable and reliable options to heat their homes and run their businesses.

For more information visit:

nationalgrid.com/fossilfree

Contents

4 **1. Mobilizing to help our customers and communities reach net zero**

6 **2. Our vision for fossil-free heat**
8 Energy efficiency in buildings
8 100% fossil-free gas network
10 Hybrid electric-gas heating systems
12 Targeted electrification and networked geothermal
15 Roadmap to achieve our vision

16 **3. How our vision benefits all our customers and society**
16 Most affordable and practical for customers to reduce emissions
20 Utilizes the skills of our existing workforce
20 A complimentary but diversified electric and gas system is safe, reliable, and resilient
21 More likely to achieve our shared net zero goals by embracing all options

22 **4. Public policies to enable our vision**
22 Scaling supply and demand for renewable fuels
22 Driving investment in building and renewable heating equipment
23 Evolving infrastructure planning and investment
23 Ensuring consistent greenhouse gas accounting methods

24 **5. Conclusion:** What our vision means for the Northeast

26 **6. Appendix:** More about the pillars of our vision

30 **7. Notes and sources**

1. Mobilizing to help our customers and communities reach net zero

At National Grid we believe achieving a fossil-free future and addressing climate change is not just a shared responsibility, but one where we must lead. We are undertaking a bold initiative that will set a new climate standard for energy companies across America. From today forward, we are working to enable a future where every time our customers use energy – to turn on their lights, charge their phones, cook for their families, or heat their homes and businesses – it can be done without fossil fuels.

National Grid is announcing our vision to fully eliminate fossil fuels from both our gas and electric systems by 2050 – sooner if possible – setting clear and measurable milestones along the way.

We are making this fossil-free announcement because we share our communities' concerns about climate change.

National Grid is proud of our role in helping Massachusetts and New York consistently rank at or near the top nationally in energy efficiency programs, electric vehicles on roads, solar installations, planned offshore wind, and environmental stewardship. In addition to this track record of success, the states in which we operate have enacted some of the most progressive climate laws in the country, establishing future requirements for net zero emissions across their economies: New York passed the Climate Leadership and Community Protection Act in 2019 and Massachusetts adopted the *Next-Generation Roadmap for Massachusetts Climate Policy* in 2021.

National Grid is mobilizing to achieve net zero emissions in a way that ensures that all customers share in the benefits of a clean energy future. Our vision builds on our previous actions:

- **2018:** We released a *Northeast 80 x 50 Pathway* white paper exploring strategies for the region to reduce economy-wide emissions at least 80% below 1990 levels by 2050
- **2019:** We announced our *Net Zero by 2050 Plan* laying out our commitment to achieve net zero greenhouse gas (GHG) emissions by 2050 from both our own operations and the emissions resulting from the sale of electricity and gas to our customers
- **2020:** We launched our *Responsible Business Charter*
- **2021:** We are proud to be a Principal Partner for the UK's Presidency of COP26 – the United Nations Climate Change Conference that brought together 120 world leaders and over 40,000 registered participants to advance national plans to keep 1.5 degrees Celsius of global warming within reach
- **Over the past five years in the Northeast US:** We have connected 2,000 megawatts (MW) of distributed solar and 180 MW of battery storage, supported $3.1 billion in energy efficiency investments for our customers, and enabled over 4,400 electric vehicle charging ports in our Northeast service areas.

Winning the fight against climate change requires us to collectively achieve emissions reductions across multiple sectors of the economy – in how we generate electricity, fuel our vehicles, heat our buildings, and power our industries.

Reducing emissions from the building sector – predominantly space and water heating, but also cooking, washers and dryers, fireplaces, outdoor grills, and industrial processes – is one of the most important and challenging problems that must be solved to achieve net zero. Heat is essential to life, and keeping it affordable and reliable is critical for economic development and wellbeing. Underscoring our vision for fossil-free heat is a sincere belief that the net zero path we take must leave no customer behind, and our actions must be bold, smart and practical to build our shared clean energy future.

This report describes National Grid's vision and the steps we are taking to leverage our existing gas and electric networks to achieve fossil-free heat fairly, affordably, and reliably for our customers and communities.



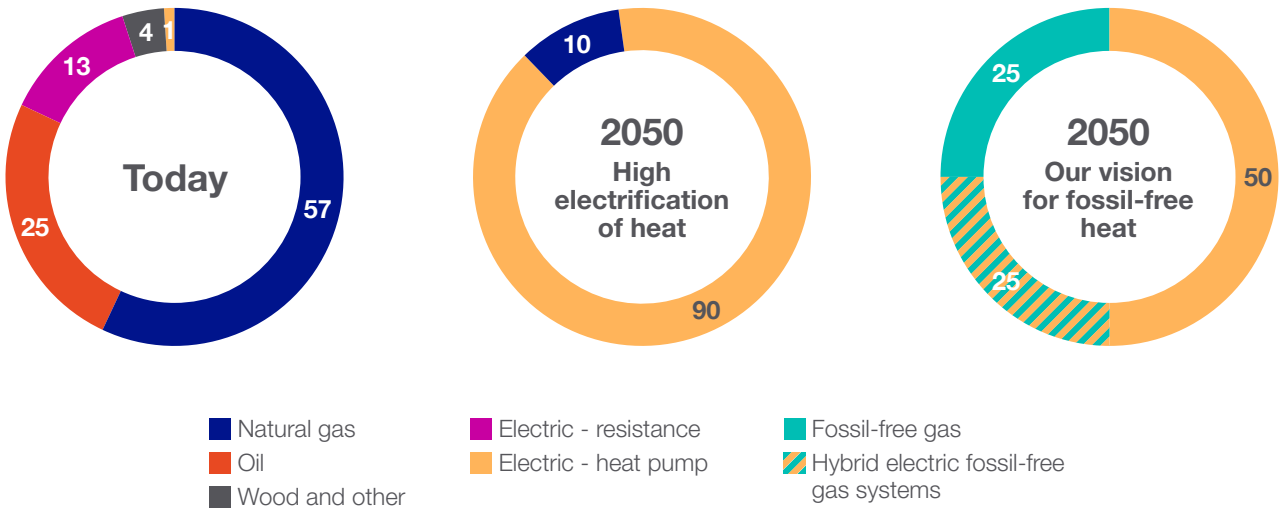


2. Our vision for fossil-free heat

Today, heating makes up about 39% of GHG emissions in Massachusetts and New York.[1] Heating is provided by a diverse set of fuels and heating systems. Of the roughly 10 million households in Massachusetts and New York, nearly 6 million (57%) have a gas furnace or boiler system, 2.5 million (25%) use a boiler fueled by oil or propane, and 1.4 million (14%) use electricity.[2] Of those using electricity, most use electric resistance, an inefficient heating source, and about 100,000 (1%) use a heat pump – a type of efficient air conditioning unit that can also provide heat.[3]

Draft climate action plans from the Massachusetts and New York state agencies propose that nearly 10 million households change their heating systems to electric heat pumps by 2050 (see Figure 1). In Massachusetts and New York, this would require over 340,000 customers per year converting their current heating source to electricity, every year until 2050.

Figure 1. Buildings heating systems in Massachusetts and New York (% of total).



National Grid's vision for fossil-free heat targets a hybrid approach. Just as we have decarbonized electricity with wind and solar, we can decarbonize the gas system with renewable natural gas and green hydrogen. This will enable customers to have choices in how to become fossil-free. And, by utilizing our existing gas and electric networks, our vision will more quickly deliver a reliable and cost-effective transition to a net zero future for the entire energy system.

At National Grid, we believe that electrification of heat will play an important role for some of our customers in achieving net zero. We also believe that we need to offer a diverse and practical range of cleaner heating solutions enabling customers to choose what best addresses their needs for performance and cost, without compromising our shared climate goals.

While converting to electric heating may be viable for some customers, others may have challenges due to the upfront cost of making this switch or the difficulty of retrofitting their building. Today, fully electrifying a typical home in Massachusetts or New York costs roughly $20,000-60,000.[4] Additionally, our analysis finds that – given the physical limitations of existing buildings – it is technically difficult to electrify 30-70% of building space in the urban areas of New York City and Boston and 5-40% outside of those major cities.[5]

National Grid's vision for fossil-free heat will enable our customers to have clean heating solutions available to them. Along the way, we can build an integrated energy network that is clean, renewable, and achieves zero emissions.

A better way to net zero

Our analysis shows that using a balanced and diversified approach to heating is the most practical, expedient, and affordable path to reducing emissions for all our customers. This is consistent with findings from Energy + Environmental Economics (E3), an energy industry consulting firm, who in their 2022 report developed for the Massachusetts "Future of Gas" proceeding state "A coordinated gas and electric decarbonization strategy, utilizing a diverse set of technologies and strategies, is likely to be better able to manage the costs and feasibility risks of decarbonization than scenarios that rely more heavily on single technologies or strategies."[6] Much like how we are creating a clean electricity network through the adoption of renewable electric generation sources, such as wind and solar, we can achieve a fossil-free gas system through the adoption of renewable natural gas and green hydrogen.

Our fossil-free vision rests on four pillars of action that will enable all homes and businesses we serve to meet their heating needs without the use of fossil fuels by 2050, if not sooner, achieving net zero emissions while keeping energy affordable and preserving customer choice.

Figure 2. Pillars of our vision for fossil-free heat.

Pillar one

Energy efficiency in buildings

We will continue to provide programs for our customers to accelerate energy efficiency improvements to buildings, including deep retrofits and measures that reduce peak gas and electric demand; and support more rigorous building codes for new buildings.

Pillar two

100% fossil-free gas network

We will eliminate fossil fuels from our existing gas network no later than 2050 by delivering renewable natural gas and green hydrogen to our customers.

Pillar three

Hybrid electric-gas heating systems

We will support our customers by providing them strategies and tools to capture and maximize the benefits of pairing electric heat pumps with their gas appliance.

Pillar four

Targeted electrification and networked geothermal

We will support cost-effective targeted electrification on our gas network, including piloting new solutions like networked geothermal. We will support customers who heat with oil and propane with strategies and tools to convert to heat pumps.

Pillar one

Energy efficiency in buildings

We will continue to provide programs for our customers to accelerate energy efficiency improvements to buildings, including deep retrofits and measures that reduce peak gas and electric demand; and support more rigorous building codes for new buildings.

Energy efficiency is a cornerstone of our vision – we expect it can deliver one-third of needed emissions reductions from heating. Wider adoption of measures to better insulate and weatherize buildings and replace old appliances with more efficient ones can significantly lower the amount of fuel required to keep buildings warm and comfortable as well as provide other services such as cooking. This not only lowers direct costs for customers but also reduces peak demand and infrastructure that needs to be built. National Grid will maximize energy efficiency's potential by continuing our nation-leading programs that provide rebates and incentives to customers to adopt these measures in existing buildings. We must also construct all new buildings to be highly energy efficient and ready for a net zero future.



Scaling up insulation (pictured) and other buildings energy efficiency improvements are foundational to keeping energy affordable and achieving our net zero future.

Pillar two

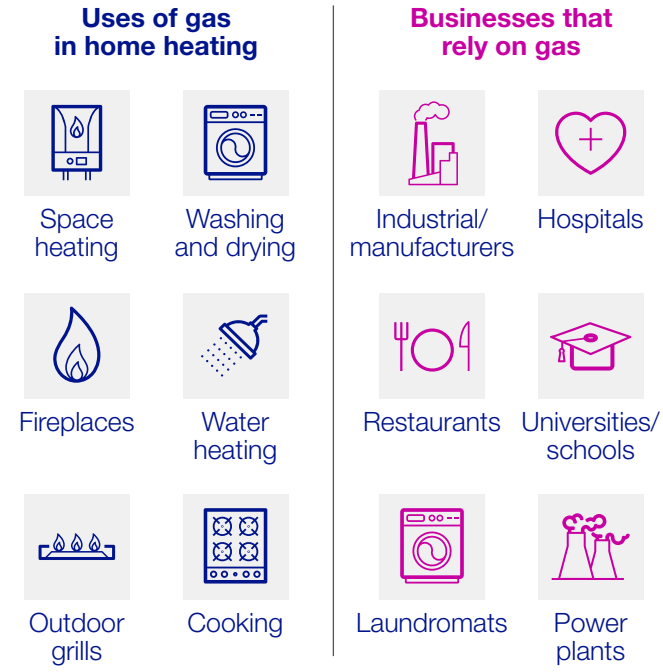
100% fossil-free gas network

We will eliminate fossil fuels from our existing gas network no later than 2050 by delivering renewable natural gas and green hydrogen to our customers.

Today, nearly 6 million households in Massachusetts and New York are heated with natural gas. In many homes, gas is also used for hot water, cooking, dryers, fireplaces, and outdoor grills (see Figure 3). Natural gas also fulfills important needs for businesses in delivering high quality goods and services: a storable and immediate source of fuel for power plants, process heat for industries and manufacturing, cooking for restaurants, washing and drying at laundromats, and others. Hospitals, for example, are required by federal law to have two sources of energy; gas plays a critical role as an emergency back-up source when the power is out. Many universities and schools have their own plants onsite that generate both power and heat from natural gas and have existing infrastructure to distribute it to their buildings on campus.

Our vision replaces fossil natural gas with fossil-free – renewable natural gas and green hydrogen – enabling all our customers to have a clean solution for heating their homes and operating their businesses in the way they determine is most affordable and practical for their needs.

Figure 3: Uses of gas in homes and businesses.



What is renewable natural gas?

Renewable natural gas (RNG) captures biogases that naturally emit from society's recurring waste streams. RNG turns a liability that negatively impacts the environment into a climate solution that can be carried in our existing gas network and used in existing customer equipment.

Society will always produce waste, which naturally emits surface-level methane as it decomposes. RNG reduces emissions by capturing this potent GHG before it can escape into the atmosphere (and contribute to our changing climate) and putting it to productive use as an energy resource. In that process, it provides a **double benefit of reducing methane emissions while providing a source of renewable energy**. By simultaneously capturing methane and displacing fossil natural gas, **RNG has the lowest lifecycle carbon intensity of any energy source available today**. Additionally, RNG is a reliable source of energy that can be easily stored and delivered to customers when and where they need it. And, it is a "drop-in" fuel – able to be transported through existing gas networks, ensuring customers will not need to make changes to their existing appliances.



RNG facilities (pictured) capture methane from waste – preventing emissions into the atmosphere – and create a valuable product to use in heating.

RNG can be sourced from a vast number of existing waste streams and renewable biomass. Sources are all around us: landfills, farms, wastewater treatment plants, food waste, and biomass waste residues (see Figure 4). As the global population grows, the World Bank projects solid waste to increase nearly 70% by 2050.[7] RNG helps improve society's response to waste and the emissions created from it.

Figure 4: Waste streams that can produce renewable natural gas.



RNG is a readily available solution and we have successfully been injecting RNG into our gas system since 1981. RNG supply is growing rapidly, increasing by more than six times nationwide since 2015. Today, there are over 150 plants in operation in the US and Canada and thousands of additional sites waiting to be developed. Using the existing gas network that connects us to the broader region, this resource can supply fossil-free gas to heat our homes and businesses. Our fossil-free vision assumes that National Grid eventually procures 10%-20% of RNG annual supply potential in the Eastern US for our customers, roughly in line with our customers' share of Eastern US residential and commercial gas demand today.

What is green hydrogen (H2)?

Hydrogen, the most abundant chemical element on earth, offers enormous potential as a source of clean energy and fossil-free heat. **When hydrogen gas is burned to release its energy, the main byproduct is water vapor**. Hydrogen produced using renewable feedstocks is known as green hydrogen. One of the most promising green hydrogen pathways is the process of electrolysis using renewable electricity from wind and solar, which is carbon free.

Hydrogen can help decarbonize multiple sectors, including heat, power generation, and transport. For heating, hydrogen can be blended with natural gas or RNG up to 20% by volume, run through our existing gas networks, and used in customer appliances without significant upgrades to infrastructure or equipment. In areas with high levels of gas demand, pure hydrogen also has potential to serve fossil-free heating and other energy needs in dedicated 100% hydrogen clusters. These may be part of "hydrogen hubs" – clusters of local hydrogen production, storage, and demand – such as those in which the US Department of Energy is investing $8 billion to establish around the country to support innovation and to scale up a hydrogen economy.[8] The Northeast is well positioned to host one of these hubs, and New York, Massachusetts, together with Connecticut and New Jersey, are developing a proposal to implement a hydrogen hub in the region [9]. Driven both by federal government support and by growing interest from companies seeking to replace fossil fuels, costs of hydrogen are expected to fall rapidly. In 2021, the US Department of Energy announced its "Hydrogen Shot" target to reduce the cost of green hydrogen by 80% over the next 10 years, which would bring the cost in line with today's costs for natural gas.

Green hydrogen complements growing renewable electricity capacity due to its ability to be stored and its flexibility to be used across different sectors. Hydrogen can be made during periods when wind or solar resources are able to produce more electricity than the grid needs, and then stored for later use, thereby maximizing the benefits of renewable energy resources. Already, a significant number of offshore wind projects are proposing hydrogen production through electrolysis as a form of long-duration storage of renewable energy.

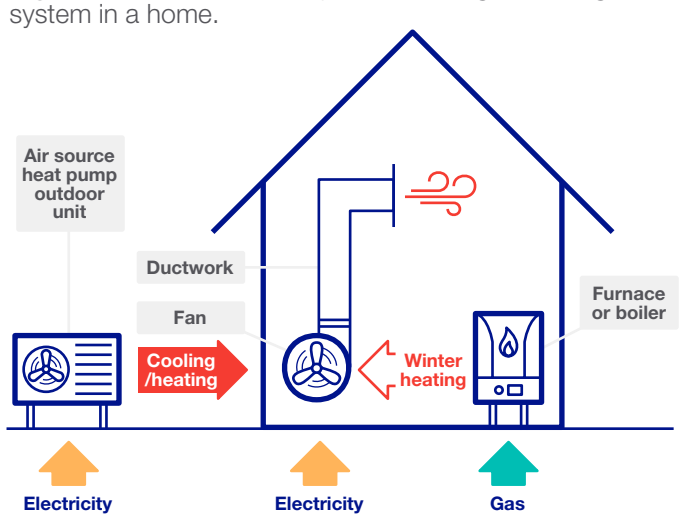
Pillar three

Hybrid electric-gas heating systems

We will support our customers by providing them strategies and tools to capture and maximize the benefits of pairing electric heat pumps with their gas appliances.

Hybrid heating systems utilize both electricity and gas for providing year-round comfort in homes and buildings. For example, heat pumps in the Northeast today are mainly installed for cooling to replace old and inefficient air-conditioning. As a result, many customers have hybrid heating systems, where the heat pump covers cooling and heating during part of the year, and a gas system provides heat during the coldest months (see Figure 5). **Of customers adopting heat pumps today, approximately 80%[10] choose hybrid configurations over full electrification**. Hybrid systems provide a more affordable and reliable mix of heating and cooling throughout the year, allowing installation of smaller and cheaper heat pump systems, while avoiding large and costly modifications to buildings and heating systems.

Figure 5: Schematic of a hybrid electric-gas heating system in a home.



Hybrid heating systems utilize both electricity and gas for providing year-round comfort in homes and buildings.



Pillar four

Targeted electrification and networked geothermal

We will support cost-effective targeted electrification on our gas network, including piloting new solutions like networked geothermal. We will support our customers who heat with oil and propane to convert to electric heat pumps.

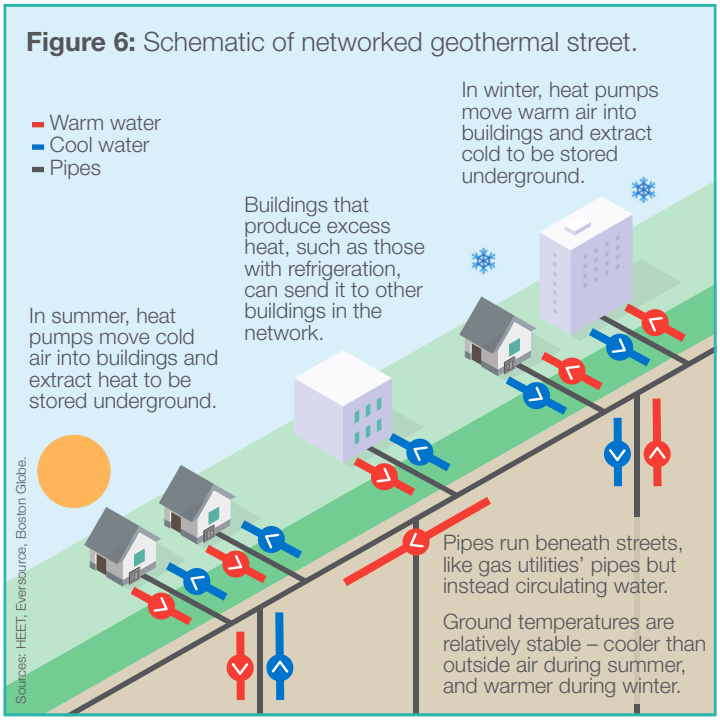
Under any heating sector net zero vision, there is an important and critical role for electrification. Air-source heat pumps move heat from the outside air into buildings, and also provide cooling. When paired with 100% renewable electricity, electric air-source heat pumps can play an important role in achieving net zero for heat, especially where cost-effective. **For example, today it is cost-effective for customers who heat with oil, propane, or old electric resistance systems to convert to heating with electric air-source heat pumps**.

Ground-source heat pumps are another electric heating technology that can help reduce GHG emissions. Ground-source heat pumps efficiently draw heat from the ground and can reduce strain on electric grids compared to air-source heat pumps, because they are able to more efficiently heat homes using consistent heat from the earth instead of variable heat from the air. Whenever possible, ground-source heat pumps should be encouraged, because they minimize electric peak impacts and can fully replace traditional heating more easily than air-source heat pumps.



Air-source heat pumps (pictured) move heat from the outside air into buildings, and also provide cooling in the summer.

Figure 6: Schematic of networked geothermal street.

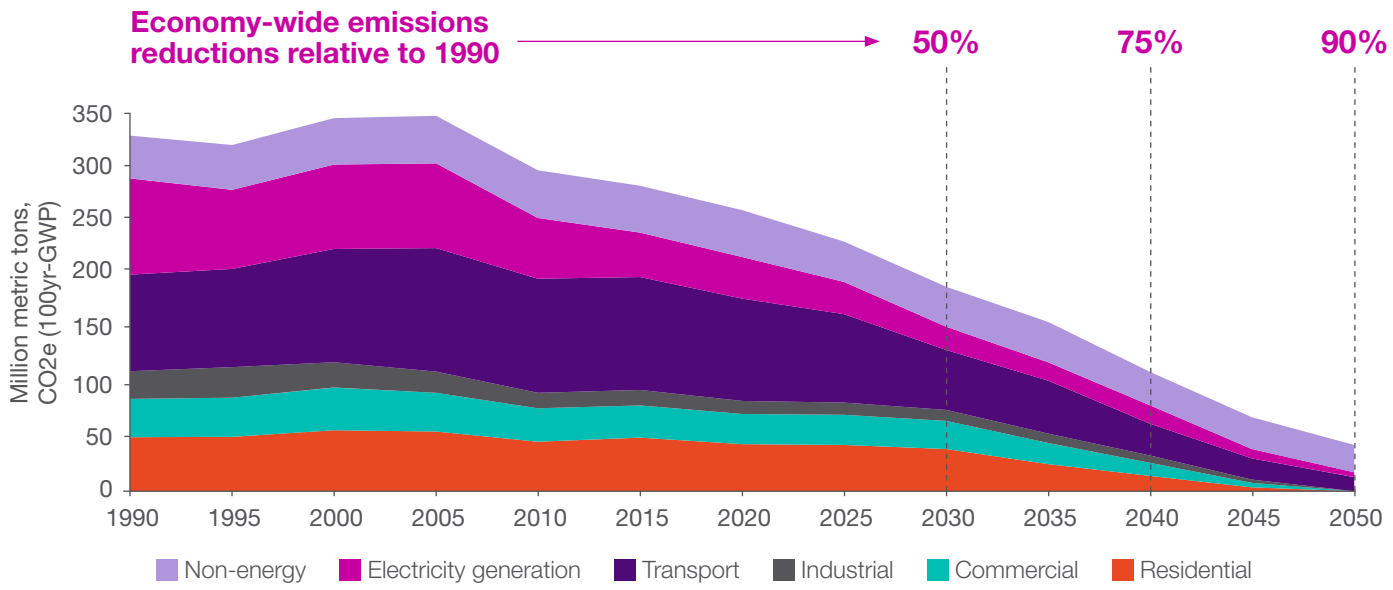


Ground-source heat pumps can also be connected via geothermal networks (see Figure 6) to allow customers to share heating and cooling, making geothermal networks even more efficient than standalone ground-source heat pumps. Installing and maintaining geothermal networks also makes sense as part of the gas utility transformation, as it utilizes our skilled workforce who understands underground construction and it leverages our existing customer relationships.

Summary of how our vision reduces emissions

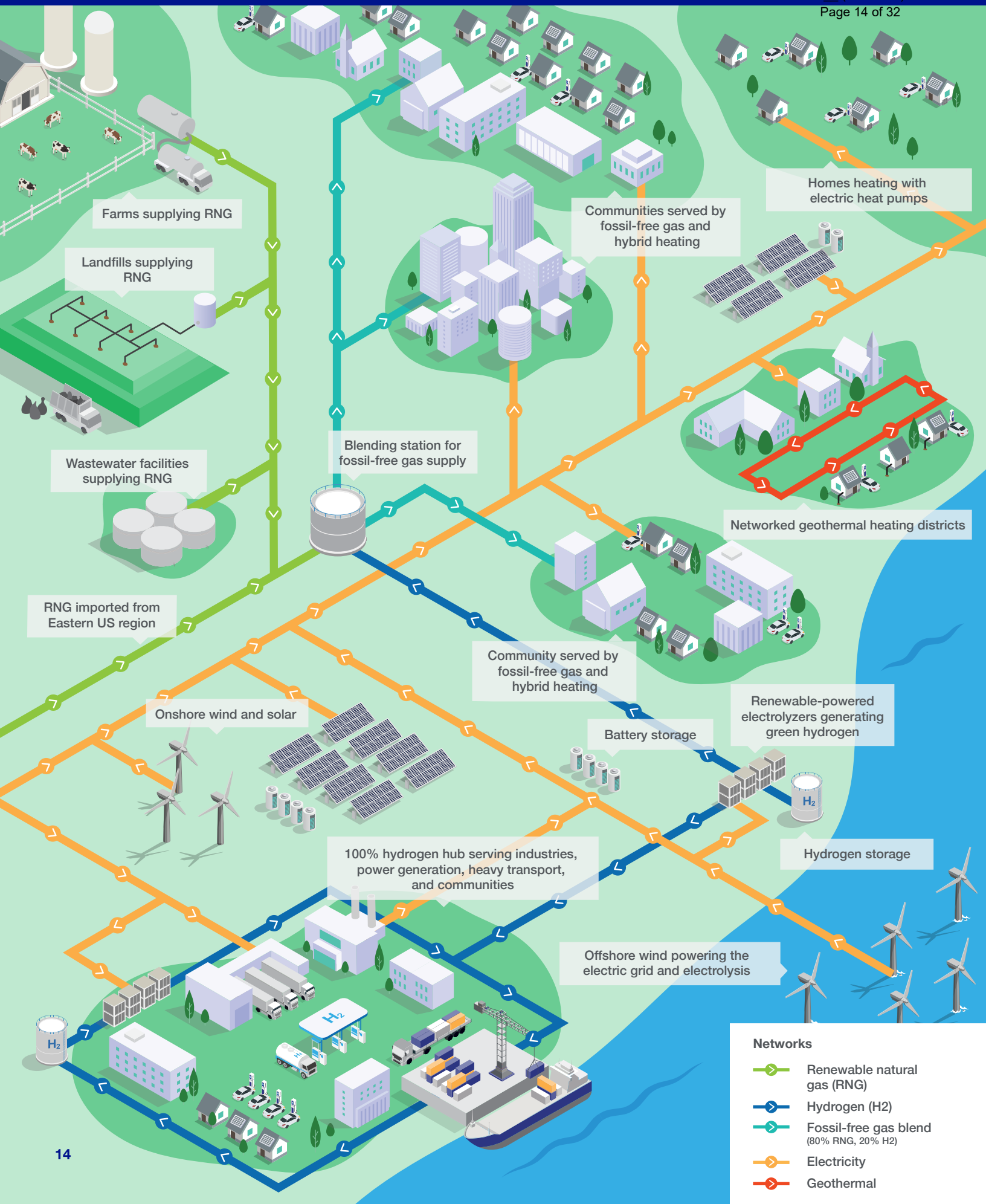
At the center of our vision is an integrated, fossil-free gas and electric system that will enable us to serve all our customers and communities with fair, affordable, and reliable heat. Our vision provides a practical path to a clean energy future, **where economy-wide emissions fall 50% by 2030 and 90% by 2050** (see Figure 7).

Figure 7: Economy-wide emissions in Massachusetts and New York in our vision for fossil free heat.



Our clean energy vision

A fossil-free future for cleanly heating homes and businesses



Roadmap to achieve our vision

2050

100% fossil-free heat

2045

Net zero economy-wide emissions

- Continue to accelerate deployment of fossil-free gas, clean electricity solutions, and energy efficiency.

2040

Wide adoption of fossil-free heating solutions

- Blend 20% green hydrogen and 30% RNG in our network
- Build 100% hydrogen hubs serving multiple uses
- Deploy hybrid heating systems widely
- Support communities with networked geothermal heating districts and targeted electrification
- Convert all customers heating with oil to electric heat pumps
- Significantly advance building efficiency.

2030

Scale a broad set of solutions

- Invest in infrastructure to deliver fossil-free gas and electric solutions
- Serve 10-20% of gas demand with RNG
- Promote adoption of hybrid heating
- Run community-scale pilots of H2 blending
- Support deployment of H2-ready appliances
- Progress neighborhood adoption of networked geothermal heating and targeted electrification
- Convert a majority of customers who heat with oil to electric heat pumps
- Continue to support states' progress toward zero-emission and renewable electricity goals.

2022-2025

Set foundation to transition to fossil-free heat

- Support legislative and regulatory policies to grow fossil-free gas, including adoption of a renewable heating portfolio standard
- Develop voluntary tariffs for customer participation in fossil-free gas offerings
- Continue to convert customers who heat with oil to electric heat pumps at pace
- Run neighborhood pilots for green H2 network blending, networked geothermal, and targeted electrification
- Promote adoption of hybrid heating
- Demonstrate technical feasibility and scalability of fossil-free gas networks
- Continue innovation in nation-leading energy efficiency programs.

Today

Delivering safe, affordable, and reliable heat

- Investing in network modernization to reduce fugitive methane emissions and increase safety and reliability
- Providing nation-leading energy efficiency and demand response programs to help customers lower energy bills and usage
- Accelerating conversion customers who heat with oil to electric heat pumps
- Partnering with NY and MA to propose a regional Hydrogen (H2) Hub
- Participating in university-led H2 blending studies
- Interconnecting renewable natural gas (RNG) projects to our network
- Seeking regulatory approval for RD&D to transition the gas network to net zero, including piloting fossil-free gas and networked geothermal.

3. How our vision benefits all our customers and society

As we build toward a fossil-free future, it is essential that we leave no customer behind. All customers must benefit from the clean energy transition. Our vision enables customers to continue choosing affordable heating options while achieving our states' emissions reduction targets and supporting safe, reliable, and equitable outcomes for society.

We believe an integrated, fossil-free gas and electric system will provide our customers and society, clear benefits:

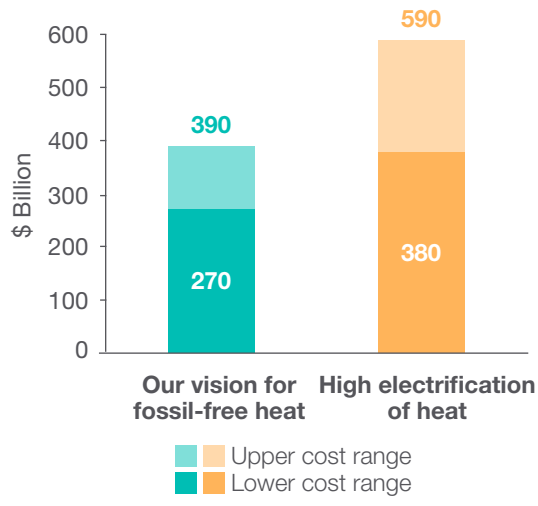
- lowest cost for society
- provides customers affordable choices to heat their homes and businesses
- results in more equitable outcomes, keeping energy affordable for everyone
- gives customers smarter, practical heating choices
- utilizes the skills of our existing workforce
- supports a reliable and resilient energy system by not having "all our eggs in one basket"
- allows a more manageable build out of electric infrastructure
- more likely to achieve our shared net zero goals by embracing all options.

Most affordable and practical for our customers to reduce emissions

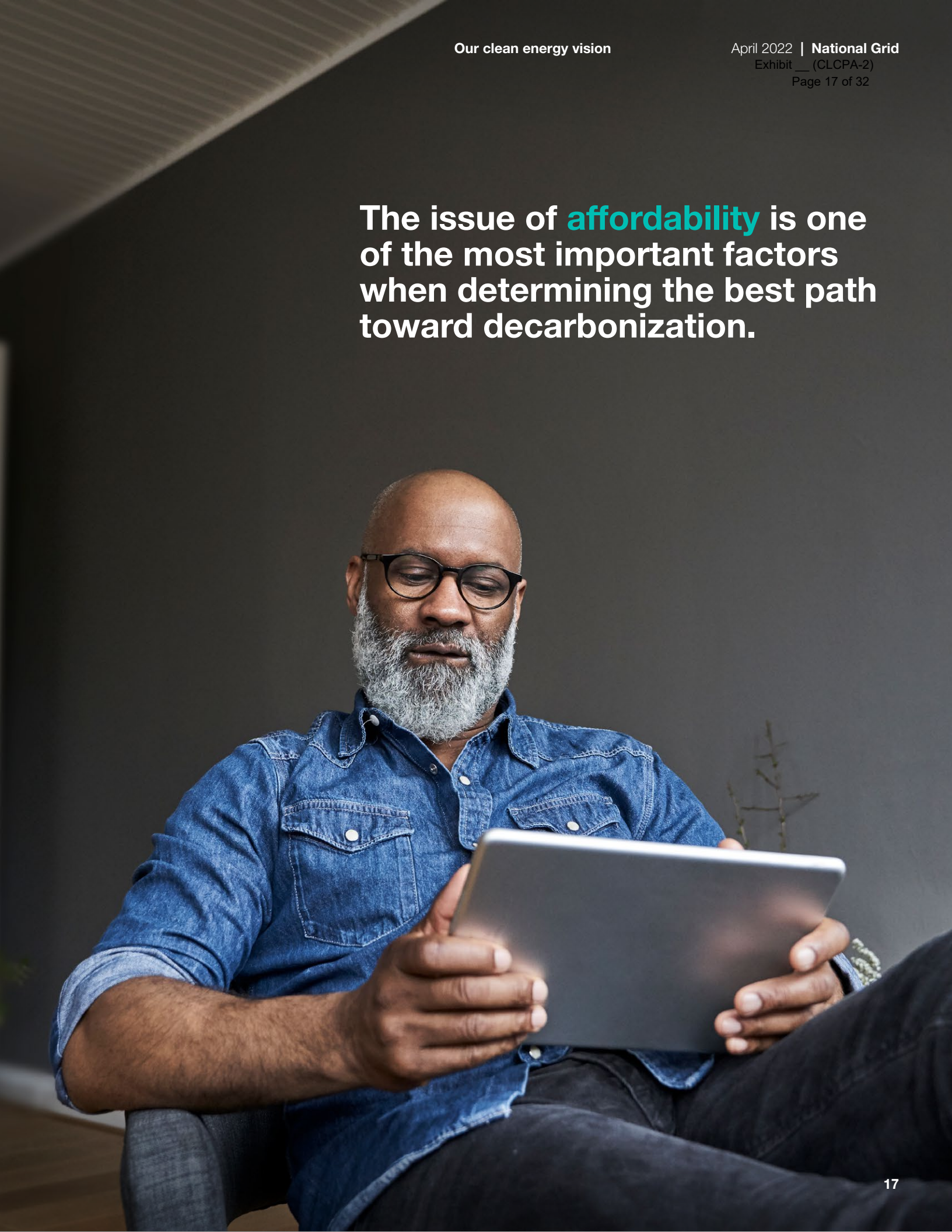
Lowest cost for society

Given concern over inflation and economic inequality, the issue of affordability is one of the most important factors when determining the best path toward net zero. On average, we estimate our vision will save our communities $110-200 billion dollars in economy-wide costs by 2050 compared to a high electrification approach, which is similar to the "All Options" pathway developed in the *Massachusetts 2050 Decarbonization Roadmap* and "Scenario 3: Accelerated Transition Away from Combustion" scenario developed in the New York State *Climate Action Council Draft Scoping Plan* (see Figure 8).[11] These cost savings are a direct result of leveraging existing gas system infrastructure, including customer heating equipment, to both achieve net zero targets and keep customers warm during the coldest times of the year.

Figure 8: Relative economy-wide costs to reduce emissions in Massachusetts and New York, 2020-2050.



The issue of affordability is one of the most important factors when determining the best path toward decarbonization.



Provides customers affordable choices to heat their homes and businesses

The answer to "what heating choice is most affordable for me?" will vary depending on the unique circumstance of each of our customers and the timing of that choice. In our service area today, heating with gas is the most cost-effective option for virtually all customers. Especially for low to moderate income customers, affordable heating is crucial.

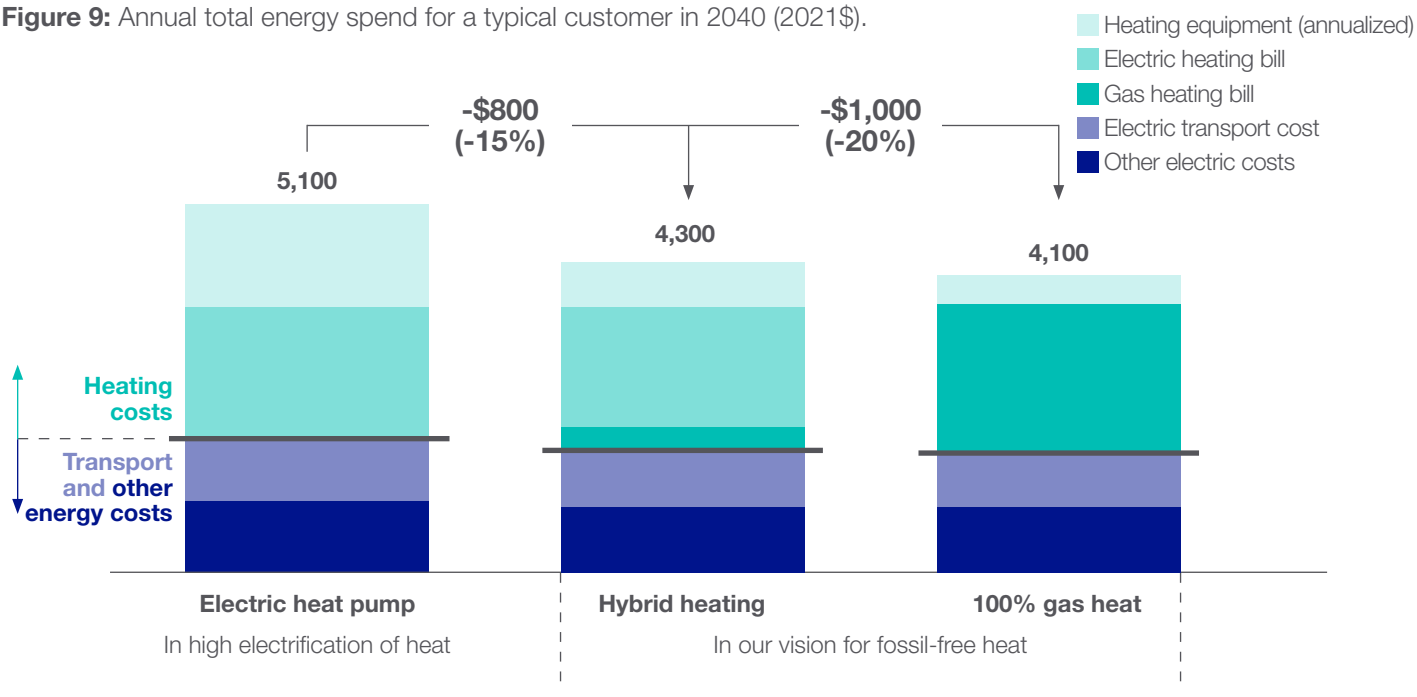
As heating technologies and net zero policies evolve, it will be possible to continue delivering affordable heating options without endangering sustainability goals.

For residential customers living in older, multi-unit buildings and for many of our commercial and industrial customers, findings from external studies show that reducing emissions through efficient, fossil-free gas heat is likely to be the most affordable option for the next three decades. Electrifying heat in these building types will require more expensive heating equipment or more significant building retrofits.[12, 13,14]

For residential customers living in single-family homes, hybrid heating systems have been shown to be the most cost-effective solution for the next decade.[15] Hybrid heating systems are able to lower customer costs by utilizing the strengths of both the gas and electric systems – they can run on renewable electricity when it's available and switch to fossil-free gas for the winter months, leveraging the gas system that has already been built to heat our customers' homes. Hybrid heating systems also significantly lower upfront equipment and building energy efficiency upgrade costs for customers who want some heating electrification.

Not only does our fossil-free vision bring lower heating costs, it also keeps electricity rates lower by avoiding significant overbuilding of generation supply and network upgrades. Compared to a high electrification pathway, electricity rates under our vision are expected to be approximately 5-10% lower in 2040 and 10-15% lower in 2050 than high electrification approaches because it requires less generation, transmission, and distribution infrastructure buildout. By keeping overall rates for electricity lower, it will make it more affordable to use heat pumps, charge electric vehicles, and power everyday appliances. As a result, the average consumer is expected to save 15-20% on their overall energy costs in a given year compared to a high electrification approach (see Figure 9). This is significant savings and real money for an average household. By saving $800-1,000 per year on energy costs, families will have more money for other essentials such as food and housing.

Figure 9: Annual total energy spend for a typical customer in 2040 (2021$).



Results in more equitable outcomes, keeping energy affordable for everyone

Ensuring energy equity must be an important goal for all of us who care about our customers and communities.

Relative to a high electrification approach, our vision to eliminate emissions from heating results in fairer outcomes for society. For example, our customers should not have to spend $20,000-60,000 in upfront costs to fully electrify their homes in order to have clean energy.[16] In our vision, access to clean heating is more equitable because more customers can have clean energy with their existing appliances and heating equipment.

In short, utilizing both renewable electricity and a fossil-free gas system to achieve net zero will significantly reduce upfront cost barriers for customers and keep customer energy bills more affordable by avoiding significant amounts of new infrastructure.

Gives customers smarter, more practical heating choices

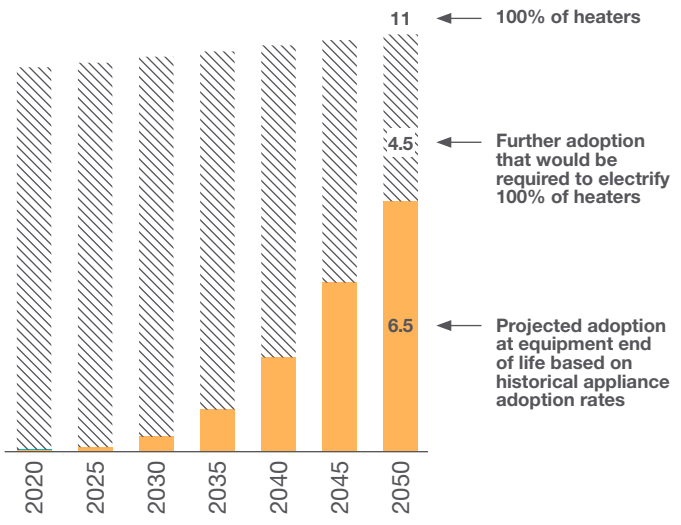
The reality is that a large share of buildings in the Northeast are difficult to electrify – and this will remain an enduring problem. **Our fossil-free vision allows customers to use existing equipment and appliances to achieve net zero and provides an invaluable option for customers who have trouble electrifying**.

In addition to the challenges mentioned above, there may also be other practical challenges that make electrification difficult, such as a renter's ability to convince their landlord to install a new heating system, the availability of equipment, or finding the skilled labor to install an electric heat pump at the time it is needed. Our vision provides more options for renters and disadvantaged communities to have access to clean energy.

Our vision allows for adoption of new cleaner and fossil-free empowered heating technologies to happen on a timeline that is more practical for customers. In contrast, a high electrification approach would require 90,000 homes to electrify per year in Massachusetts and 250,000 homes to electrify per year in New York. What does this mean in reality? This means roughly 2,000 full electric conversions per week in Massachusetts and 5,000 in New York, every week from now until 2050. Either household electrification rates would need to increase by roughly 15-20 times, or half of households would need to replace their gas appliances and heaters before their useful end of life in order to achieve net zero by 2050.[17]

At normal levels of heating equipment turnover, Massachusetts and New York cannot come close to electrifying everything by 2050. Adoption rates similar to historical home appliance adoption levels imply a 40-50% shortfall (see Figure 10). A strategy that relies on customers' willingness to spend thousands of dollars to replace heating equipment that is currently working well is unlikely to succeed. Instead, a better and more equitable way to achieve net zero emissions by 2050 is to transition the gas network to fossil-free fuels. This ensures all our customers will have fossil-free heating regardless of their choice in heating appliances.

Figure 10: Heating appliances that could be electrified in Massachusetts and New York with normal levels of turnover (millions).



Utilizes the skills of our existing workforce

Our fossil-free vision puts the existing skill set of our gas workforce, including over four thousand union workers, at the heart of the clean energy transition. Empowering our gas workers to play a crucial role in ensuring states achieve their net zero targets will keep important jobs available and leverage the skill sets we already have within our talent pool today. Our workforce is uniquely skilled to transform our existing gas system to deliver RNG and green hydrogen, as well as install new heating technologies such as networked geothermal loops.

It will also help retain the talent necessary for ensuring the safe and reliable operation of the gas network throughout the energy transition while providing robust training opportunities for individuals interested in transitioning to roles within the electric industry. Maintaining a role for the gas network in a net zero future reinforces the enduring value of our frontline union gas workforce as we build this new energy future.



The gas industry provides tens of thousands of good-paying union jobs across the Northeast.

A complementary but diversified electric and gas system is safe, reliable, and more resilient

Supports a reliable and resilient energy system by not having "all our eggs in one basket"

Relying on one system for everything – heat, transport, and power – is riskier than a system with redundancy. An underground pipeline system doesn't have storm-related outages, and can be leveraged to improve the reliability of the overall energy system.

Utilizing both a fossil-free gas system and electric network can reduce the risk of relying on a single energy system for the region's heating capacity, and the fundamental health, security, and safety issues that result from heating service interruptions, such as storms. Our fossil-free vision supports reliability by reducing dependence on high gas volumes during peak times – E3's *Technical Analysis of Decarbonization Pathways* report for Massachusetts showed that a high electrification scenario would need 50% more total gas volume on a peak day to generate electricity as compared to scenarios that utilize the gas network for heat and 15% more gas volume than peak day gas consumption today.[18] This is because gas generation is required on days when there is little sunshine for solar and variable amounts of wind for renewable electricity generation. We expect similar trends to hold true for New York, given similar levels of electrification needed.

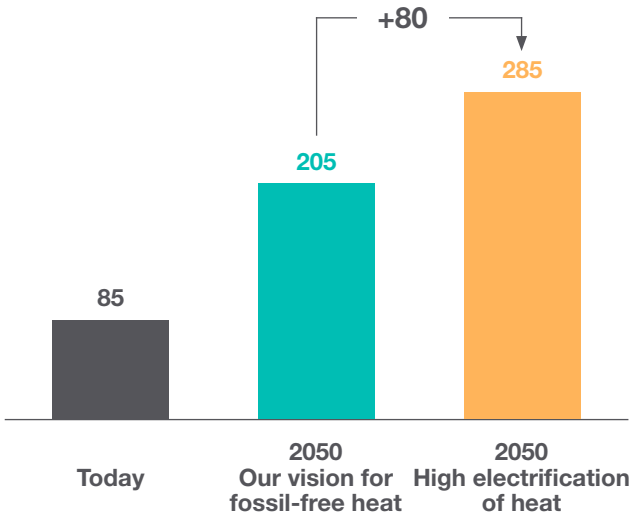
Allows a more manageable build out of electric infrastructure

By utilizing an existing energy network, our fossil-free vision supports a more manageable energy transition by relieving pressure to site, permit, and build electric infrastructure at unprecedented rates. The reality is that we would need more electric infrastructure to electrify transport and heating than we have today. Why would such infrastructure be needed? In a high electrification future, winter peak demand would be double today's summer peak and 2.5 times greater than the current peak demand for electricity in the winter.[19] Due to a shift in reliance toward renewables that vary with the sun and the wind, some of which do not perform well in winter (*e.g.* solar), we would need to **increase the amount of electric supply capacity by about 3.5 times** to cover a cold winter day in an all-electric scenario (see Figure 11). This would also require approximately 6,500 miles of new or upgraded high voltage electric transmission lines across New York and New England – requiring the siting and permitting of approximately 40 to 50 new high voltage transmission lines in the Northeast before 2050 (as pictured).[20]



A high electrification of heat approach will require building significant additional electric infrastructure, such as high voltage transmission lines (pictured) to supply greater amounts of electricity at peak times.

Figure 11: Electricity supply capacity in New England and New York (GW)



In contrast, using fossil-free gas avoids the additional costs and challenges of 80 gigawatts (GW) of new generation and associated transmission and distribution networks – 25 GW in New England and 55 GW in New York. **This avoids the siting, permitting, and construction of one third of these challenging projects, which typically take 4-8 years to permit and build.**

More likely to achieve our shared net zero goals by embracing all options

Achieving net zero targets over the next few decades will be challenging, but we have a higher chance of succeeding if we embrace all options versus picking winners and losers. Although net zero models and scenarios help us make informed decisions, we will undoubtedly face a number of unknowns that will shape our path to net zero that are out of our control. For example, what impact will recent global and geopolitical events have on energy prices or energy security? How vulnerable is an energy system that is fully electric to outside attack by a foreign actor? Reducing our dependence on fossil fuels further insulates us from geopolitical volatility.

Our fossil-free vision improves reliability by providing a critical redundancy and preserves and provides options for customers without sacrificing or jeopardizing our net zero goals. In our vision, there are clear 'offramps' available. For example, if low carbon gas prices evolve differently than expected, higher electrification could be leveraged. In contrast, taking steps that remove clean gas options today would shut down a wide range of options that would be difficult to re-open later.

4. Public policies to enable our vision

Many of the policies to enable our vision for fossil-free heat are already in place, such as renewable electricity portfolio standard (RPS), energy efficiency programs and targets, and heat pump incentives. To complete the picture, a few additional state and federal policies would allow greater progress towards net zero, such as:

Scaling supply and demand for renewable fuels

New York and Massachusetts should drive toward net zero gas by establishing renewable gas procurement standards for utilities. Following the successful examples of state renewable electricity portfolio standards, renewable gas standards should:

- require gas utilities to procure a growing proportion of their gas from RNG and renewable green hydrogen over time – by creating a RPS for the gas network, as was done for the electric network, or evolving the way regulators evaluate gas supply plans to include consideration of environmental benefits
- allow further voluntary options to allow customers seeking to decarbonize faster to choose higher blends of renewable gas (up to 100%), much like they can choose varying percentages of renewable electricity.

Complementing these portfolio standards, state and federal incentives should be put in place to support growing investment in equipment to capture biogas (*e.g.* "digesters") and clean and condition it to RNG to make hydrogen (*e.g.* "electrolyzers"). Other states with emissions reduction goals have already implemented these complementary policies, such as California and Washington.

Driving investment in buildings and renewable heating equipment

State and local building codes are essential drivers of efficient and fossil-free-ready new construction, and should prioritize highly-efficient building envelope improvements to reduce the amount of energy required to provide comfortable heat. Energy efficiency policies should increasingly incentivize measures that reduce both annual heating demand as well as peak demand on the gas network, including in targeted gas network locations.

Existing utility incentives for renewable heating equipment should be expanded, supporting more customers with lower-carbon options best suited to their needs – including hybrid/dual-fuel, all-electric, and high-efficiency gas configurations. Where possible, ground-source heat pumps should be encouraged, including networked geothermal (or "district") systems, because they minimize peak electric impacts and can fully replace traditional heating more easily than air-source heat pumps. Appliance standards for high-hydrogen gas blends should likewise be developed. Finally, market-based carbon pricing mechanisms should be established to provide long-term price signals for cleaner heating and provide an additional source of revenue for efficiency and heating incentive programs.

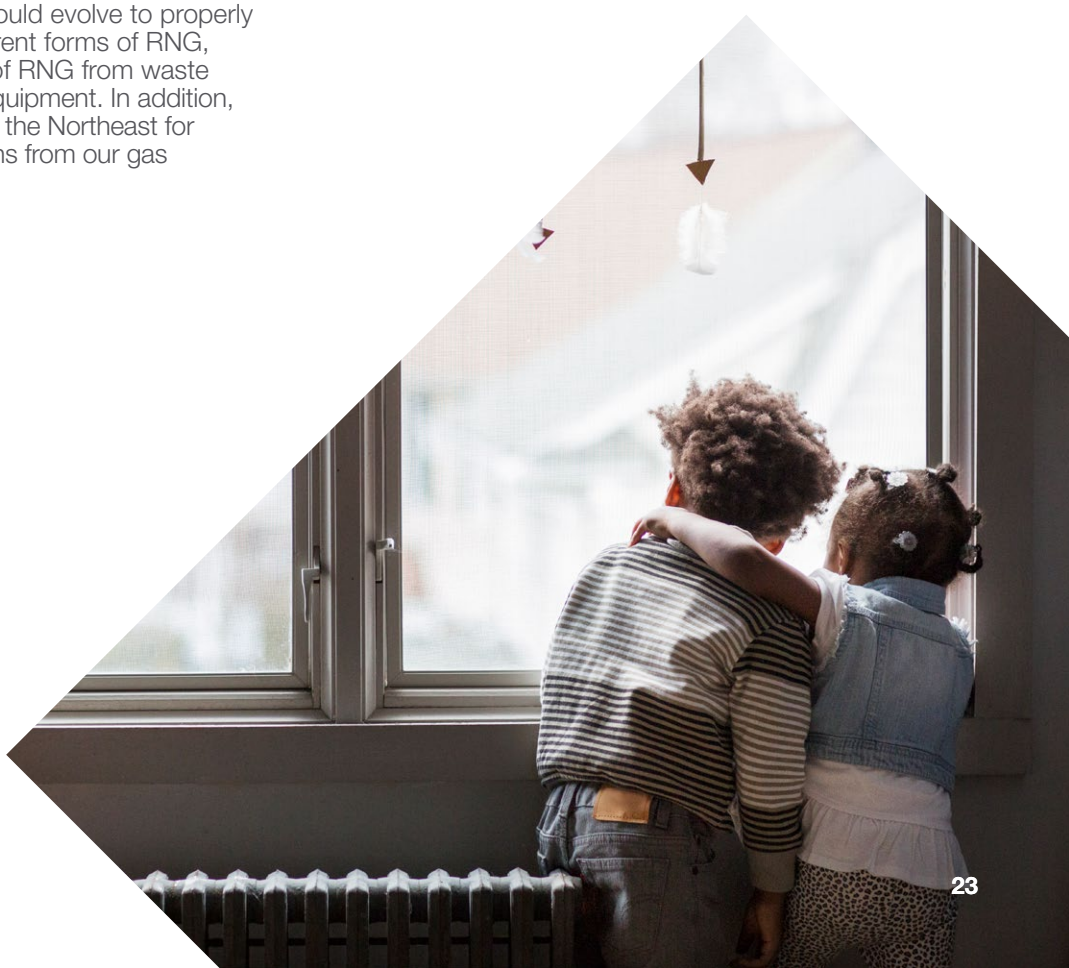
Evolving infrastructure planning and investment

Regulatory planning processes should support long-term network strategies that promote the most efficient use of the gas and electric systems for heating, including geographically targeted fossil-free strategies (*e.g.* hydrogen clusters or electrified neighborhoods). Current planning processes only look out 2-5 years. This horizon should be expanded for gas and electric utilities.

Gas planning practices, service regulations and cost recovery should support both the delivery of renewable gas and the reduction in overall gas demand over time. Sufficient regulatory funding should be provided for the research, development, and demonstration (RD&D) of innovative technologies required to meet state climate targets. In addition, policy frameworks should be established to guide utility investment in renewable district thermal networks, providing an additional tool for achieving fossil-free heat.

Ensuring consistent greenhouse gas accounting methods

Transparent and consistent GHG accounting methods are essential across all sectors of the economy, including the energy sector. Accounting methods should be standardized across our state and federal jurisdictions, and they should be as consistent as possible with international standards. New York and Massachusetts accounting should evolve to properly account for the life cycle carbon of different forms of RNG, recognizing the environmental benefits of RNG from waste and of displacing fossil gas in end-use equipment. In addition, standard approaches are needed across the Northeast for measuring the impact of fugitive emissions from our gas distribution network.



5. Conclusion: What our vision means for the Northeast

How does our vision meet state and regional decarbonization goals?

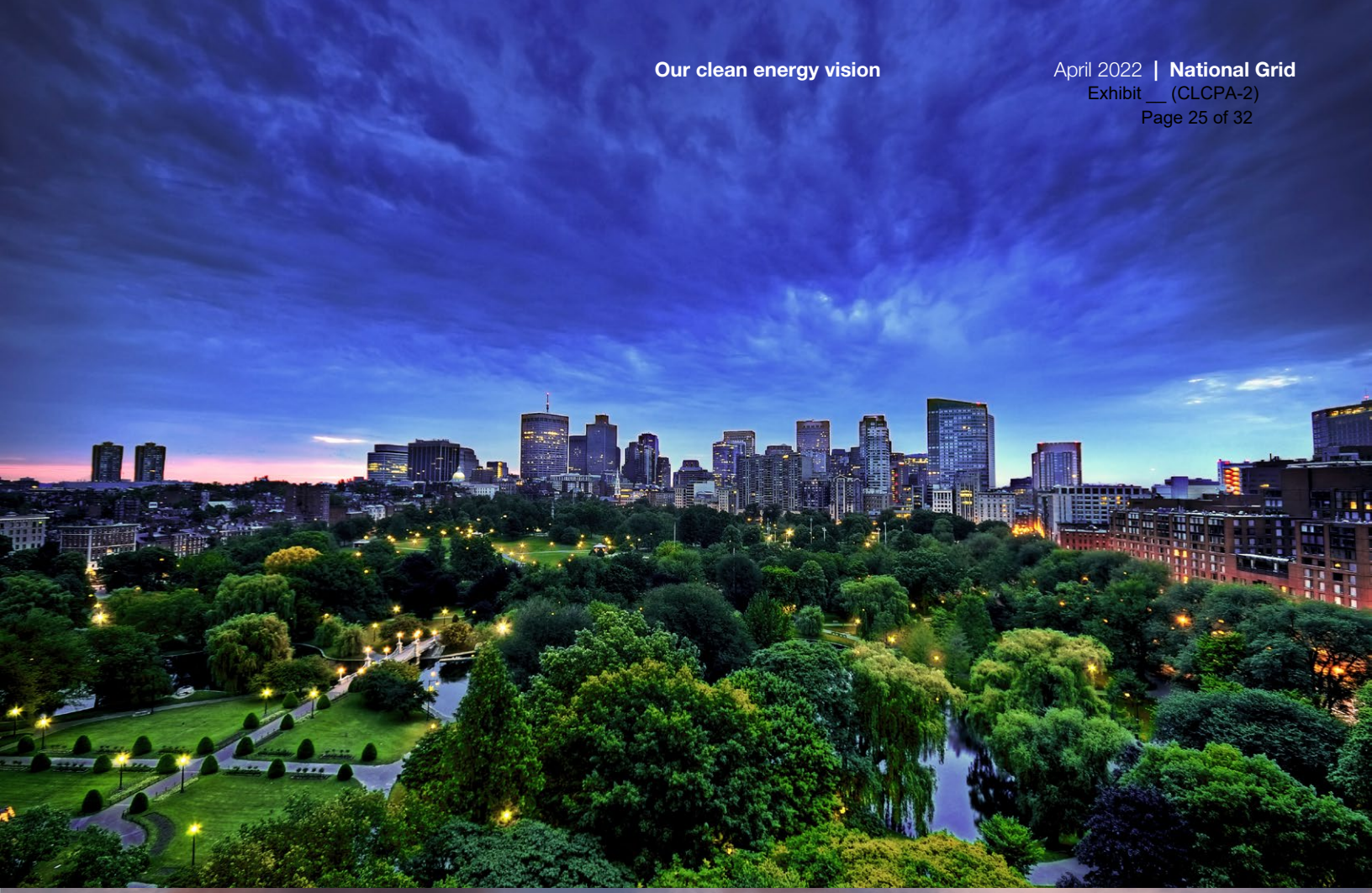
- **It fully eliminates reliance on fossil fuels for heat** and achieves this at lower cost and hassle for customers than high electrification proposals under consideration in Massachusetts and New York.
- **It improves our likelihood of achieving our climate goals** by reducing the amount of new infrastructure required and enabling customers to have multiple options for clean energy and heating their homes and businesses, instead of being mandated to switch to all-electric.

How does our Pathway better achieve the climate goals of the region?

- **Lower heating costs:** It delivers 15-20% lower home heating costs than a high-electrification approach. This will help keep energy costs more affordable.
- **Leaves no customer behind:** It supports equitable outcomes for disadvantaged communities. By avoiding large upfront investments, our vision enables all customers to have access to clean energy.
- **Preserves customer choice:** Our vision is a practical approach that gives customers choice in their heating options to reach net zero, and suits the diversity of buildings in the region. Under our pathway, customers will not have to purchase all new appliances.
- **Leverages existing Infrastructure:** It requires significantly less generation, transmission, and distribution infrastructure. By leveraging existing infrastructure, it reduces the challenge of achieving net zero.
- **Puts the existing workforce at the heart of the clean energy transition:** Our workforce is uniquely skilled to transform our existing gas system to deliver RNG and green hydrogen, as well as install new technologies such as networked geothermal.
- **Strengthens resiliency and reliability:** Our vision is a more resilient and reliable solution. A high electrification approach puts all our eggs in one basket, relying on one system for power, transport, and heating. By decarbonizing the gas system in parallel with the electric system, it creates a more resilient energy system.
- **Practical and achievable pathway to net zero:** Our vision includes tangible and near-term milestones to make clean heat solutions accessible to all our customers.

What we will do in the near-term to enable a fossil-free future?

- **Provide input to Massachusetts and New York state climate plans.** We will engage in the public feedback sessions for our state climate action plans, highlighting the benefits of our fossil-free vision for all residents and businesses.



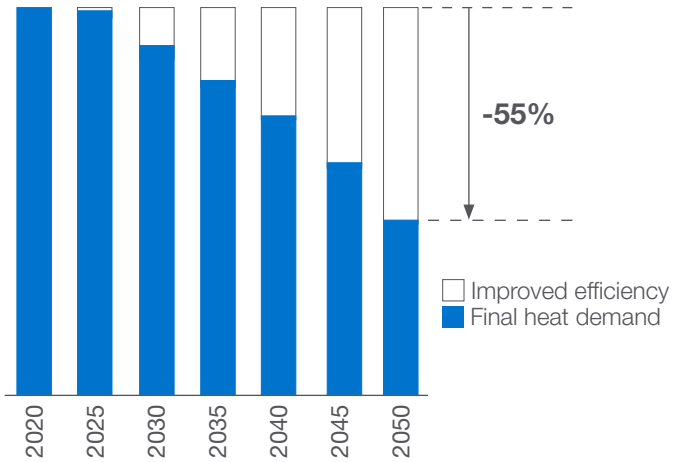


6. Appendix: More about the pillars of our vision

Energy efficiency in buildings

Continued deployment and scaling up of energy efficiency measures will be foundational to keeping energy affordable, avoiding the need to build additional infrastructure and achieving our shared net zero energy future. In our vision, improved energy efficiency reduces final demand for heat by more than half relative to today – through a combination of buildings thermal envelope efficiency, heat pump efficiency, and gas appliance efficiency (see Figure 12).

Figure 12: Final demand for heat in buildings in our vision for fossil-free heat (relative to today).



100% fossil-free gas network

Renewable natural gas

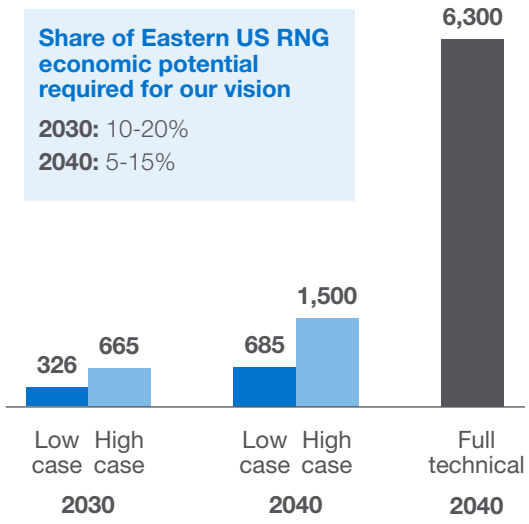
Renewable natural gas (RNG) is surface methane captured from the natural decomposition of organic waste. It has all the same characteristics of and can be used interchangeably with traditional natural gas – to heat buildings, generate electricity or fuel vehicles. Whereas traditional natural gas is sourced from fossilized organic material underground, RNG can be sourced from a vast number of existing waste streams. Potential sources of supply are in every community, including at our landfills, farms, dairies, and wastewater treatment plants, and in everyday sources like our food waste and biomass waste residues, like lawn clippings.

RNG contains biogenic carbon that is part of the planet's natural carbon cycle and does not release additional carbon into the atmosphere when consumed, unlike traditional natural gas which contains geologic carbon. By capturing RNG, we eliminate emissions at the source and, by using it, we displace emissions of geologic carbon. Because biogenic carbon is part of the natural carbon cycle, emissions from RNG use are considered carbon neutral.[21, 22]

RNG offers value in converting waste into a climate solution that can reliably serve diverse energy needs using existing infrastructure. It reduces emissions by providing an economic incentive for local communities to capture methane before it can escape into the atmosphere and exploits an otherwise unused energy source to offset use of traditional natural gas.[23] RNG is a reliable source of renewable energy that can be easily stored and delivered to customers when and where they need it. And it is also considered a "drop-in" fuel – able to be transported through existing gas infrastructure and used in existing customer equipment without modifications.

Large accessible RNG resources are untapped and could be developed to meet the residential and commercial fossil-free heating needs of the Northeast. The most comprehensive study to date on the availability and costs of RNG in the United States, completed by ICF for the American Gas Foundation in 2019 [24], estimated a maximum technical potential – with no economic constraints – of 6,300 TBtu in the Eastern US. With economic considerations taken into account, the supply potential was estimated to be between 685 – 1,500 TBtu in 2040 in US regions already connected by pipeline to the Northeast (see Figure 13).

Figure 13: Eastern US renewable natural gas (RNG) resource potential (Tbtu).

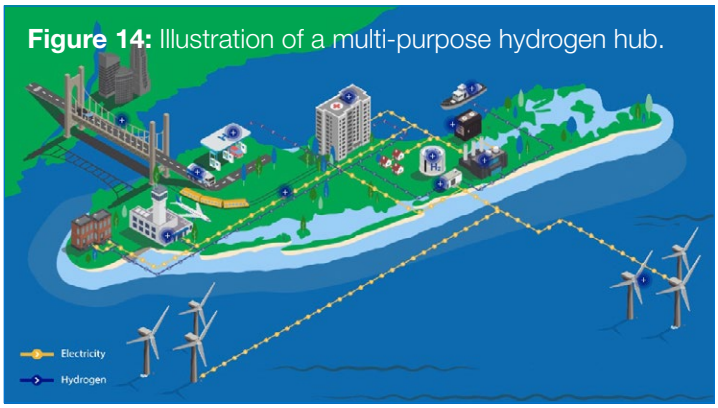


Our fossil-free vision assumes National Grid eventually procures 10-20% of RNG annual supply potential in the Eastern US for our customers, roughly in line with our customers' share of gas demand today in the supplying geography.[25] Sources of RNG supply are rapidly developing and being contracted as utilities and individual customers seek fossil-free options to decarbonize heating and other uses. In the US and Canada, over 250 RNG facilities are in operation today, more than 110 are under construction and growing interest by end-users and developers indicate an accelerating pipeline of supply ahead.

Green hydrogen

Hydrogen, the most abundant chemical element on earth, offers enormous potential as a source of clean energy. When hydrogen is converted to useable energy in a fuel cell or burned to release its energy, the only byproduct is water vapor. And when hydrogen is produced with renewable feedstocks it is known as green hydrogen, and is carbon free.

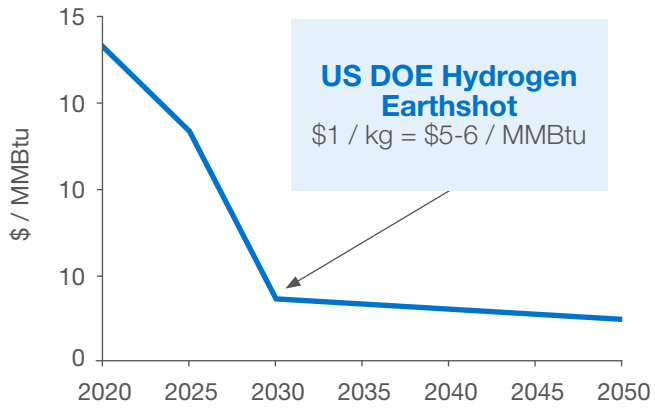
Hydrogen has the potential to help decarbonize multiple sectors, including power generation, transport, and heating. For heating, hydrogen can be blended with natural gas or RNG up to 20% by volume and run through gas networks and used in customer appliances without significant upgrades to infrastructure or equipment. In areas with high levels of gas demand, pure hydrogen also has potential to serve fossil-free heating and other energy needs in dedicated 100% hydrogen clusters. These may be part of hydrogen hubs – clusters of local hydrogen production, storage, and demand (see Figure 14).



Hydrogen is a valuable option for its complementarity to growing renewable electricity capacity, its ability to be stored and its flexibility to be used across different sectors. Green hydrogen can be made during periods when wind or solar resources are able to produce more electricity than the grid needs, and then stored for later use, thereby maximizing the benefits of renewable energy resources. Already, a significant number of offshore wind projects are proposing green hydrogen production through electrolysis as a form of long-duration storage of renewable electricity. The gas network itself can serve as a large storage reserve by carrying hydrogen.

Hydrogen development to enable the energy transition is receiving strong government support and interest from industry. The US Department of Energy's Earthshot Initiative aims to reduce the cost of green hydrogen to $6-7 per MMBtu in the next 10 years [26] – a target that we assume is reached and would bring the cost in line with today's costs for natural gas (see Figure 15). The Bipartisan Infrastructure Investment and Jobs Act passed by Congress in February 2022 allocated $8 billion to establish regional clean hydrogen hubs, one or more of which could be sited in the Northeast; $1 billion for RD&D to reduce costs of hydrogen produced from clean electricity; and another $500 million to support hydrogen equipment manufacturing and domestic supply chains.[27]

Figure 15. Hydrogen commodity cost estimate 2020-2050.



Pilot projects are proliferating across the US – covering production, storage, pipeline transmission and distribution, end uses, and use in power generation.[28] In New York, National Grid has launched the HyGrid Project to build one of the first and largest clean hydrogen projects in the country to blend green hydrogen into existing gas networks, and Plug Power is investing in a large green hydrogen production facility to supply fuel for multiple uses.[29]

Hydrogen to supply our customers could be sourced from a mix of renewable generation and electrolyzer capacity in the Northeast as well as imports. If we relied only on dedicated offshore windfarms to produce the amount of hydrogen in our vision, it would require 7-11 GWs of additional capacity, compared to existing plans in the Northeast to build about 40 GW of offshore wind by 2050.

While existing customer heating and cooking appliances could use low blends of hydrogen, higher blends or 100% hydrogen require new units with modifications to burners, fittings and sensors. 100% hydrogen-ready appliances are already being demonstrated commercially in the UK, where manufacturers are supporting a hydrogen-ready standard for all equipment sold by 2025 and have pledged to match the price to conventional gas equipment.

Hybrid electric-gas heating systems

Hybrid – or dual-fuel – heating systems pair an electric heat pump and gas heater in a building. The electric heat pump provides cooling in the summer and heating during the more temperate "shoulder" seasons. The gas heater provides heating during the coldest months of the year, when temperatures fall below a threshold temperature. Depending on gas during these coldest periods allows customers to install a smaller, more affordable heat pump than would otherwise be needed if the heat pump was sized to cover the customer's full heat load. Heating needs served by the gas component of the hybrid system vary according to heat pump specifications (size and balance point temperature) and climate.

For many customers, hybrid systems can offer a clean solution that is more comfortable, cost-effective and practical for serving cold weather heating needs compared with an all-electric installation. They avoid using inefficient and costly electric resistance as the supplemental heating source during the coldest months. They also allow customers who use gas today to continue using their existing gas heater and avoid larger building changes that might be required for an all-electric system, while still capturing the summer cooling and shoulder season heating benefits of heat pumps. For the broader energy system, hybrid heating adoption across the customer base can help to lessen electric supply and network infrastructure challenges by avoiding high electric peak loads during winter.

Hybrid heating systems are already being adopted in our region today. Nearly all gas customers in the Northeast who have installed an electric heat pump to date have retained their existing heater, effectively gaining cooling and hybridizing their heating. Sustainability advisors like E3[30] and McKinsey & Co. have suggested hybrid systems as an important option to decarbonize heat in the Northeast and other regions.

Targeted electrification and networked geothermal

National Grid's vision embraces our role as a leader in evaluating opportunities for non-pipe alternatives, including consideration of targeted electrification and networked geothermal systems. In specific instances, it may be possible to coordinate electrification of heat across multiple streets or neighborhoods. The main difference between networked geothermal and targeted electrification through air-source heat pumps is that the former has better winter performance and places lower strain on the electric grid. This is because networked geothermal systems exchange heat with the ground (which stays at a constant temperature below a certain depth) instead of with the outside air, which fluctuates more widely. For this reason, ground-source heat pumps have higher efficiency and lower electric needs.

This targeted approach can have positive impacts, provided that the benefits of avoided gas infrastructure outweigh the incremental electric system upgrade costs. While this approach has not yet been piloted at scale in the Northeast, and there are likely to be a range of regulatory issues, there are affordability, safety, and reliability criteria that can guide our approach.

For example:

Reliability and critical loads

Targeted electrification should minimize impact on gas network performance. For example, mains that supply multiple lower-pressure distribution networks or critical customers (*e.g.* hospitals) would be difficult to electrify, while smaller segments that do not serve critical system requirements would be easier.

System economics

Targeted electrification should optimize the economics of avoided gas network investment and additional electric grid investment. For example, it would be preferable to target electrification in areas where substantial leak-prone-pipe investment is required, rather than in areas where new main has been recently installed. Similarly, it should aim to minimize significant electric grid investment, for example where costs are high or siting issues are acute. In such areas, for example, upgrading to hydrogen-readiness is likely to be preferable. This optimization would require understanding key gas and electric network characteristics, for example: capacity 'headroom' on the electric system, cost of electric upgrades in different parts of the network, cost and expected timing of pipe replacement in different parts of the network, and customer density (*e.g.* meters per mile) as a proxy for customer contributions to system investment. In sum, a paradigm shift toward whole system optimization is likely to be required, to minimize costs on the path to net zero.

Customer economics and practicality

Targeted electrification should minimize customer cost and disruption. This could be achieved by focusing on areas with a prevalence of building types that can be electrified at relatively lower cost (*e.g.* newer, detached homes) and where customer disruption could be minimized.

7. Notes and sources

1 Sum of Residential, Commercial, and Industrial million metric tons of energy-related carbon dioxide emissions divided by economy-wide Total emissions from U.S. Energy Information Administration, "Summary Table 4. State energy-related carbon dioxide emissions by sector," 2018, available at https://www.eia.gov/environment/emissions/state/

2 U.S. Census, "American Community Survey B25040 House Heating Fuel" for Massachusetts and New York, 2020, available at https://data.census.gov/cedsci/table?t=Heating%20and%20Air%20Conditioning%20%28HVAC%29&g=0400000US25,36&tid=ACSDT5Y2020.B25040

3 Estimated 1% electric heat pump adoption in Massachusetts from low historical adoption rate from Mass Save, "Massachusetts Energy Efficiency Program Administrators Quarterly Report" section titled Bi-Annual Data at 3, 2019-2021, available at https://ma-eeac.org/results-reporting/quarterly-reports/. Estimated 1% electric heat pump adoption in New York from VEIC, "Ramping Up Heat Pump Adoption in New York State" at 4, September 2018, available at https://www.veic.org/Media/default/documents/resources/reports/veic-ramping-up-heat-pump-adoption-in-new-york-state.pdf

4 Upfront air-source heat pump and building retrofit costs estimated as $29,000 - $36,000 in Massachusetts from Energy + Environmental Economics ("E3"), "The Role of Gas Distribution Companies in Achieving the Commonwealth's Climate Goals: Technical Analysis of Decarbonization Pathways" ("E3 Decarbonization Pathways Report") at 102, March 2022, available at https://thefutureofgas.com/content/downloads/2022-03-21/3.18.22%20-%20Independent%20Consultant%20Report%20-%20Decarbonization%20Pathways.pdf. Upfront air-source heat pump and building retrofit costs estimated as $21,087 (Residential Single Family Basic Shell + Air Source Heat Pump) - $59,814 (Residential Single Family Deep Shell + Air Source Heat Pump) from Climate Action Council Draft Scoping Plan, "Appendix G: Annex 1: Inputs and Assumptions [XLSX]" at Bldg_Res Device Cost tab, February 2022, available at https://climate.ny.gov/Our-Climate-Act/Draft-Scoping-Plan

5 National Grid estimate based on Arup analysis of existing state and municipal building data.

6 E3 Decarbonization Pathways Report at 19.

7 World Bank, "Brief: Solid Waste Management," February 2022, available at https://www.worldbank.org/en/topic/urbandevelopment/brief/solid-waste-management

8 U.S. Department of Energy, "DOE Establishes Bipartisan Infrastructure Law's $9.5 Billion Clean Hydrogen Initiatives," February 2022, available at https://www.energy.gov/articles/doe-establishes-bipartisan-infrastructure-laws-95-billion-clean-hydrogen-initiatives

9 New York State, "Governor Hochul Announces Multi-State Agreement Signed with Major Hydrogen Ecosystem Partners to Propose a Regional Clean Energy Hydrogen Hub," March 2022, available at https://www.governor.ny.gov/news/governor-hochul-announces-multi-state-agreement-signed-major-hydrogen-ecosystem-partners

10 (16,235 total ASHP - 2,933 total full ASHP replacements) / 16,235 total ASHP customers = 82% partial ASHP installations from Mass Save, "Massachusetts Energy Efficiency Program Administrators Quarterly Report, Fourth Quarter, 2021", section titled Bi-Annual Data, at 3 of 75, 2021, available at https://ma-eeac.org/results-reporting/quarterly-reports/

11 Massachusetts values for "Our vision for fossil-free heat" based on cost range for "Hybrid Electrification" and "Efficient Gas Equipment" scenarios and "High electrification" based on "High Electrification," "2030 CECP," and "100% Gas Decommissioning" scenarios from E3 Decarbonization Pathways Report at 12-13. New York values for "High electrification" based on NPV of Net Direct Costs for "Accelerated Transition" from Climate Action Council Draft Scoping Plan, "Appendix G: Integration Analysis Technical Supplement" at 66, February 2022, available at https://climate.ny.gov/Our-Climate-Act/Draft-Scoping-Plan. A fossil-free gas pathway has not been publicly modeled for New York, so values for "Our vision for fossil-free heat" scale New York costs proportionally based on scenario costs calculated for Massachusetts.

12 E3 Decarbonization Pathways Report at 41-42.

13 New York City Mayor's Office of Sustainability, "Pathways to Carbon-Neutral NYC: Modernize, Reimagine, Reach" at 38, April 2021, available at https://www1.nyc.gov/assets/sustainability/downloads/pdf/publications/Carbon-Neutral-NYC.pdf

14 BloombergNEF, "New Energy Outlook 2020," at 59, available at https://about.bnef.com/new-energy-outlook-2020/

15 E3 Decarbonization Pathways Report at 102.

16 Upfront air-source heat pump and building retrofit costs estimated as $29,000 - $36,000 in Massachusetts from E3 Decarbonization Pathways Report at 102, March 2022, available at https://thefutureofgas.com/content/downloads/2022-03-21/3.18.22%20-%20Independent%20Consultant%20Report%20-%20Decarbonization%20Pathways.pdf. Upfront air-source heat pump and building retrofit costs estimated as $21,087 (Residential Single Family Basic Shell + Air Source Heat Pump) - $59,814 (Residential Single Family Deep Shell + Air Source Heat Pump) from Climate Action Council Draft Scoping Plan, "Appendix G: Annex 1: Inputs and Assumptions [XLSX]" at Bldg_Res Device Cost tab, February 2022, available at https://climate.ny.gov/Our-Climate-Act/Draft-Scoping-Plan

[17] 4,000 homes estimated to have adopted air-source heat pumps in Massachusetts from E3 Decarbonization Pathways Report at 98. Massachusetts would need to install roughly 90,000 heat pumps a year to electrify 2.5 million buildings that aren't already electrically heated in the next 28 years. Of 43,418 homes estimated to have adopted air-source heat pumps in New York, half estimated to be for NEEP's Cold Climate Specification and assumed to be maximum number of full-home heaters from VEIC, "Ramping Up Heat Pump Adoption in New York State" at 4, September 2018, available at https://www.veic.org/Media/default/documents/resources/reports/veic-ramping-up-heat-pump-adoption-in-new-york-state.pdf. New York would need to install roughly 250,000 heat pumps a year to electrify 7.1 million buildings that aren't already electrically heated in the next 28 years.

[18] E3 Decarbonization Pathways Report at 65.

[19] ISONE 22 GW summer peak and 20 GW winter peak in 2020 and 59 GW winter peak in the "High Electrification" scenario in 2050 from E3 Decarbonization Pathways Report at 59. NYISO 31 GW summer peak and 24 GW winter peak in 2020 and 50 GW winter peak in 2050 in the "Accelerated Transition Away from Combustion" pathway from Climate Action Council Draft Scoping Plan, "Appendix G: Annex 2: Key Drivers and Outputs [XLSX]" at All_Peak by Scenario tab, February 2022, available at https://climate.ny.gov/Our-Climate-Act/Draft-Scoping-Plan

[20] Estimated 4,500 miles overloaded from ISO New England, "2050 Transmission Study" at 15, available at https://www.iso-ne.com/static-assets/documents/2022/03/a4_2050_transmission_study_preliminary_n_1_and_n_1_1_thermal_results_presentation.pdf. Internal estimate of 2,000 miles for NYISO assumes build out of 1,400 miles of West – East and North – South transmission and 600 miles of local transmission.

[21] Intergovernmental Panel on Climate Change, "2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories," Chapter 2: Stationary Combustion, Section 2.3.3.4 Treatment of Biomass, 2019, available at https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/2_Volume2/19R_V2_2_Ch02_Stationary_Combustion.pdf

[22] CDP,"Guidance & questionnaires," available at https://guidance.cdp.net/

[23] MJ Bradley, "Renewable Natural Gas: The RNG Opportunity for Natural Gas Utilities" at 1, April 2017, available at https://www.mjbradley.com/sites/default/files/MJB%26A_RNG_Final.pdf

[24] ICF, "Renewable Sources of Natural Gas: Supply and Emission Reduction Assessment" at 68-69, December 2019, available at https://gasfoundation.org/2019/12/18/renewable-sources-of-natural-gas/

[25] U.S. Energy Information Administration, "Natural Gas Consumption by End Use" for Residential and Commercial, available at https://www.eia.gov/dnav/ng/ng_cons_sum_a_EPG0_vgt_mmcf_a.htm

[26] U.S. Department of Energy, "Hydrogen Shot," available at https://www.energy.gov/eere/fuelcells/hydrogen-shot

[27] U.S. Department of Energy, "DOE Establishes Bipartisan Infrastructure Law's $9.5 Billion Clean Hydrogen Initiatives," February 2022, available at https://www.energy.gov/articles/doe-establishes-bipartisan-infrastructure-laws-95-billion-clean-hydrogen-initiatives

[28] S&P Global Market Intelligence, "US hydrogen pilot projects build up as gas utilities seek low-carbon future," October 2021, available at https://www.spglobal.com/marketintelligence/en/news-insights/latest-news-headlines/us-hydrogen-pilot-projects-build-up-as-gas-utilities-seek-low-carbon-future-65570349

[29] Plug Power, "Plug Power To Build North America's Largest Green Hydrogen Production Facility In Western New York," February 2021, available at https://www.ir.plugpower.com/Press-Releases/Press-Release-Details/2021/Plug-Power-to-Build-North-Americas-Largest-Green-Hydrogen-Production-Facility-in-Western-New-York/default.aspx

[30] E3 Decarbonization Pathways Report. E3, "Philadelphia Gas Works Diversification Study," December 2021, available at https://www.phila.gov/media/20211207134817/PGW-Business-Diversification-Study-2021-12.pdf. E3, "New England Reliability Under Deep Decarbonization," November 2020, available at https://www.ethree.com/wp-content/uploads/2020/11/E3-EFI_Report-New-England-Reliability-Under-Deep-Decarbonization_Full-Report_November_2020.pdf

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April 2022. NG13251/WS.

Exhibit __ (CLCPA-3)

Assessment of KEDNY Estimated Greenhouse Gas Emissions Impacts Rate

Year – Data Year 3

KEDNY Emissions Reductions in Metric Tons of CO2e						
Workstream	**Initiatives**	**Rate Year**	**Data Year 1**	**Data Year 2**	**Data Year 3**	**Cumulative Emissions Rate Year-Date Year 3**
GIOP	Large Diameter Pipe Rehabilitation	740	1,481	2,221	2,962	7,404
	Large Diameter Pipe Rehabilitation - CISBOT	926	1,851	2,777	3,702	9,255
	Main Replacement - LPP - Proactive, CSC, Reliability	13,237	27,398	42,174	57,258	140,068
	Replace Pipe on Bridges	5	10	16	21	52
	Service Replacement - Atmospheric Corrosion Inside Inspections (Remediation)	665	1,331	1,996	2,662	6,654
	CNG fill station retirement - Brooklyn - Greenpoint				1,401	1,401
	CNG fill station retirement - Brooklyn - Canarsie				52	52
	LNG - Tail Gas Compressor Upgrade				-6	-6
	LNG - Relocate Maint. Area & New Control Building				-40	-40
	LNG - Tank 2 Foundation Heaters				-55	-55
	LNG - Salt Water Pump House Upgrade				-557	-557
	RNG Project - KEDNY 1 (avoided upstream emissions only)				6,103	6,103
	RNG Project - KEDNY 2 (avoided upstream emissions only)				22,490	22,490
	Fleet and Facilities	899	2,404	4,416	7,078	14,797
Customer	Energy Efficiency Programs	67,857	135,714	203,571	271,428	678,571
Total		84,330	170,189	257,171	374,499	886,190

List of Acronyms
CI: Cast Iron
CNG: Certified Natural Gas
CO2e: Carbon Dioxide Equivalent
CSC: City State Construction
LNG: Liquid Natural Gas
LPP: Leak Prone Pipe
PS: Protected Steel
RNG: Renewable Natural Gas
UPS: Unprotected Steel

KEDNY Programs with Non-Quantified Emissions Benefits			
Panel	**Program**	**Description**	**How This Program Will Reduce Emissions**
GIOP	Gas System Reinforcement- Belmont Gate Station	This project rebuilds the regulator station to increase capacity to maintain pressure above system minimum on the gas network during periods of peak demand.	Replacing old and aging station equipment with new equipment and technologies should reduce emissions.
	Corrosion	This program is annual corrosion work on the Company's gas system. Cathodic protection is installed on the pipe.	Cathodic protection increases the pipe's useful life, reducing future emissions associated with leaks.
	Remote Controlled Valve (RCV)	This program will install RCV on select pipe segments.	In the event of an incident on an RCV-equipped seqment, the RCV will reduce the time of gas blowing to atmosphere as the result of third party or other damage. The ability to control these events remotely will also reduce the number of truck rolls required to respond to an incident.
	Main Replacement- Reactive-Maintenance	This program will replace sections of main segments and associated services identified during leak repairs that leak clamps could not repair.	This program will reduce fugitive emissions from leaking connections, pipe damage, or faulty equipment.
	Automatic Meter Reading (AMR)	This program includes the purchase and installation of AMR equipment to support KEDLI's mandated Meter Test/Replacement program.	AMR allows the Company to remote read meters and reduces GHG emissions by reducing vehicle trips to read meters.
	Pipeline Integrity- IMP	This program enables select transmission pipes to be enabled to use in-line inspection (ILI) technology. This will allow for the enhanced operational safety of the Company's transmission system.	This is a mandated project, and also has the potential to avoid future leaks and failures on the system and therefore tends to reduce GHG reductions.
	Pipeline Integrity-IVP	This program will replace pipe with insufficient records.	This project will proactively review assets and has the potential to avoid future leaks and failures on the system and therefore tends to reduce GHG reductions.
	Service Replacements	This program is non-discretionary work randomly generated through public leak reports, programmed leak surveys, mandated activities, and customer generated requests.	This program will reduce fugitive emissions from leaking connections, pipe damage, or faulty equipment.
	Transmission Station Integrity	This program will repair or replace transmission stations or parts at the transmission stations.	This project will indirectly reduce GHG emissions by replacing old, antiquated equipment with new and more reliable equipment.
	System Automation & Control	This program will install Remote Terminal Units (RTUs), which will provide temperature, pressure, and flow data back to the Gas Control Room.	This project will indirectly reduce GHG emissions by replacing old, antiquated equipment with new and more reliable equipment.
	Gas System Control	This program supports necessary upgrades to the Gas Control Room.	This project will proactively review assets and has the potential to avoid future leaks and failures on the system and therefore tends to reduce GHG reductions.
	Heater Installation Program	Gas heaters are designed to safely heat gas to mitigate problems caused by low gas temperatures.	Newly installed heater equipment will perform more efficiently than existing equipment and may result in nominal emissions reductions.
	Instrumentation and Regulation (I&R)-Reactive	This program funds the capital repairs of deficiencies at pressure regulating facilities found during inspection and maintenance.	This program will reduce fugitive emissions from leaking connections, pipe damage, or faulty equipment.
	Main Protection- Storm Hardening	This program installs storm hardening measures, including ultrasonic meters and valves, to mitigate the public risk caused by flood damages to the gas equipment and service regulator.	The valve will provide the ability to remotely shutoff the flow of gas to services which will stop gas leaks located after the valve more quickly than previously possible.
	Proactive Regulator Stations	Pressure regulating stations are repaired or replaced.	This project will indirectly reduce GHG emissions by replacing old, antiquated equipment with new and more reliable equipment.
	Transmission Mains- Greenpoint	This project will replace a 16x12 tee and replace with a straight piece of pipe to avoid possible liquids from pooling in the spur.	This project will indirectly reduce GHG emissions by replacing old, antiquated equipment with new and more reliable equipment.
	Transmission Service Integrity	This program will ensure the integrity of the transmission customer services.	This project will indirectly reduce GHG emissions by replacing old, antiquated equipment with new and more reliable equipment.
	Telecomm- Reactive	This program funds capital repairs and replacements of telecom equipment and installations.	This program helps provide for effective and reliable long distance communication services so that field information can be obtained without personnel travel, reducing emissions associated with additional truck rolls.

Customer	Economic Development and the Future of Heat	This program invests in technologies that enable customers to be more efficient and productive while contributing to New York's ongoing energy transformation.	This program invests in projects that involve non-pipeline alternatives to traditional natural gas delivery.
	Sustainable Gas and Economic Development	This program will fund projects that demonstrate sustainable gas technologies	This program funds projects that could reduce methane emissions from naturally occurring sources (landfills, waste water treatment plants, farms, etc.) .
	Weatherization Health and Safety	This program will enable remediation of health and safety barriers to weatherization, such as carbon monoxide hazards, mold, insufficient wiring or ventilation, etc.	This program will encourage more weatherization projects, which reduce emissions, in Disadvantaged Communities.
	Disadvantaged Community Outreach	This program identifies key barriers to engagement in energy efficiency programs and strategies to overcome those barriers, while identifying the most effective ways to promote the energy efficiency programs.	This program will increase participation by customers in Disadvantaged Communities in energy efficiency programs, which will reduce emissions.
	Energy Efficiency Language Access Program	This program provides a mix of language accessibility service to apply to websites, flyers, advertisements, and other customer facing materials.	This program will remove a significant barrier to customer participation in energy efficiency programs and should increase participation.
	Digital Building Replica Pilot (DAC School Building)	This program creates a data model of a school building that functions identically to a the physical building being modeled.	The model will be created to predict energy savings from participation in the Companies' energy efficiency programs and drive increased participation.
	Clean Energy Advisors	This program will help customers navigate the various clean energy programs.	This program will help customers, particularly LMI customers reduce their emissions through participation in clean energy programs
CLCPA	Gas Low and Moderate Income EAM	This metric encourages the Companies' to exceed the LMI EE targets set by NE:NY by finding innovative ways to achieve more energy savings.	Exceeding the LMI EE targets will reduce GHG emissions beyond expected GHG emission reductions in NE:NY.
Gas Safety	Advanced Communications Infrastructure Programs	This program provides real time gas system feedback to allow for more effective emergency response and potentially minimize the impact and duration of emerging leaks.	Minimizing the impact and duration of emerging gas leaks will reduce methane emissions.
	Advanced Leak Detection Program	This program surveys service territories for possible high emitting leaks and uses this information to prioritize high emitting leaks for repair.	Helps reduce system methane leaks.
	Inside Service Line Inspections Program	This program inspects pipes within buildings.	Helps the Companies identify and repair gas leaks.
Information Technology	Clean Energy 2.0	This program will support the Companies' energy efficiency programs by replacing an older, more inefficient software system.	This program will allow the Companies manage the energy efficiency programs more effectively and efficiently.
	Green Button Connect	This program will enable increased access to products and services to energy data.	Will facilitate customer sharing of energy usage information to enhance overall energy savings.

Additional information on the listed programs is provided in the testimony of the respective indicated panel.

Exhibit __ (CLCPA-4)

Assessment of KEDLI Estimated Greenhouse Gas Emissions Impacts Rate

Year – Data Year 3

KEDLI Emissions Reductions in Metric Tons of CO2e						
Workstream	**Initiatives**	**Rate Year 1**	**Data Year 1**	**Data Year 2**	**Data Year 3**	**Cumulative Emissions Rate Year-Date Year 3**
GIOP	Large Diameter Pipe Rehabilitation	173	345	518	691	1,727
	LPP - Proactive, CSC, Reinforcement, Reliability	19,861	40,412	61,136	82,205	203,614
	Replace Pipe on Bridges	5	10	16	21	52
	CNG fill station retirement - Riverhead				52	52
	LNG - Holtsville Plant Modernization				-2,289	-2,289
	RNG Project - KEDLI 1 (avoided upstream emissions only)				4,841	4,841
	RNG Project - KEDLI 2 (avoided upstream emissions only)				4,284	4,284
	HyGrid				2,163	2,163
	Fleet and Facilities	444	1,320	2,311	3,734	7,809
Customer	Energy Efficiency Programs	57,128	114,256	171,384	228,512	571,281
Total		77,611	156,343	235,364	324,214	793,533

List of Acronyms

CI: Cast Iron
CNG: Certified Natural Gas
CO2e: Carbon Dioxide Equivalent
CSC: City State Construction
LNG: Liquid Natural Gas
LPP: Leak Prone Pipe
PS: Protected Steel
RNG: Renewable Natural Gas
UPS: Unprotected Steel

KEDLI Programs with Non-Quantified Emissions Benefits			
Panel	**List of Programs**	**Description**	**How This Program Will Impact Emissions**
GIOP	Corrosion	This program is annual corrosion work on the Company's gas system. Cathodic protection is installed on the pipe to help prevent corrosion.	Cathodic protection increases the pipe's useful life, reducing future emissions associated with leaks.
	Remote Controlled Valve (RCV)	This program will install RCV on select pipe segments.	In the event of an incident on an RCV-equipped seqment, the RCV will reduce the time of gas blowing to atmosphere as the result of third party or other damage. The ability to control these events remotely will also reduce the number of truck rolls required to respond to an incident.
	Main Replacement- Reactive-Maintenance	This program will replace sections of main segments and associated services identified during leak repairs that leak clamps could not repair.	This program will reduce fugitive emissions from leaking connections, pipe damage, or faulty equipment.
	Advanced Meter Reading (AMR)	This program includes the purchase and installation of AMR equipment to support KEDLI's mandated Meter Test/Replacement program.	AMR allows the Company to remotely read meters and reduces GHG emissions by reducing vehicle trips to read meters.
	Pipeline Integrity-IVP	This program will replace pipe with insufficient records.	This project will proactively review assets and has the potential to avoid future leaks and failures on the system and therefore tends to reduce GHG reductions.
	Integrity- Meter Pitts	This program will repair and remediate gas piping components, regulators, and meters exposed to atmospheric conditions.	This program will remediate current asset conditions where future leaks could occur.
	Service Replacement	This program is non-discretionary work randomly generated through public leak reports, programmed leak surveys, mandated activities, and customer generated requests.	This program will reduce fugitive emissions from leaking connections, pipe damage, or faulty equipment.
	Take Station	This program will install over pressurization protection at select take stations.	This project will indirectly reduce GHG emissions by replacing old, antiquated equipment with new and more reliable equipment.
	Gas System Control	This program supports necessary upgrades to the Gas Control Room.	This project will proactively review assets and has the potential to avoid future leaks and failures on the system and therefore tends to reduce GHG reductions.
	Transmission Station Integrity	This program will repair or replace transmission stations or parts at the transmission stations.	This project will indirectly reduce GHG emissions by replacing old, antiquated equipment with new and more reliable equipment.
	Heater Installation Program	This program will install gas heaters, which are designed to safely heat gas to mitigate problems caused by low gas temperatures.	Newly installed heater equipment will perform more efficiently than existing equipment and may result in nominal emissions reductions.
	Instrumentation and Regulation (I&R)- Reactive	This program funds the capital repairs of deficiencies at pressure regulating facilities found during inspection and maintenance.	This program will reduce fugitive emissions from leaking connections, pipe damage, or faulty equipment.
	Main Protection- Storm Hardening	This program installs storm hardening measures, including ultrasonic meters and valves, to mitigate the public risk caused by flood damages to the gas equipment and service regulator.	The valve will provide the ability to remotely shutoff the flow of gas to services which will stop gas leaks located after the valve more quickly than previously possible.
	Proactive Regulator Stations	Pressure regulating stations are repaired or replaced.	This project will indirectly reduce GHG emissions by replacing old, antiquated equipment with new and more reliable equipment.

	Transmission Service Integrity	This program will ensure the integrity of the transmission customer services.	This project will indirectly reduce GHG emissions by replacing old, antiquated equipment with new and more reliable equipment.
	Telecom- Reactive	This program funds capital repairs and replacements of telecom equipment and installations.	This program helps provide for effective and reliable long distance communication services so that field information can be obtained without personnel travel, reducing emissions associated with additional truck rolls.
Customer	Economic Development and the Future of Heat	This program invests in technologies that enable customers to be more efficient and productive while contributing to New York's ongoing energy transformation.	This program invests in projects that involve non-pipeline alternatives to traditional natural gas delivery.
	Sustainable Gas and Economic Development	This program will fund projects that demonstrate sustainable gas technologies	This program funds projects that could reduce methane emissions from naturally occurring sources (landfills, waste water treatment plants, farms, etc.) .
	Weatherization Health and Safety	This program will enable remediation of health and safety barriers to weatherization, such as carbon monoxide hazards, mold, insufficient wiring or ventilation, etc.	This program will encourage more weatherization projects, which reduce emissions, in Disadvantaged Communities.
	Disadvantaged Community Outreach	This program identifies key barriers to engagement in energy efficiency programs and strategies to overcome those barriers, while identifying the most effective ways to promote the energy efficiency programs.	This program will increase participation by customers in Disadvantaged Communities in energy efficiency programs, which will reduce emissions.
	Energy Efficiency Language Access Program	This program provides a mix of language accessibility service to apply to websites, flyers, advertisements, and other customer facing materials.	This program will remove a significant barrier to customer participation in energy efficiency programs and should increase participation.
	Digital Building Replica Pilot (DAC School Building)	This program creates a data model of a school building that functions identically to a the physical building being modeled.	The model will be created to predict energy savings from participation in the Companies' energy efficiency programs and drive increased participation.
	Clean Energy Advisors	This program will help customers navigate the various clean energy programs.	This program will help customers, particularly LMI customers reduce their emissions through participation in clean energy programs
CLCPA	Gas Low and Moderate Income EAM	This metric encourages the Companies to exceed the LMI EE targets set by NE:NY by finding innovative ways to achieve more energy savings.	Exceeding the LMI EE targets will reduce GHG emissions beyond expected GHG emission reductions in NE:NY.
Gas Safety	Advanced Communications Infrastructure Programs	This program provides real time gas system feedback to allow for more effective emergency response and potentially minimize the impact and duration of emerging leaks.	Minimizing the impact and duration of emerging gas leaks will reduce methane emissions.
	Advanced Leak Detection Program	This program surveys service territories for possible high emitting leaks and uses this information to prioritize high emitting leaks for repair.	Helps reduce system methane leaks.
	Inside Service Line Inspections Program	This program inspects pipes within buildings.	Helps the Companies identify and repair gas leaks.
Information Technology	Clean Energy 2.0	This program will support the Companies' energy efficiency programs by replacing an older, more inefficient software system.	This program will allow the Companies manage the energy efficiency programs more effectively and efficiently.
	Green Button Connect	This program will enable increased access to products and services to energy data.	Will facilitate customer sharing of energy usage information to enhance overall energy savings.

Additional information on the listed programs is provided in the testimony of the respective indicated panel.

Exhibit __ (CLCPA-5)

Proposals Supporting Disadvantaged Communities

KEDNY and KEDLI Rate Case Proposals
Supporting Disadvantaged Communities

Initiative	Disadvantaged Communities Benefits	Sponsoring Rate Case Panel
Leak Prone Pipe Retirement Programs	Significant proportion of LPP located in Disadvantaged Communities, and retiring such facilities will promote safety and improve local air quality.	GIOP
Fleet electrification	Proposed EV roll out located in and near Disadvantaged Communities, reducing local ambient noise, improving air quality, and reducing GHG emissions.	GIOP
KEDNY Greenpoint Master Plan	To use an overall design that improves local air quality and mitigates potential adverse environmental factors that affect Disadvantaged Communities in the area.	GIOP
Weatherization Health & Safety Program	Will help address prerequisite home conditions to enable LMI customers and customers in Disadvantaged Communities to access weatherization services more equitably because weatherization often involves significant home construction projects.	Customer
Disadvantaged Community Outreach Program	Multiple strategies will engage customers within Disadvantaged Communities to participate in energy efficiency programs.	Customer
Language Access Program	Language accessibility services will be applied to a variety of customer-facing public information in focused effort to increase participation by non-English speaking customers in energy savings programs.	Customer
Digital Building Replica Pilot (Disadvantaged Communities School Buildings)	Will enable schools in Disadvantaged Communities to understand the energy benefits of different energy-savings measures without, or in advance of, hiring an external vendor, to make more cost-effective decisions.	Customer
Economic Development	The economic development programs are being re-framed to deliver greater benefits to Disadvantaged Communities, including adjusting eligibility requirements reprioritizing to provide additional funding for eligible projects located in Disadvantaged Communities.	Customer
Low & Moderate Income (“LMI”) Customer Energy Efficiency (“EE”) Savings EAM	Encourages the Companies to exceed on the LMI EE targets set by NE:NY and to direct EE savings towards Disadvantaged Communities in support of the 35% benefits target established in the CLCPA.	CLCPA
Energy Affordability Program (“EAP”) Enrollment EAM	Would drive greater EAP participation by providing a bill discount to more income eligible customers. Encourages the Companies to increase investment and identify and advance innovative approaches to identify, reach, and enroll customers in EAP.	CLCPA

Exhibit___(CLCPA-6)

Gas Share the Savings Metrics and Targets Calculations

Gas Share the Savings Metrics and Targets

GSTS performance and associated incentives will be calculated by determining: (i) a baseline energy efficiency ("EE") unit cost savings; (ii) the difference between that baseline and the actual acquired EE unit cost savings; and (iii) applying a percent share (30%) to the result. The formula is as follows:

$$STS\ EAM\ (\$) = (CYx\ Baseline\ LMMBtu\ Unit\ Cost - CYx\ Acquired\ LMMBtu\ Unit\ Cost) * CYx\ Acquired\ LMMBtu * 30\%$$

where:
x = 1, 2, 3, etc. for Calendar Year 1 (2024), Calendar Year 2 (2025), Calendar Year 3 (2026), etc.

CYx Baseline LMMBtu ("lifetime MMBtu") Unit Cost = $\frac{CYx\ Baseline\ Budget}{CYx\ Baseline\ LMMBtu}$

CYx Baseline Budget = The Company's total non-LMI energy efficiency budgets approved in the January 2020 New Efficiency:New York ("NE:NY") Order

CYx Baseline LMMBtu = CYx Baseline AMMBtu * Portfolio EUL

CYx Baseline AMMBtu ("annual MMBtu") = The Company's non-LMI gas energy annual savings target approved in the January 2020 NE:NY Order

Portfolio EUL = The Company's portfolio-level Effective Useful Life in the year prior to CYx, as determined by the applicable Technical Resources Manual ("TRM"), and utilizing a verified gross savings methodology2

CYx Acquired LMMBtu Unit Cost = $\frac{CYx\ Actual\ Expenditures}{CYx\ Acquired\ LMMBtu}$

CYx Actual Expenditures = The Company's total expenditures on its gas energy efficiency portfolio in CYx exclusive of LMI programs

CYx Acquired LMMBtu = CYx Acquired AMMBtu * CYx Portfolio EUL

CYx Acquired AMMBtu = The Company's acquired annual verified gross gas energy savings, exclusive of savings from LMI programs, as reported in the Company's year-end scorecard

CYx Portfolio EUL = The Company's portfolio EUL, as determined by the applicable TRM at the time the savings are acquired, and utilizing a verified gross savings methodology

LMI programs = the portfolio of programs established in the Statewide Implementation plan

This EAM is computed formulaically, per the Metric above, and as such has no minimum, midpoint, or maximum targets.

Achievement is based on the formula detailed in the section above. To account for the shifting of funds between years, the threshold for qualifying to collect the GSTS EAM shall be each year's NE:NY annual savings target (excluding LMI programs) plus the cumulative over- or underperformance the NE:NY period, as follows:

$$Threshold = (Over/underperformance\ CYx{-}1 + Over/underperformance\ CYx{-}2 + \ldots) \div (2026 - CYx)$$

where:
$Over$/underperformance = $Actual\ annual\ MMBtu\ savings\ achieved\ via\ the\ Company's\ market\text{-}rate\ energy\ efficiency\ portfolio - Annual\ MMBtu\ market\text{-}rate\ annual\ savings\ target\ as\ set\ in\ the\ NE{:}\ NY\ Order$
and where the first CY in which over/underperformance is measured is 2019. If the Commission authorizes additional budgets and targets, the EAM calculation, including threshold shall be updated accordingly.

Exhibit___(CLCPA-7)

Earnings Adjustment Mechanism ("EAM") Targets

KEDNY Earning Adjustment Mechanism Targets

	a	b	c	d	e	f	g	h

System Efficiency EAM

Metric	Measure		CY 2024[3]	CY 2025[3]	CY 2026[3]	CY 2027[3]	Term
Gas Demand Response	Dth/Hr Reduction[1,2]	Min	to be calculated[4]	to be calculated[4]	to be calculated[4]	to be calculated[4]	-
		Mid					-
		Max					-

Energy Efficiency EAMs

Metric	Measure		CY 2024	CY 2025	CY 2026	CY 2027	Term
Share the Savings[5]	Gross Verified Annual Dth	Min	1,027,021	1,271,190	1,271,190	1,271,190	-

Metric	Measure		CY 2024	CY 2025	CY 2026	CY 2027	Term
Low and Moderate Income Customer Savings[5]	Gross Verified Annual Dth	Min	58,505	80,378	80,378	80,378	299,637
		Mid	64,077	88,033	88,033	88,033	328,174
		Max	80,793	110,998	110,998	110,998	413,785

Disadvantaged Community EAM

Metric	Measure		CY 2024	CY 2025	CY 2026	CY 2027	Term
Energy Affordability Program Enrollment	Customers Enrolled	Min	to be calculated	to be calculated	to be calculated	to be calculated	-
		Mid					-
		Max					-

Notes

1. Measures the maximum average Dth/Hr reduced during each winter's DR events - see Customer panel testimony for more details
2. Targets for the Gas DR EAM are not separate by OpCo, i.e., targets extend across both KEDNY and KEDLI
3. In the case of the Gas DR EAM, each CY refers to the winter season that begins during that CY and extends into the next (e.g. "CY 2024" measures achievement during winter 2024-25)
4. Gas DR EAM targets are calculated based on previous winter's achievement, and so cannot be calculated until the previous winter has concluded
5. Targets are derived from the Commission's NE:NY Order, utilizing 2025 levels

KEDLI Earning Adjustment Mechanism Targets

	a	b	c	d	e	f	g	h

System Efficiency EAM

Metric	Measure		CY 2024[3]	CY 2025[3]	CY 2026[3]	CY 2027[3]	Term
Gas Demand Response	Dth/Hr Reduction[1,2]	Min	to be calculated[4]	to be calculated[4]	to be calculated[4]	to be calculated[4]	-
		Mid					-
		Max					-

Energy Efficiency EAMs

Metric	Measure		CY 2024	CY 2025	CY 2026	CY 2027	Term
Share the Savings[5]	Gross Verified Annual Dth	Min	906,621	1,070,197	1,070,197	1,070,197	-

Metric	Measure		CY 2024	CY 2025	CY 2026	CY 2027	Term
Low and Moderate Income Customer Savings[5]	Gross Verified Annual Dth	Min	49,560	62,605	62,605	62,605	237,376
		Mid	54,280	68,568	68,568	68,568	259,983
		Max	68,440	86,455	86,455	86,455	327,804

Disadvantaged Community EAM

Metric	Measure		CY 2024	CY 2025	CY 2026	CY 2027	Term
Energy Affordability Program Enrollment	Customers Enrolled	Min	to be calculated	to be calculated	to be calculated	to be calculated	-
		Mid					-
		Max					-

Notes

1. Measures the maximum average Dth/Hr reduced during each winter's DR events - see Customer panel testimony for more details
2. Targets for the Gas DR EAM are not separate by OpCo, i.e., targets extend across both KEDNY and KEDLI
3. In the case of the Gas DR EAM, each CY refers to the winter season that begins during that CY and extends into the next (e.g. "CY 2024" measures achievement during winter 2024-25)
4. Gas DR EAM targets are calculated based on previous winter's achievement, and so cannot be calculated until the previous winter has concluded
5. Targets are derived from the Commission's NE:NY Order, utilizing 2025 levels

Exhibit___(CLCPA-8)

Gas Demand Response EAM Methodology

Gas Demand Response EAM Methodology

Table 1: Actual performance during gas DR events in KEDNY-KEDLI, winters 2021-22 & 2022-23, by event type

Gas day beginning	Winter	Program	Event Type	Event Hours	Dth/Hr Reduction
11/30/2021	2021-22	Load Shedding	Test	3	1,142
11/30/2021	2021-22	Load Shifting	Test	3	26
1/11/2022	2021-22	Load Shedding	Test	3	1,341
1/11/2022	2021-22	Load Shifting	Test	3	25
1/11/2022	2021-22	BYOT	Test	4	41
1/30/2022	2021-22	BYOT	Test	4	47
12/1/2022	2022-23	Load Shedding	Test	3	1,463
12/1/2022	2022-23	Load Shifting	Test	3	61
12/24/2022	2022-23	Load Shedding	Emergency	8	994
12/24/2022	2022-23	Load Shifting	Emergency	4	52
12/24/2022	2022-23	BYOT	Emergency	8	142
2/3/2023	2022-23	Load Shedding	Actual	4	1,863
2/3/2023	2022-23	Load Shifting	Actual	4	42
2/3/2023	2022-23	BYOT	Actual	4	242

Table 2: Actual performance during gas DR events in KEDNY-KEDLI , winters 2021-22 & 2022-23, aggregated by gas day

Gas day beginning	Winter	Dth/Hr Reduction
11/30/2021	2021-22	1,167
1/11/2022	2021-22	1,407
1/30/2022	2021-22	47
12/1/2022	2022-23	1,524
12/24/2022	2022-23	1,189
2/3/2023	2022-23	2,146

Table 3: Maximum dth/hr reduction among each winter's gas DR events, winters 2021-22 through 2023-24

Winter	Dth/Hr Reduction	Notes
2021-22	1,407	*actual value*
2022-23	2,146	*actual value*
2023-24	2,747	*forecast value (to be updated with actual after close of winter 2023-24 season)*

Winter	Dth/Hr Reduction	Notes
2021-22	1,407	actual value
2022-23	2,146	actual value

2023-24	2,500	placeholder value (to be updated with actual value after close of winter 2023-24 season)

Chart 1: Baseline values (based on winters 2021-22 through 2023-24, placeholder value for winter 2023-24)

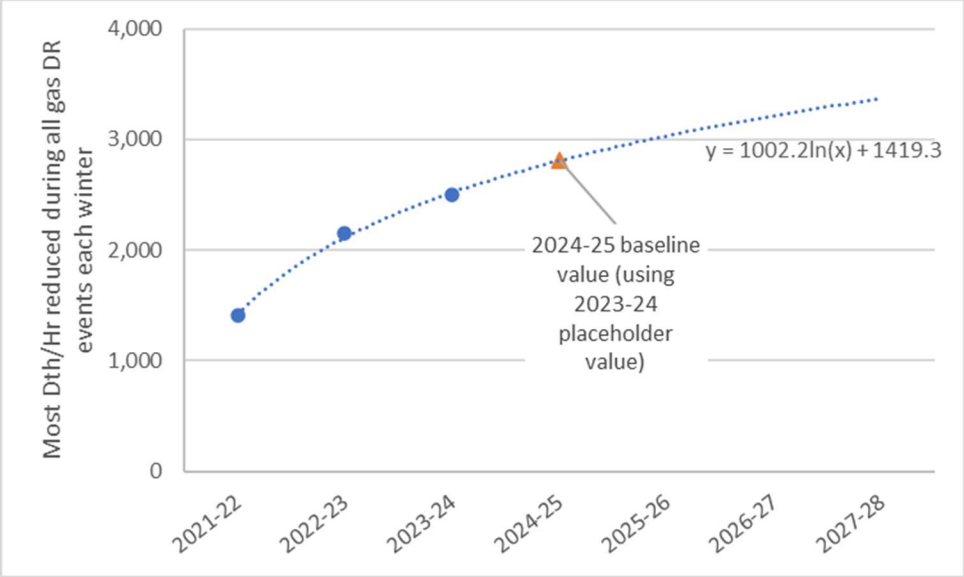


Table 4: Baseline and target values (using 2023-24 placeholder value; to be updated after close of winter season)

Level	% above baseline	Rate Year 1 value (dth/hr)
Baseline	n/a	2,809
Minimum	1%	2,837
Midpoint	2%	2,865
Maximum	5%	2,949

Exhibit___(CLCPA-9)

EAM Basis Points and Incentives

KEDNY Earning Adjustment Mechanism Basis Points & Incentives

	a	b	c	d	e	f	g	h	i	j	k	l

System Efficiency EAM

			Basis Points					$ Million				
	Metric		CY 2024	CY 2025	CY 2026	CY 2027	Term	CY 2024	CY 2025	CY 2026	CY 2027	Term
1	Gas Demand Response	Min	2.0	2.0	2.0	2.0	-	$ 1.0	$ 1.0	$ 1.1	$ 1.2	-
2		Mid	5.0	5.0	5.0	5.0	-	$ 2.4	$ 2.6	$ 2.9	$ 3.0	-
3		Max	10.0	10.0	10.0	10.0	-	$ 4.8	$ 5.2	$ 5.7	$ 6.0	-

Energy Efficiency EAMs

			Basis Points					$ Million				
	Metric		CY 2024	CY 2025	CY 2026	CY 2027	Term	CY 2024	CY 2025	CY 2026	CY 2027	Term
4	Share the Savings	Min	-	-	-	-	-	-	-	-	-	-
5		Mid	-	-	-	-	-	-	-	-	-	-
6		Max	-	-	-	-	-	-	-	-	-	-
7	Low and Moderate Income Customer Savings	Min	0.5	0.5	0.5	0.5	2.0	$ 0.2	$ 0.3	$ 0.3	$ 0.3	$0.9
8		Mid	1.0	1.0	1.0	1.0	4.0	$ 0.5	$ 0.5	$ 0.6	$ 0.6	$1.9
9		Max	2.0	2.0	2.0	2.0	8.0	$ 1.0	$ 1.0	$ 1.1	$ 1.2	$3.7

Disadvantaged Community Engagement EAM

			Basis Points					$ Million				
	Metric		CY 2024	CY 2025	CY 2026	CY 2027	Term	CY 2024	CY 2025	CY 2026	CY 2027	Term
10	Energy Affordability Program Enrollment	Min	0.5	0.5	0.5	0.5	-	$ 0.2	$ 0.3	$ 0.3	$ 0.3	-
11		Mid	1.0	1.0	1.0	1.0	-	$ 0.5	$ 0.5	$ 0.6	$ 0.6	-
12		Max	2.0	2.0	2.0	2.0	-	$ 1.0	$ 1.0	$ 1.1	$ 1.2	-

		Basis Points				$ Million			
		CY 2024	CY 2025	CY 2026	CY 2027	CY 2024	CY 2025	CY 2026	CY 2027
13	Min	3.0	3.0	3.0	3.0	$ 1.4	$ 1.6	$ 1.7	$ 1.8
14	KEDNY EAM Total Mid	7.0	7.0	7.0	7.0	$ 3.3	$ 3.7	$ 4.0	$ 4.2
15	Max	14.0	14.0	14.0	14.0	$ 6.7	$ 7.3	$ 8.0	$ 8.4

KEDLI Earning Adjustment Mechanism Basis Points & Incentives

	a	b	c	d	e	f	g	h	i	j	k	l

System Efficiency EAM

			Basis Points					$ Million				
	Metric		CY 2024	CY 2025	CY 2026	CY 2027	Term	CY 2024	CY 2025	CY 2026	CY 2027	Term
1	Gas Demand Response	Min	2.0	2.0	2.0	2.0	-	$ 0.6	$ 0.7	$ 0.7	$ 0.8	-
2		Mid	5.0	5.0	5.0	5.0	-	$ 1.6	$ 1.7	$ 1.8	$ 2.0	-
3		Max	10.0	10.0	10.0	10.0	-	$ 3.1	$ 3.4	$ 3.7	$ 3.9	-

Energy Efficiency EAMs

			Basis Points					$ Million				
	Metric		CY 2024	CY 2025	CY 2026	CY 2027	Term	CY 2024	CY 2025	CY 2026	CY 2027	Term
4	Share the Savings	Min	-	-	-	-	-	-	-	-	-	-
5		Mid	-	-	-	-	-	-	-	-	-	-
6		Max	-	-	-	-	-	-	-	-	-	-
7	Low and Moderate Income Customer Savings	Min	0.5	0.5	0.5	0.5	2.0	$ 0.2	$ 0.2	$ 0.2	$ 0.2	$0.9
8		Mid	1.0	1.0	1.0	1.0	4.0	$ 0.3	$ 0.3	$ 0.4	$ 0.4	$1.9
9		Max	2.0	2.0	2.0	2.0	8.0	$ 0.6	$ 0.7	$ 0.7	$ 0.8	$3.7

Disadvantaged Community Engagement EAM

			Basis Points					$ Million				
	Metric		CY 2024	CY 2025	CY 2026	CY 2027	Term	CY 2024	CY 2025	CY 2026	CY 2027	Term
10	Energy Affordability Program Enrollment	Min	0.5	0.5	0.5	0.5	-	$ 0.2	$ 0.2	$ 0.2	$ 0.2	-
11		Mid	1.0	1.0	1.0	1.0	-	$ 0.3	$ 0.3	$ 0.4	$ 0.4	-
12		Max	1.5	1.5	1.5	1.5	-	$ 0.6	$ 0.7	$ 0.7	$ 0.8	-

		Basis Points				$ Million			
		CY 2021	CY 2022	CY 2023	CY 2024	CY 2021	CY 2022	CY 2023	CY 2024
13	Min	3.0	3.0	3.0	3.0	$ 0.9	$ 1.0	$ 1.1	$ 1.2
14	KEDLI EAM Total Mid	7.0	7.0	7.0	7.0	$ 2.2	$ 2.4	$ 2.6	$ 2.7
15	Max	23.5	13.5	13.5	13.5	$ 4.4	$ 4.7	$ 5.2	$ 5.5

Exhibit___(CLCPA-10) EAM Net

Benefits Calculations

Summary of Gas KEDNY Net Benefits

	a	b	c	d	e	f	g
1	Summary of EAM Net Benefits	Portfolio Net Benefits	4-year EAM Incentive Value	1-year EAM Incentive Value	Savings to Customers	EAM Incentive	Savings to Customers
2		NPV ($M)	NPV ($M)	NPV ($M)	NPV ($M)	%	%
3	Minimum	$466.8	$9.3	$2.33	$457.5	2%	98%
4	Midpoint	$498.1	$24.9	$6.23	$473.2	5%	95%
5	Maximum	$564.5	$56.5	$14.11	$508.1	10%	90%

Benefits (NPV $M)								
Benefits (Gas EAM Metric)		**Minimum**		**Midpoint**		**Maximum**		
Avoided Dth	**$**	**545.87**	**$**	**570.91**	**$**	**624.39**		
Avoided Dth (Share the Savings)	$	432.80	$	454.44	$	497.72		
Avoided Dth (Gas Demand Response)[1]	$	86.35	$	87.20	$	89.77		
Avoided Dth (Low-and-Moderate Income)	$	26.73	$	29.27	$	36.91		
Avoided Dth (Energy Affordability Program Enrollment)								
Avoided CO_2	**$**	**125.12**	**$**	**131.71**	**$**	**145.58**		
Avoided CO_2 (Share the Savings)	$	117.94	$	123.84	$	135.63		
Avoided CO_2 (Gas Demand Response)[1]	$	(0.11)	$	(0.11)	$	(0.12)		
Avoided CO_2 (Low-and-Moderate Income)	$	7.29	$	7.98	$	10.07		
Avoided CO_2 (Energy Affordability Program Enrollment)	$	-	$	-	$	-		
TOTAL BENEFITS (NPV $M)	$	670.99	$	702.62	$	769.97		

Costs (NPV $M)								
Costs (Gas EAM Metric)		**Minimum**		**Midpoint**		**Maximum**		
Share the Savings	$	146.57	$	146.57	$	146.57		
Gas Demand Response[1]	$	31.77	$	32.08	$	33.03		
Low-and-Moderate Income	$	25.85	$	25.85	$	25.85		
Energy Affordability Program Enrollment	$	-	$	-	$	-		
TOTAL COSTS (NPV $M)	$	204.19	$	204.51	$	205.45		

	Minimum		**Midpoint**		**Maximum**	
NET BENEFITS (NPV $M)	**$**	**466.80**	**$**	**498.11**	**$**	**564.52**

	Minimum		**Midpoint**		**Maximum**	
Total Benefits (NPV $M)						
Share the Savings	$	550.74	$	578.28	$	633.35
Gas Demand Response	$	86.23	$	87.09	$	89.65
Low-and-Moderate Income	$	34.02	$	37.26	$	46.97
Energy Affordability Program Enrollment	$	-	$	-	$	-
Total Costs (NPV $M)						
Share the Savings	$	146.57	$	146.57	$	146.57
Gas Demand Response	$	31.77	$	32.08	$	33.03
Low-and-Moderate Income	$	25.85	$	25.85	$	25.85
Energy Affordability Program Enrollment	$	-	$	-	$	-
Total Net Benefits (NPV $M)						
Share the Savings	$	404.17	$	431.70	$	486.78
Gas Demand Response	$	54.47	$	55.00	$	56.62
Low-and-Moderate Income	$	8.17	$	11.40	$	21.12
Energy Affordability Program Enrollment	$	-	$	-	$	-
Estimated 1-Year Incentive Levels (NPV $M)						
Share the Savings	$	2.02	$	5.40	$	12.17
Gas Demand Response	$	0.27	$	0.69	$	1.42
Low-and-Moderate Income	$	0.04	$	0.14	$	0.53
Energy Affordability Program Enrollment	$	-	$	-	$	-
	$	2.33	$	6.23	$	14.11
	Minimum		Midpoint		Maximum	

Notes

1. Since the Gas DR EAM spans both KEDNY and KEDLI, benefits and costs for it have been split 70/30 between the respective operating companies, based on approximate historical figures of load reductions and enrollment in each.

Summary of Gas KEDLI Net Benefits

	a	b	c	d	e	f	g
1	Summary of EAM Net Benefits	Portfolio Net Benefits	4-year EAM Incentive Value	1-year EAM Incentive Value	Savings to Customers	EAM Incentive	Savings to Customers
2		NPV ($M)	NPV ($M)	NPV ($M)	NPV ($M)	%	%
3	Minimum	$392.5	$7.8	$1.96	$384.6	2%	98%
4	Midpoint	$418.7	$20.9	$5.23	$397.8	5%	95%
5	Maximum	$474.0	$47.4	$11.85	$426.6	10%	90%

6	Benefits (NPV $M)			
7	**Benefits (Gas EAM Metric)**	**Minimum**	**Midpoint**	**Maximum**
8	**Avoided Dth**	**$ 426.75**	**$ 447.57**	**$ 491.58**
9	Avoided Dth (Share the Savings)	$ 368.54	$ 386.96	$ 423.82
10	Avoided Dth (Gas Demand Response)[1]	$ 37.01	$ 37.37	$ 38.47
11	Avoided Dth (Low-and-Moderate Income)	$ 21.21	$ 23.23	$ 29.30
12	Avoided Dth (Energy Affordability Program Enrollment)			
13	**Avoided CO_2**	**$ 106.13**	**$ 111.70**	**$ 123.39**
14	Avoided CO_2 (Share the Savings)	$ 100.39	$ 105.41	$ 115.45
15	Avoided CO_2 (Gas Demand Response)[1]	$ (0.05)	$ (0.05)	$ (0.05)
16	Avoided CO_2 (Low-and-Moderate Income)	$ 5.78	$ 6.33	$ 7.98
17	Avoided CO_2 (Energy Affordability Program Enrollment)	$ -	$ -	$ -
18	TOTAL BENEFITS (NPV $M)	$ 532.88	$ 559.27	$ 614.97

19	Costs (NPV $M)			
20	**Costs (Gas EAM Metric)**	**Minimum**	**Midpoint**	**Maximum**
21	**Implementation Costs**	**$ 140.42**	**$ 140.56**	**$ 140.96**
22	Share the Savings	$ 106.29	$ 106.29	$106.29
23	Gas Demand Response[1]	$ 13.61	$ 13.75	$ 14.15
24	Low-and-Moderate Income	$ 20.52	$ 20.52	$ 20.52
25	Energy Assistance Program Enrollment	$ -	$ -	$ -
26	TOTAL COSTS (NPV $M)	$ 140.42	$ 140.56	$ 140.96

27		**Minimum**	**Midpoint**	**Maximum**
28	**NET BENEFITS (NPV $M)**	**$ 392.46**	**$ 418.71**	**$ 474.01**

29		**Minimum**	**Midpoint**	**Maximum**
30	**Total Benefits (NPV $M)**			
31	Share the Savings	$ 468.93	$ 492.38	$ 539.27
32	Gas Demand Response	$ 36.96	$ 37.32	$ 38.42
33	Low-and-Moderate Income	$ 27.00	$ 29.57	$ 37.28
34	Energy Affordability Program Enrollment	$ -	$ -	$ -
35	**Total Costs (NPV $M)**			
36	Share the Savings	$ 106.29	$ 106.29	$ 106.29
37	Gas Demand Response	$ 13.61	$ 13.75	$ 14.15
38	Low-and-Moderate Income	$ 20.52	$ 20.52	$ 20.52
39	Energy Affordability Program Enrollment	$ -	$ -	$ -
40	**Total Net Benefits (NPV $M)**			
41	Share the Savings	$ 362.64	$ 386.09	$ 432.98
42	Gas Demand Response	$ 23.34	$ 23.57	$ 24.27
43	Low-and-Moderate Income	$ 6.48	$ 9.05	$ 16.76
44	Energy Affordability Program Enrollment	$ -	$ -	$ -
45	Estimated 1-Year Incentive Levels (NPV $M)			
46	Share the Savings	$ 1.81	$ 4.83	$ 10.82
47	Gas Demand Response	$ 0.12	$ 0.29	$ 0.61
48	Low-and-Moderate Income	$ 0.03	$ 0.11	$ 0.42
49	Energy Affordability Program Enrollment	$ -	$ -	$ -
50		$ 1.96	$ 5.23	$ 11.85
51		Minimum	Midpoint	Maximum

Notes

1. Since the Gas DR EAM spans both KEDNY and KEDLI, benefits and costs for it have been split 70/30 between the respective operating companies, based on approximate historical figures of load reductions and enrollment in each.

KEDNY Energy Efficiency EAM: Share the Savings Net Benefits Calculation

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p		
Gross Annual MMBtu or Dth		**CY 2024**	**CY 2025**	**CY 2026**	**CY 2027**		**Assumptions**										
	Min	1,027,021	1,271,190	1,271,190	1,271,190		Portfolio EUL (yrs)		11								
	Mid	1,078,372	1,334,750	1,334,750	1,334,750		lbs of CO2 per Dth		117								
	Max	1,181,074	1,461,869	1,461,869	1,461,869		lbs per ton		2,205								
							WACC		7.09%								

	Benefits - Minimum	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**	**2036**	**NPV**
Gross Annual Dth Saved	Natural Gas Savings - 2024 (Dth)	1,027,021	1,027,021	1,027,021	1,027,021	1,027,021	1,027,021	1,027,021	1,027,021	1,027,021	1,027,021				
Gross Annual Dth Saved	Natural Gas Savings - 2025 (Dth)		1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190			
Gross Annual Dth Saved	Natural Gas Savings - 2026 (Dth)			1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190		
Gross Annual Dth Saved	Natural Gas Savings - 2027 (Dth)				1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	1,271,190	
Sum of Gross Annual Dth Saved	Total Annual Natural Gas (Dth)	1,027,021	2,298,211	3,569,401	4,840,591	4,840,591	4,840,591	4,840,591	4,840,591	4,840,591	4,840,591	3,813,570	2,542,380	1,271,190	
(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =	CO2 Savings - 2024 (MT)	54,494	54,494	54,494	54,494	54,494	54,494	54,494	54,494	54,494	54,494				
(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =	CO2 Savings - 2025 (MT)		67,449	67,449	67,449	67,449	67,449	67,449	67,449	67,449	67,449	67,449			
(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =	CO2 Savings - 2026 (MT)			67,449	67,449	67,449	67,449	67,449	67,449	67,449	67,449	67,449	67,449		
(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =	CO2 Savings - 2027 (MT)				67,449	67,449	67,449	67,449	67,449	67,449	67,449	67,449	67,449	67,449	
Sum of CO2 Saved	Total Annual CO2 (MT)	54,494	121,943	189,392	256,841	256,841	256,841	256,841	256,841	256,841	256,841	202,348	134,899	67,449	
Estimated	Marginal Cost of Gas ($/Dth)	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	
Per NYS "E-Value"	Net Marginal Social Cost of Carbon ($/MT)	$ 60.09	$ 62.54	$ 65.06	$ 67.66	$ 70.34	$ 73.11	$ 75.96	$ 78.99	$ 82.11	$ 85.34	$ 72.23	$ 74.92	$ 77.60	
Sum of Gross Annual Dth Saved × Marginal Cost of Gas =	Avoided Dth Savings ($M)	$ 14.58	$ 32.63	$ 50.69	$ 68.74	$ 68.74	$ 68.74	$ 68.74	$ 68.74	$ 68.74	$ 68.74	$ 54.15	$ 36.10	$ 18.05	**$ 432.80**
Sum of CO2 Savings × Cost of CO2 $/Dth =	Avoided CO2 Savings ($M)	$ 3.27	$ 7.63	$ 12.32	$ 17.38	$ 18.07	$ 18.78	$ 19.51	$ 20.29	$ 21.09	$ 21.92	$ 14.62	$ 10.11	$ 5.23	**$ 117.94**
Avoided Dth Savings + Avoided CO2 Savings =	**Total Benefits ($M)**	**$ 17.86**	**$ 40.26**	**$ 63.01**	**$ 86.12**	**$ 86.80**	**$ 87.51**	**$ 88.25**	**$ 89.02**	**$ 89.83**	**$ 90.65**	**$ 68.77**	**$ 46.21**	**$ 23.28**	**$ 550.74**

	Costs - Minimum	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**	**2036**	**NPV**
	Implementation Costs ($M)	$ 36.85	$ 45.85	$ 45.85	$ 45.85	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	**$ 146.57**
	Total Costs ($M)	**$ 36.85**	**$ 45.85**	**$ 45.85**	**$ 45.85**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ 146.57**
Total Benefits - Total Costs =	**Net Benefit at Minimum ($M)**	$ (18.99)	$ (5.59)	$ 17.16	$ 40.27	$ 86.80	$ 87.51	$ 88.25	$ 89.02	$ 89.83	$ 90.65	$ 68.77	$ 46.21	$ 23.28	**$ 404.17**

	Benefits - Midpoint	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**	**2036**	**NPV**
Gross Annual Dth Saved	Natural Gas Savings - 2024 (Dth)	1,078,372	1,078,372	1,078,372	1,078,372	1,078,372	1,078,372	1,078,372	1,078,372	1,078,372	1,078,372				
Gross Annual Dth Saved	Natural Gas Savings - 2025 (Dth)		1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750			
Gross Annual Dth Saved	Natural Gas Savings - 2026 (Dth)			1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750		
Gross Annual Dth Saved	Natural Gas Savings - 2027 (Dth)				1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	1,334,750	
Sum of Gross Annual Dth Saved	Total Annual Natural Gas (Dth)	1,078,372	2,413,122	3,747,871	5,082,621	5,082,621	5,082,621	5,082,621	5,082,621	5,082,621	5,082,621	4,004,249	2,669,499	1,334,750	
(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =	CO2 Savings - 2024 (MT)	57,218	57,218	57,218	57,218	57,218	57,218	57,218	57,218	57,218	57,218				
(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =	CO2 Savings - 2025 (MT)		70,822	70,822	70,822	70,822	70,822	70,822	70,822	70,822	70,822	70,822			
(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =	CO2 Savings - 2026 (MT)			70,822	70,822	70,822	70,822	70,822	70,822	70,822	70,822	70,822	70,822		
(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =	CO2 Savings - 2027 (MT)				70,822	70,822	70,822	70,822	70,822	70,822	70,822	70,822	70,822	70,822	
Sum of CO2 Saved	Total Annual CO2 (MT)	57,218	128,040	198,862	269,684	269,684	269,684	269,684	269,684	269,684	269,684	212,465	141,643	70,822	
Estimated	Marginal Cost of Gas ($/Dth)	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	
Per NYS "E-Value"	Net Marginal Social Cost of Carbon ($/MT)	$ 60.09	$ 62.54	$ 65.06	$ 67.66	$ 70.34	$ 73.11	$ 75.96	$ 78.99	$ 82.11	$ 85.34	$ 72.23	$ 74.92	$ 77.60	
Sum of Gross Annual Dth Saved × Marginal Cost of Gas =	Avoided Dth Savings ($M)	$ 15.31	$ 34.27	$ 53.22	$ 72.17	$ 72.17	$ 72.17	$ 72.17	$ 72.17	$ 72.17	$ 72.17	$ 56.86	$ 37.91	$ 18.95	**$ 454.44**
Sum of CO2 Savings × Cost of CO2 $/Dth =	Avoided CO2 Savings ($M)	$ 3.44	$ 8.01	$ 12.94	$ 18.25	$ 18.97	$ 19.72	$ 20.48	$ 21.30	$ 22.14	$ 23.01	$ 15.35	$ 10.61	$ 5.50	**$ 123.84**
Avoided Dth Savings + Avoided CO2 Savings =	**Total Benefits**	**$ 18.75**	**$ 42.27**	**$ 66.16**	**$ 90.42**	**$ 91.14**	**$ 91.89**	**$ 92.66**	**$ 93.48**	**$ 94.32**	**$ 95.19**	**$ 72.21**	**$ 48.52**	**$ 24.45**	**$ 578.28**

	Costs - Midpoint	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**	**2036**	**NPV**
	Implementation Costs	$ 36.85	$ 45.85	$ 45.85	$ 45.85	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	**$ 146.57**
	Total Costs	**$ 36.85**	**$ 45.85**	**$ 45.85**	**$ 45.85**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ 146.57**
Total Benefits - Total Costs =	**Net Benefit at Midpoint**	$ (18.09)	$ (3.57)	$ 20.31	$ 44.57	$ 91.14	$ 91.89	$ 92.66	$ 93.48	$ 94.32	$ 95.19	$ 72.21	$ 48.52	$ 24.45	**$ 431.70**

	Benefits - Maximum	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**	**2036**	**NPV**
Gross Annual Dth Saved	Natural Gas Savings - 2024 (Dth)	1,181,074	1,181,074	1,181,074	1,181,074	1,181,074	1,181,074	1,181,074	1,181,074	1,181,074	1,181,074				
Gross Annual Dth Saved	Natural Gas Savings - 2025 (Dth)		1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869			
Gross Annual Dth Saved	Natural Gas Savings - 2026 (Dth)			1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869		
Gross Annual Dth Saved	Natural Gas Savings - 2027 (Dth)				1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	1,461,869	
Sum of Gross Annual Dth Saved	Total Annual Natural Gas (Dth)	1,181,074	2,642,943	4,104,811	5,566,680	5,566,680	5,566,680	5,566,680	5,566,680	5,566,680	5,566,680	4,385,606	2,923,737	1,461,869	
(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =	CO2 Savings - 2024 (MT)	62,668	62,668	62,668	62,668	62,668	62,668	62,668	62,668	62,668	62,668				
(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =	CO2 Savings - 2025 (MT)		77,567	77,567	77,567	77,567	77,567	77,567	77,567	77,567	77,567	77,567			
(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =	CO2 Savings - 2026 (MT)			77,567	77,567	77,567	77,567	77,567	77,567	77,567	77,567	77,567	77,567		
(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =	CO2 Savings - 2027 (MT)				77,567	77,567	77,567	77,567	77,567	77,567	77,567	77,567	77,567	77,567	
Sum of CO2 Saved	Total Annual CO2 (MT)	62,668	140,234	217,801	295,368	295,368	295,368	295,368	295,368	295,368	295,368	232,700	155,133	77,567	
Estimated	Marginal Cost of Gas ($/Dth)	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	
Per NY BCA Handbook	Net Marginal Damage Cost of Carbon ($/MT)	$ 60.09	$ 62.54	$ 65.06	$ 67.66	$ 70.34	$ 73.11	$ 75.96	$ 78.99	$ 82.11	$ 85.34	$ 72.23	$ 74.92	$ 77.60	
Sum of Gross Annual Dth Saved × Marginal Cost of Gas =	Avoided Dth Savings ($M)	$ 16.77	$ 37.53	$ 58.29	$ 79.05	$ 79.05	$ 79.05	$ 79.05	$ 79.05	$ 79.05	$ 79.05	$ 62.28	$ 41.52	$ 20.76	**$ 497.72**
Sum of CO2 Savings × Cost of CO2 $/MWh =	Avoided CO2 Savings ($M)	$ 3.77	$ 8.77	$ 14.17	$ 19.99	$ 20.78	$ 21.59	$ 22.44	$ 23.33	$ 24.25	$ 25.21	$ 16.81	$ 11.62	$ 6.02	**$ 135.63**
Avoided Dth Savings + Avoided CO2 Savings =	**Total Benefits**	**$ 20.54**	**$ 46.30**	**$ 72.46**	**$ 99.03**	**$ 99.82**	**$ 100.64**	**$ 101.48**	**$ 102.38**	**$ 103.30**	**$ 104.25**	**$ 79.08**	**$ 53.14**	**$ 26.78**	**$ 633.35**

	Costs - Maximum	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**	**2036**	**NPV**
	Implementation Costs	$ 36.85	$ 45.85	$ 45.85	$ 45.85	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	**$ 146.57**
	Total Costs	**$ 36.85**	**$ 45.85**	**$ 45.85**	**$ 45.85**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ 146.57**
Total Benefits - Total Costs =	**Net Benefit at Maximum**	$ (16.31)	$ 0.45	$ 26.61	$ 53.19	$ 99.82	$ 100.64	$ 101.48	$ 102.38	$ 103.30	$ 104.25	$ 79.08	$ 53.14	$ 26.78	**$ 486.78**

KEDLI Energy Efficiency EAM: Share the Savings Net Benefits Calculation

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
1		**Gross Annual MMBtu or Dth**	**CY 2024**	**CY 2025**	**CY 2026**	**CY 2027**		**Assumptions**								
2			**Min**	906,621	1,070,197	1,070,197	1,070,197		Portfolio EUL (yrs)	11						
3			**Mid**	951,952	1,123,707	1,123,707	1,123,707		lbs of CO2 per Dth	117						
4			**Max**	1,042,614	1,230,727	1,230,727	1,230,727		lbs per ton	2,205						
5									WACC	7.09%						
6																

7		**Benefits - Minimum**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**	**2036**	**NPV**
8	*Gross Annual Dth Saved*	Natural Gas Savings - 2024 (Dth)	906,621	906,621	906,621	906,621	906,621	906,621	906,621	906,621	906,621	906,621				
9	*Gross Annual Dth Saved*	Natural Gas Savings - 2025 (Dth)		1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197			
10	*Gross Annual Dth Saved*	Natural Gas Savings - 2026 (Dth)			1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197		
11	*Gross Annual Dth Saved*	Natural Gas Savings - 2027 (Dth)				1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	1,070,197	
12	*Sum of Gross Annual Dth Saved*	Total Annual Natural Gas (Dth)	906,621	1,976,818	3,047,015	4,117,212	4,117,212	4,117,212	4,117,212	4,117,212	4,117,212	4,117,212	3,210,591	2,140,394	1,070,197	
13	*(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =*	CO2 Savings - 2024 (MT)	48,105	48,105	48,105	48,105	48,105	48,105	48,105	48,105	48,105	48,105				
14	*(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =*	CO2 Savings - 2025 (MT)		56,785	56,785	56,785	56,785	56,785	56,785	56,785	56,785	56,785	56,785			
15	*(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =*	CO2 Savings - 2026 (MT)			56,785	56,785	56,785	56,785	56,785	56,785	56,785	56,785	56,785	56,785		
16	*(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =*	CO2 Savings - 2027 (MT)				56,785	56,785	56,785	56,785	56,785	56,785	56,785	56,785	56,785	56,785	
17	*Sum of CO2 Saved*	Total Annual CO2 (MT)	48,105	104,890	161,674	218,459	218,459	218,459	218,459	218,459	218,459	218,459	170,354	113,569	56,785	
18	*Estimated*	Marginal Cost of Gas ($/Dth)	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	
19	*Per NYS "E-Value"*	Net Marginal Social Cost of Carbon ($/MT)	$ 60.09	$ 62.54	$ 65.06	$ 67.66	$ 70.34	$ 73.11	$ 75.96	$ 78.99	$ 82.11	$ 85.34	$ 72.23	$ 74.92	$ 77.60	
20	*Sum of Gross Annual Dth Saved × Marginal Cost of Gas =*	Avoided Dth Savings ($M)	$ 12.87	$ 28.07	$ 43.27	$ 58.46	$ 58.46	$ 58.46	$ 58.46	$ 58.46	$ 58.46	$ 58.46	$ 45.59	$ 30.39	$ 15.20	**$ 368.54**
21	*Sum of CO2 Savings × Cost of CO2 $/Dth =*	Avoided CO2 Savings ($M)	$ 2.89	$ 6.56	$ 10.52	$ 14.78	$ 15.37	$ 15.97	$ 16.59	$ 17.26	$ 17.94	$ 18.64	$ 12.30	$ 8.51	$ 4.41	**$ 100.39**
22	*Avoided Dth Savings + Avoided CO2 Savings =*	**Total Benefits ($M)**	**$ 15.76**	**$ 34.63**	**$ 53.79**	**$ 73.25**	**$ 73.83**	**$ 74.44**	**$ 75.06**	**$ 75.72**	**$ 76.40**	**$ 77.11**	**$ 57.90**	**$ 38.90**	**$ 19.60**	**$ 468.93**
23																

24		**Costs - Minimum**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**	**2036**	**NPV**
25		Implementation Costs ($M)	$ 27.56	$ 32.93	$ 32.93	$ 32.93	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	**$ 106.29**
26		**Total Costs ($M)**	**$ 27.56**	**$ 32.93**	**$ 32.93**	**$ 32.93**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ 106.29**
27	*Total Benefits - Total Costs =*	**Net Benefit at Minimum ($M)**	$ (11.79)	$ 1.70	$ 20.86	$ 40.32	$ 73.83	$ 74.44	$ 75.06	$ 75.72	$ 76.40	$ 77.11	$ 57.90	$ 38.90	$ 19.60	**$ 362.64**
28																

29		**Benefits - Midpoint**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**	**2036**	**NPV**
30	*Gross Annual Dth Saved*	Natural Gas Savings - 2024 (Dth)	951,952	951,952	951,952	951,952	951,952	951,952	951,952	951,952	951,952	951,952				
31	*Gross Annual Dth Saved*	Natural Gas Savings - 2025 (Dth)		1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707			
32	*Gross Annual Dth Saved*	Natural Gas Savings - 2026 (Dth)			1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707		
33	*Gross Annual Dth Saved*	Natural Gas Savings - 2027 (Dth)				1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	1,123,707	
34	*Sum of Gross Annual Dth Saved*	Total Annual Natural Gas (Dth)	951,952	2,075,659	3,199,366	4,323,073	4,323,073	4,323,073	4,323,073	4,323,073	4,323,073	4,323,073	3,371,121	2,247,414	1,123,707	
35	*(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =*	CO2 Savings - 2024 (MT)	50,511	50,511	50,511	50,511	50,511	50,511	50,511	50,511	50,511	50,511				
36	*(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =*	CO2 Savings - 2025 (MT)		59,624	59,624	59,624	59,624	59,624	59,624	59,624	59,624	59,624	59,624			
37	*(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =*	CO2 Savings - 2026 (MT)			59,624	59,624	59,624	59,624	59,624	59,624	59,624	59,624	59,624	59,624		
38	*(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =*	CO2 Savings - 2027 (MT)				59,624	59,624	59,624	59,624	59,624	59,624	59,624	59,624	59,624	59,624	
39	*Sum of CO2 Saved*	Total Annual CO2 (MT)	50,511	110,134	169,758	229,382	229,382	229,382	229,382	229,382	229,382	229,382	178,871	119,248	59,624	
40	*Estimated*	Marginal Cost of Gas ($/Dth)	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	
41	*Per NYS "E-Value"*	Net Marginal Social Cost of Carbon ($/MT)	$ 60.09	$ 62.54	$ 65.06	$ 67.66	$ 70.34	$ 73.11	$ 75.96	$ 78.99	$ 82.11	$ 85.34	$ 72.23	$ 74.92	$ 77.60	
42	*Sum of Gross Annual Dth Saved × Marginal Cost of Gas =*	Avoided Dth Savings ($M)	$ 13.52	$ 29.47	$ 45.43	$ 61.39	$ 61.39	$ 61.39	$ 61.39	$ 61.39	$ 61.39	$ 61.39	$ 47.87	$ 31.91	$ 15.96	**$ 386.96**
43	*Sum of CO2 Savings × Cost of CO2 $/Dth =*	Avoided CO2 Savings ($M)	$ 3.04	$ 6.89	$ 11.04	$ 15.52	$ 16.14	$ 16.77	$ 17.42	$ 18.12	$ 18.84	$ 19.57	$ 12.92	$ 8.93	$ 4.63	**$ 105.41**
44	*Avoided Dth Savings + Avoided CO2 Savings =*	**Total Benefits**	**$ 16.55**	**$ 36.36**	**$ 56.48**	**$ 76.91**	**$ 77.52**	**$ 78.16**	**$ 78.81**	**$ 79.51**	**$ 80.22**	**$ 80.96**	**$ 60.79**	**$ 40.85**	**$ 20.58**	**$ 492.38**
45																

46		**Costs - Midpoint**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**	**2036**	**NPV**
47		Implementation Costs	$ 27.56	$ 32.93	$ 32.93	$ 32.93	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	**$ 106.29**
48		**Total Costs**	**$ 27.56**	**$ 32.93**	**$ 32.93**	**$ 32.93**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ 106.29**
49	*Total Benefits - Total Costs =*	**Net Benefit at Midpoint**	$ (11.01)	$ 3.44	$ 23.55	$ 43.98	$ 77.52	$ 78.16	$ 78.81	$ 79.51	$ 80.22	$ 80.96	$ 60.79	$ 40.85	$ 20.58	**$ 386.09**
50																

51		**Benefits - Maximum**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**	**2036**	**NPV**
52	*Gross Annual Dth Saved*	Natural Gas Savings - 2024 (Dth)	1,042,614	1,042,614	1,042,614	1,042,614	1,042,614	1,042,614	1,042,614	1,042,614	1,042,614	1,042,614				
53	*Gross Annual Dth Saved*	Natural Gas Savings - 2025 (Dth)		1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727			
54	*Gross Annual Dth Saved*	Natural Gas Savings - 2026 (Dth)			1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727		
55	*Gross Annual Dth Saved*	Natural Gas Savings - 2027 (Dth)				1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	1,230,727	
56	*Sum of Gross Annual Dth Saved*	Total Annual Natural Gas (Dth)	1,042,614	2,273,341	3,504,068	4,734,794	4,734,794	4,734,794	4,734,794	4,734,794	4,734,794	4,734,794	3,692,180	2,461,453	1,230,727	
57	*(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =*	CO2 Savings - 2024 (MT)	55,321	55,321	55,321	55,321	55,321	55,321	55,321	55,321	55,321	55,321				
58	*(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =*	CO2 Savings - 2025 (MT)		65,302	65,302	65,302	65,302	65,302	65,302	65,302	65,302	65,302	65,302			
59	*(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =*	CO2 Savings - 2026 (MT)			65,302	65,302	65,302	65,302	65,302	65,302	65,302	65,302	65,302	65,302		
60	*(Gross Annual Dth Saved × Lbs of CO2 per Dth) ÷ Lbs per Ton =*	CO2 Savings - 2027 (MT)				65,302	65,302	65,302	65,302	65,302	65,302	65,302	65,302	65,302	65,302	
61	*Sum of CO2 Saved*	Total Annual CO2 (MT)	55,321	120,623	185,926	251,228	251,228	251,228	251,228	251,228	251,228	251,228	195,907	130,605	65,302	
62	*Estimated*	Marginal Cost of Gas ($/Dth)	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	$ 14.20	
63	*Per NY BCA Handbook*	Net Marginal Damage Cost of Carbon ($/MT)	$ 60.09	$ 62.54	$ 65.06	$ 67.66	$ 70.34	$ 73.11	$ 75.96	$ 78.99	$ 82.11	$ 85.34	$ 72.23	$ 74.92	$ 77.60	
64	*Sum of Gross Annual Dth Saved × Marginal Cost of Gas =*	Avoided Dth Savings ($M)	$ 14.81	$ 32.28	$ 49.76	$ 67.23	$ 67.23	$ 67.23	$ 67.23	$ 67.23	$ 67.23	$ 67.23	$ 52.43	$ 34.95	$ 17.48	**$ 423.82**
65	*Sum of CO2 Savings × Cost of CO2 $/MWh =*	Avoided CO2 Savings ($M)	$ 3.32	$ 7.54	$ 12.10	$ 17.00	$ 17.67	$ 18.37	$ 19.08	$ 19.84	$ 20.63	$ 21.44	$ 14.15	$ 9.78	$ 5.07	**$ 115.45**
66	*Avoided Dth Savings + Avoided CO2 Savings =*	**Total Benefits**	**$ 18.13**	**$ 39.83**	**$ 61.85**	**$ 84.23**	**$ 84.91**	**$ 85.60**	**$ 86.32**	**$ 87.08**	**$ 87.86**	**$ 88.67**	**$ 66.58**	**$ 44.74**	**$ 22.54**	**$ 539.27**
67																

68		**Costs - Maximum**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**	**2036**	**NPV**
69		Implementation Costs	$ 27.56	$ 32.93	$ 32.93	$ 32.93	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	**$ 106.29**
70		**Total Costs**	**$ 27.56**	**$ 32.93**	**$ 32.93**	**$ 32.93**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	**$ 106.29**
71	*Total Benefits - Total Costs =*	**Net Benefit at Maximum**	$ (9.43)	$ 6.90	$ 28.93	$ 51.31	$ 84.91	$ 85.60	$ 86.32	$ 87.08	$ 87.86	$ 88.67	$ 66.58	$ 44.74	$ 22.54	**$ 432.98**

Gas System Efficiency: Gas Demand Response EAM Net Benefits Calculation

	a	b	c	d	e	f
1	**Demand Response Reduction Targets (Dth)***	**Winter 2024-25**	**Winter 2025-26**	**Winter 2026-27**	**Winter 2027-28**	
2	Minimum	2,837	3,063	3,247	3,403	
3	Midpoint	2,865	3,093	3,279	3,437	
4	Maximum	2,949	3,184	3,376	3,538	
5						
6	**Constant assumptions**		**Source**			
7	Events called per year	3				
8	Average event duration (hrs)	8				
11	% reductions derived from Load Shedding program	85%	*based on approximate amounts provided during winter 2022-23 emergency and actual events*			
12	Natural gas emission rate (lb CO2/MMBtu)	116.7	*EIA.gov*			
13	Fuel oil emission rate (lb CO2/MMBtu)	163.5				
14	Replacement ratio of fuel oil for natural gas (gallons/dth)	7.2				
15	mmBTU per gallon, #2 fuel oil	0.1385	*EIA.gov*			
16	Implementation costs per hourly dth saved (includes program administration, incentives, and EM&V)	$3,778.41	*2021-25 average annual costs used for 2022 Annual DSM filing*			
17	Discount rate	7.09%				
18						
19	**Variable assumptions**	**Winter 2024-25**	**Winter 2025-26**	**Winter 2026-27**	**Winter 2027-28**	**Source**
20	Estimated annualized marginal capacity cost of gas ($/dth/hr) (KEDNY)	$ 11,350	$ 11,350	$ 11,350	$ 11,350	*Exhibit __(GSP-9), CNG supply row, multiplied by 20 to convert to an hourly rate*
22	Peaking services - commodity cost ($/dth) (KEDNY)	$ 14.20	$ 14.20	$ 14.20	$ 14.20	*Exhibit __(GSP-8)*
23	Social cost of carbon	$ 60.09	$ 62.54	$ 65.06	$ 67.66	*Per NYS "E-Value"*
24	Fuel oil cost ($/mmBTU)	$ 24.73	$ 25.35	$ 26.01	$ 26.76	*EIA AEO 2022, Table 3.2, Middle Atlantic Region, Reference Case*
25						
26	**Benefits at Minimum**	**Winter 2024-25**	**Winter 2025-26**	**Winter 2026-27**	**Winter 2027-28**	**NPV**
27	Capacity savings ($M)	$ 32.20	$ 34.76	$ 36.86	$ 38.63	
28	Avoided annual dth	68,082	73,502	77,932	81,676	
29	Peaking commodity savings ($M)	$ 0.97	$ 1.04	$ 1.11	$ 1.16	
30	*Total avoided dth savings ($M)*	*$ 33.16*	*$ 35.80*	*$ 37.96*	*$ 39.79*	*$123.35*
31	Fuel oil savings (mmBTU/yr)	(57,707)	(62,302)	(66,056)	(69,231)	
32	Peak gas CO2 savings (tons CO2)	3,602	3,888	4,123	4,321	
33	Fuel oil CO2 savings (tons CO2)	(4,278)	(4,618)	(4,897)	(5,132)	
34	Total CO2 savings (tons CO2)	(676)	(730)	(774)	(811)	
35	*Total avoided CO2 savings ($M)*	*$ (0.04)*	*$ (0.05)*	*$ (0.05)*	*$ (0.05)*	*($0.16)*
36	**Total benefits ($M)**	**$ 33.12**	**$ 35.76**	**$ 37.91**	**$ 39.73**	**$123.19**
37						
38	**Costs at Minimum**	**Winter 2024-25**	**Winter 2025-26**	**Winter 2026-27**	**Winter 2027-28**	**NPV**
39	Total fuel oil commodity costs ($M)	$ 1.43	$ 1.58	$ 1.72	$ 1.85	
40	Implementation costs ($M)	$ 10.72	$ 11.57	$ 12.27	$ 12.86	
41	**Total costs ($M)**	**$ 12.15**	**$ 13.15**	**$ 13.99**	**$ 14.71**	**$45.38**
42						
43		**Winter 2024-25**	**Winter 2025-26**	**Winter 2026-27**	**Winter 2027-28**	**NPV**
44	**Net benefits at minimum ($M)**	**$ 20.98**	**$ 22.61**	**$ 23.93**	**$ 25.02**	**$77.81**
45	**BCA**	**2.73**	**2.72**	**2.71**	**2.70**	**2.71**
46						
47	**Benefits at Midpoint**	**Winter 2024-25**	**Winter 2025-26**	**Winter 2026-27**	**Winter 2027-28**	**NPV**
48	Capacity savings ($M)	$ 32.52	$ 35.11	$ 37.22	$ 39.01	
49	Avoided annual dth	68,756	74,230	78,703	82,485	
50	Peaking commodity savings ($M)	$ 0.98	$ 1.05	$ 1.12	$ 1.17	
51	*Total avoided dth savings ($M)*	*$ 33.49*	*$ 36.16*	*$ 38.34*	*$ 40.18*	*$124.57*
52	Fuel oil savings (mmBTU/yr)	(58,279)	(62,919)	(66,710)	(69,916)	
53	Peak gas CO2 savings (tons CO2)	3,637	3,927	4,164	4,364	
54	Fuel oil CO2 savings (tons CO2)	(4,320)	(4,664)	(4,945)	(5,183)	
55	Total CO2 savings (tons CO2)	(683)	(737)	(781)	(819)	
56	*Total avoided CO2 savings ($M)*	*$ (0.04)*	*$ (0.05)*	*$ (0.05)*	*$ (0.06)*	*($0.16)*
57	**Total benefits ($M)**	**$ 33.45**	**$ 36.11**	**$ 38.29**	**$ 40.13**	**$124.41**
58						
59	**Costs at Midpoint**	**Winter 2024-25**	**Winter 2025-26**	**Winter 2026-27**	**Winter 2027-28**	**NPV**
60	Total fuel oil commodity costs ($M)	$ 1.44	$ 1.59	$ 1.73	$ 1.87	
61	Implementation costs ($M)	$ 10.82	$ 11.69	$ 12.39	$ 12.99	
62	**Total costs ($M)**	**$ 12.27**	**$ 13.28**	**$ 14.13**	**$ 14.86**	**$45.83**
63						
64		**Winter 2024-25**	**Winter 2025-26**	**Winter 2026-27**	**Winter 2027-28**	**NPV**
65	**Net benefits at midpoint ($M)**	**$ 21.19**	**$ 22.83**	**$ 24.16**	**$ 25.27**	**$78.58**
66	**BCA**	**2.73**	**2.72**	**2.71**	**2.70**	**2.71**
67						
68	**Benefits at Maximum**	**Winter 2024-25**	**Winter 2025-26**	**Winter 2026-27**	**Winter 2027-28**	**NPV**
69	Capacity savings ($M)	$ 33.47	$ 36.14	$ 38.32	$ 40.16	
70	Avoided annual dth	70,778	76,413	81,018	84,911	
71	Peaking commodity savings ($M)	$ 1.01	$ 1.09	$ 1.15	$ 1.21	
72	*Total avoided dth savings ($M)*	*$ 34.48*	*$ 37.22*	*$ 39.47*	*$ 41.36*	*$128.24*
73	Fuel oil savings (mmBTU/yr)	(59,993)	(64,770)	(68,673)	(71,972)	
74	Peak gas CO2 savings (tons CO2)	3,744	4,042	4,286	4,492	
75	Fuel oil CO2 savings (tons CO2)	(4,447)	(4,801)	(5,090)	(5,335)	
76	Total CO2 savings (tons CO2)	(703)	(759)	(804)	(843)	
77	*Total avoided CO2 savings ($M)*	*$ (0.04)*	*$ (0.05)*	*$ (0.05)*	*$ (0.06)*	*($0.17)*
78	**Total benefits ($M)**	**$ 34.44**	**$ 37.18**	**$ 39.41**	**$ 41.31**	**$128.07**
79						
80	**Costs at Maximum**	**Winter 2024-25**	**Winter 2025-26**	**Winter 2026-27**	**Winter 2027-28**	**NPV**
81	Total fuel oil commodity costs ($M)	$ 1.48	$ 1.64	$ 1.79	$ 1.93	
82	Implementation costs ($M)	$ 11.14	$ 12.03	$ 12.75	$ 13.37	
83	**Total costs ($M)**	**$ 12.63**	**$ 13.67**	**$ 14.54**	**$ 15.29**	**$47.18**
84						
85		**Winter 2024-25**	**Winter 2025-26**	**Winter 2026-27**	**Winter 2027-28**	**NPV**
86	**Net benefits at maximum ($M)**	**$ 21.81**	**$ 23.50**	**$ 24.87**	**$ 26.01**	**$80.89**
87	**BCA**	**2.73**	**2.72**	**2.71**	**2.70**	**2.71**

* Minimum targets based on logarithmic regression derived from historical values for winters 2021-22 and 2022-23 and placeholder value of 2,500 for winter 2023-24

KEDNY Energy Efficiency EAM: Low-and-Moderate Income Net Benefits Calculation

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KEDLI Energy Efficiency EAM: Low-and-Moderate Income Net Benefits Calculation

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